Measure Theory and Probability

Lecture notes, Chapters 1–5

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Chapter 1. Measure Spaces, Measurable Functions and Integration

1.1 Systems of sets and measures

Motivation. Measures live on families closed under countable operations, and are determined by any $\cap$-stable generator.

Definition 1.1 (σ-algebra)

Let Ω be a set and $\mathcal{P}(\Omega) := \{A \mid A \subset \Omega\}$ its power set. A family $\mathcal{A} \subset \mathcal{P}(\Omega)$ is a **σ-algebra** (or σ-field) on Ω if

(i) $\Omega \in \mathcal{A}$;

(ii) $A \in \mathcal{A} \Rightarrow A^c := \Omega \setminus A \in \mathcal{A}$;

(iii) $A_i \in \mathcal{A},\ i \in \mathbb{N} \Rightarrow \bigcup_{i\in\mathbb{N}} A_i \in \mathcal{A}$.

"σ" signals *countable* operations.

Example 1.2

(i) $\mathcal{P}(\Omega)$ is a σ-algebra (natural for countable Ω, otherwise too large).

(ii) If $\mathcal{A}_i$, $i \in I$ (I arbitrary, possibly uncountable), are σ-algebras on Ω, then $\bigcap_{i\in I} \mathcal{A}_i = \{A \subset \Omega \mid A \in \mathcal{A}_i\ \forall i\}$ is a σ-algebra.

(iii) (*Generated σ-algebra*) For $\emptyset \neq \mathcal{A}_0 \subset \mathcal{P}(\Omega)$,

$$\sigma(\mathcal{A}_0) := \bigcap_{\mathcal{B}\ \sigma\text{-algebra},\ \mathcal{A}_0\subset\mathcal{B}} \mathcal{B}$$

is a σ-algebra by (ii) and is the smallest σ-algebra containing $\mathcal{A}_0$.

(iv) (*Borel sets*) If Ω is a topological space and $\mathcal{A}_0$ the family of open sets, $\mathcal{B}(\Omega) := \sigma(\mathcal{A}_0)$ is the **Borel σ-algebra**. Open sets, closed sets and countable unions of closed sets are Borel; nevertheless $\mathcal{B}(\mathbb{R}) \neq \mathcal{P}(\mathbb{R})$.

(v) (*Trace*) For a σ-algebra $\mathcal{A}$ on Ω and $\Omega' \subset \Omega$, $\Omega' \cap \mathcal{A} := \{A \cap \Omega' \mid A \in \mathcal{A}\}$ is a σ-algebra on Ω'. If $\Omega' \in \mathcal{A}$, then $\Omega' \cap \mathcal{A} = \{A \in \mathcal{A} \mid A \subset \Omega'\}$.

Definition 1.3 (Measure)

Let $\Omega \neq \emptyset$ and $\mathcal{A}$ a σ-algebra on Ω. A map $\mu : \mathcal{A} \to [0, \infty]$ is a **measure** on $(\Omega, \mathcal{A})$ if

(i) $\mu(\emptyset) = 0$;

(ii) (σ-additivity) $\mu(\dot{\bigcup}_{i\in\mathbb{N}} A_i) = \sum_{i=1}^{\infty} \mu(A_i)$ for pairwise disjoint $A_i \in \mathcal{A}$.

$(\Omega, \mathcal{A}, \mu)$ is a **measure space**, $(\Omega, \mathcal{A})$ a **measurable space**, and the elements of $\mathcal{A}$ are **measurable sets**. If moreover $\mu(\Omega) = 1$, μ is a **probability measure**; we then write $P := \mu$, call $(\Omega, \mathcal{A}, P)$ a **probability space** and its measurable sets **events**.

Notation. $\dot{\bigcup}_{i\in\mathbb{N}} A_i$ (or $A_1 \dot{\cup} A_2$) denotes a union whose members are pairwise disjoint.

Example 1.4

(i) (*Dirac measure*) For $x \in \mathbb{R}^d$, $\delta_x(A) := 1$ if $x \in A$ and 0 otherwise defines a probability measure on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$.

(ii) (*Restriction*) If $(\Omega, \mathcal{A}, \mu)$ is a measure space and $\Omega' \in \mathcal{A}$, then $(\Omega', \mathcal{A} \cap \Omega', \mu)$ is again a measure space.

(iii) (*Counting measure*) On $(\mathbb{N}_0, \mathcal{P}(\mathbb{N}_0))$, $\mu(A) := |A| \in [0, \infty]$ (number of elements) is a measure.

Dynkin systems (λ-systems) are easier to verify and serve uniqueness arguments.

Definition 1.5 (Dynkin system)

$\mathcal{D} \subset \mathcal{P}(\Omega)$ is a **Dynkin system** if (i) $\Omega \in \mathcal{D}$;
(ii) $A \in \mathcal{D} \Rightarrow A^c \in \mathcal{D}$;
(iii) $A_i \in \mathcal{D}$, $i \in \mathbb{N}$, pairwise disjoint $\Rightarrow \dot{\bigcup}_i A_i \in \mathcal{D}$.

Unlike in Definition 1.1, condition (iii) is for disjoint sequences only.

Example 1.6

(i) Every σ-algebra is a Dynkin system.

(ii) If P_1, P_2 are probability measures on $(\Omega, \mathcal{A})$, then $\mathcal{D} := \{A \in \mathcal{A} \mid P_1(A) = P_2(A)\}$ is a Dynkin system (total mass 1, $P_i(A^c) = 1 - P_i(A)$, σ-additivity), in general *not* a σ-algebra.

Remark 1.7

(i) If $\mathcal{D}$ is a Dynkin system and $A, B \in \mathcal{D}$ with $A \subset B$, then $B \setminus A = (B^c \dot{\cup} A)^c \in \mathcal{D}$.

(ii) A Dynkin system that is $\cap$-stable (closed under finite intersections) is a σ-algebra.

(a) For $A, B \in \mathcal{D}$: $A \cup B = A \dot{\cup} (B \setminus (A \cap B)) \in \mathcal{D}$, by (i) and $\cap$-stability.

(b) For $A_i \in \mathcal{D}$: $\bigcup_i A_i = \dot{\bigcup}_i \left(A_i \cap \left(\bigcup_{n<i} A_n\right)^c\right)$, and each piece lies in $\mathcal{D}$ by (a), $\cap$-stability and closure under complements.

Theorem 1.8 (π–λ theorem)

Let $\mathcal{A}_0 \subset \mathcal{P}(\Omega)$ be $\cap$-stable. Then $\sigma(\mathcal{A}_0) = \mathcal{D}(\mathcal{A}_0)$, where

$$\mathcal{D}(\mathcal{A}_0) := \bigcap_{\mathcal{D} \text{ Dynkin system},\ \mathcal{A}_0 \subset \mathcal{D}} \mathcal{D}$$

is the Dynkin system generated by $\mathcal{A}_0$.

Proof.

Step 1. $\sigma(\mathcal{A}_0)$ is a Dynkin system containing $\mathcal{A}_0$, so $\mathcal{D}(\mathcal{A}_0) \subset \sigma(\mathcal{A}_0)$.

Step 2. For the converse it suffices, by Remark 1.7(ii), to show that $\mathcal{D}(\mathcal{A}_0)$ is $\cap$-stable. For $D \in \mathcal{D}(\mathcal{A}_0)$ put

$$\mathcal{D}_D := \{Q \subset \Omega \mid Q \cap D \in \mathcal{D}(\mathcal{A}_0)\}.$$

It is a Dynkin system: $\Omega \cap D = D$; $Q^c \cap D = D \setminus (Q \cap D)$ (Remark 1.7(i)); disjoint unions pass through $\cap D$.

Step 3. For $E \in \mathcal{A}_0$, $\mathcal{A}_0 \subset \mathcal{D}_E$ ($\cap$-stability), so $\mathcal{D}(\mathcal{A}_0) \subset \mathcal{D}_E$: $E \cap D \in \mathcal{D}(\mathcal{A}_0)$ for all $D \in \mathcal{D}(\mathcal{A}_0)$.

Step 4. So $\mathcal{A}_0 \subset \mathcal{D}_D$, hence $\mathcal{D}(\mathcal{A}_0) \subset \mathcal{D}_D$ for every $D \in \mathcal{D}(\mathcal{A}_0)$: $\cap$-stability. ■

Theorem 1.9 (Uniqueness of measures)

Let μ_1, μ_2 be measures on $(\Omega, \mathcal{A})$ and $\mathcal{A}_0 \subset \mathcal{A}$ a $\cap$-stable family with $\sigma(\mathcal{A}_0) = \mathcal{A}$ and $\mu_1 = \mu_2$ on $\mathcal{A}_0$. Suppose there are $\Omega_n \in \mathcal{A}_0$ with $\Omega_n \nearrow \Omega$ (i.e. $\Omega_n \subset \Omega_{n+1}$, $\bigcup_n \Omega_n = \Omega$) and $\mu_1(\Omega_n) = \mu_2(\Omega_n) < \infty$. Then $\mu_1 = \mu_2$. If μ_1, μ_2 are probability measures, the sets Ω_n are not needed.

Proof.

Step 1 (probability measures). By Example 1.6(ii), $\mathcal{D} := \{A \in \mathcal{A} \mid \mu_1(A) = \mu_2(A)\}$ is a Dynkin system containing $\mathcal{A}_0$. By Theorem 1.8, $\mathcal{A} = \sigma(\mathcal{A}_0) = \mathcal{D}(\mathcal{A}_0) \subset \mathcal{D}$, so $\mu_1 = \mu_2$.

Step 2 (localisation). For $E \in \mathcal{A}_0$ with $\mu_1(E) = \mu_2(E) < \infty$, $\mathcal{D}_E := \{A \in \mathcal{A} \mid \mu_1(A \cap E) = \mu_2(A \cap E)\}$ is a Dynkin system ($\mu_i(A^c \cap E) = \mu_i(E) - \mu_i(A \cap E)$ by finiteness) containing $\mathcal{A}_0$; Theorem 1.8 gives $\mathcal{D}_E = \mathcal{A}$.

Step 3. $E = \Omega_n$ and continuity from below (Theorem 1.15(ii)): $\mu_1(A) = \lim_n \mu_1(A \cap \Omega_n) = \lim_n \mu_2(A \cap \Omega_n) = \mu_2(A)$ for every $A \in \mathcal{A}$. ■

Measures are first defined on rings, then extended.

Definition 1.10 (Rings, measures on rings, σ-finiteness)

(i) $\mathcal{R} \subset \mathcal{P}(\Omega)$ is a **ring** if $A, B \in \mathcal{R} \Rightarrow A \setminus B \in \mathcal{R}$ and $A \cup B \in \mathcal{R}$. Then $\emptyset \in \mathcal{R}$ and $A \cap B = A \setminus (A \setminus B) \in \mathcal{R}$, so rings are $\cap$-stable. A ring is a **σ-ring** if it is closed under countable unions; then also $\bigcap_n A_n = A_1 \setminus \bigcup_n (A_1 \setminus A_n) \in \mathcal{R}$. A ring containing Ω is an **algebra**, and a σ-ring containing Ω is a σ-algebra.

(ii) A **measure on a ring** $\mathcal{R}$ is defined as in Definition 1.3, with σ-additivity required only for disjoint $A_i \in \mathcal{R}$ whose union lies in $\mathcal{R}$.

(iii) A measure μ on a ring $\mathcal{R}$ is **σ-finite** if there are $\Omega_n \in \mathcal{R}$ with $\Omega_n \nearrow \Omega$ and $\mu(\Omega_n) < \infty$.

Theorem 1.11 (Carathéodory's extension theorem)

Let μ_0 be a σ-finite measure on a ring $\mathcal{R}$. Then there is a unique σ-finite measure μ on $\sigma(\mathcal{R})$ with $\mu = \mu_0$ on $\mathcal{R}$.

Proof [Supplement].

Step 1 (uniqueness). $\mathcal{R}$ is $\cap$-stable and $\mu_1(\Omega_n) = \mu_2(\Omega_n) = \mu_0(\Omega_n) < \infty$; apply Theorem 1.9.

Step 2 (outer measure). For $Q \subset \Omega$ put

$$\mu^*(Q) := \inf\Big\{ \sum_{n=1}^{\infty} \mu_0(A_n) \;\Big|\; A_n \in \mathcal{R},\ Q \subset \bigcup_n A_n \Big\}.$$

Covers exist ($\Omega_n \nearrow \Omega$); $\mu^*(\emptyset) = 0$, μ^* is monotone and σ-subadditive: given $Q = \bigcup_k Q_k$ with $\mu^*(Q_k) < \infty$ and $\varepsilon > 0$, choose covers $(A_{kn})_n$ of Q_k with $\sum_n \mu_0(A_{kn}) \le \mu^*(Q_k) + \varepsilon 2^{-k}$; then $(A_{kn})_{k,n}$ covers Q and $\mu^*(Q) \le \sum_k \mu^*(Q_k) + \varepsilon$.

Step 3 ($\mu^* = \mu_0$ on $\mathcal{R}$). $\mu^*(A) \le \mu_0(A)$ (cover by A). If $A \subset \bigcup_n A_n$, $A_n \in \mathcal{R}$, the disjoint $B_n := A \cap A_n \setminus \bigcup_{k<n} A_k \in \mathcal{R}$ have union A, so $\mu_0(A) = \sum_n \mu_0(B_n) \le \sum_n \mu_0(A_n)$. Hence $\mu_0(A) \le \mu^*(A)$.

Step 4 (Carathéodory-measurable sets). Let $\mathcal{M}^* := \{A \subset \Omega \mid \mu^*(Q) = \mu^*(Q \cap A) + \mu^*(Q \setminus A)\ \forall Q \subset \Omega\}$. Only "$\ge$" for $\mu^*(Q) < \infty$ needs checking. $\Omega \in \mathcal{M}^*$, and complements by symmetry. For

$A, B \in \mathcal{M}^*$,

$$\mu^*(Q) = \mu^*(Q \cap A) + \mu^*(Q \cap A^c \cap B) + \mu^*(Q \cap A^c \cap B^c) \geq \mu^*(Q \cap (A \cup B)) + \mu^*(Q \setminus (A \cup B)),$$

by $Q \cap (A \cup B) = (Q \cap A) \cup (Q \cap A^c \cap B)$: an algebra. For disjoint $A_1, \ldots, A_n \in \mathcal{M}^*$, by induction $\mu^*(Q \cap \bigcup_{i \leq n} A_i) = \sum_{i \leq n} \mu^*(Q \cap A_i)$. Now let $A_i \in \mathcal{M}^*$ be disjoint, $A := \bigcup_i A_i$, $S_n := \bigcup_{i \leq n} A_i \in \mathcal{M}^*$. Then

$$\mu^*(Q) = \mu^*(Q \cap S_n) + \mu^*(Q \setminus S_n) \geq \sum_{i \leq n} \mu^*(Q \cap A_i) + \mu^*(Q \setminus A),$$

and as $n \to \infty$, by σ-subadditivity, $\mu^*(Q) \geq \sum_i \mu^*(Q \cap A_i) + \mu^*(Q \setminus A) \geq \mu^*(Q \cap A) + \mu^*(Q \setminus A)$. So $A \in \mathcal{M}^*$, and disjointifying (Remark 1.7(ii)(b)) makes $\mathcal{M}^*$ a σ-algebra; $Q = A$ gives σ-additivity.

Step 5 ($\mathcal{R} \subset \mathcal{M}^*$). For $A \in \mathcal{R}$, $\mu^*(Q) < \infty$, and a cover $(A_n) \subset \mathcal{R}$ of Q with $\sum \mu_0(A_n) \leq \mu^*(Q) + \varepsilon$: $(A_n \cap A)$ covers $Q \cap A$, $(A_n \setminus A)$ covers $Q \setminus A$, and $\mu_0(A_n) = \mu_0(A_n \cap A) + \mu_0(A_n \setminus A)$, so $\mu^*(Q \cap A) + \mu^*(Q \setminus A) \leq \mu^*(Q) + \varepsilon$.

Step 6. $\sigma(\mathcal{R}) \subset \mathcal{M}^*$, and $\mu := \mu^*|_{\sigma(\mathcal{R})}$ extends μ_0 (Step 3), σ-finite as $\mu(\Omega_n) < \infty$. ■

Remark 1.12

The lectures construct Lebesgue measure directly (Section 1.2, after Rudin, *Principles*, Ch. 11): the same outer-measure idea on the ring of elementary sets, using compactness, with measurability defined by approximation and σ-additivity obtained only on the resulting σ-ring.

1.2 Construction of the Lebesgue measure

Motivation. Extend box volume σ-additively: outer measure, a pseudo-metric on sets, then sets approximable by elementary sets.

Definition 1.13 (Additive and σ-additive set functions)

Let $\varphi : \mathcal{R} \to [0, \infty]$ on a ring $\mathcal{R}$ with $\varphi(\emptyset) = 0$. φ is **additive** if

$$A, B \in \mathcal{R},\ A \cap B = \emptyset \implies \varphi(A \cup B) = \varphi(A) + \varphi(B), \tag{1.1}$$

and **σ-additive** (a measure on $\mathcal{R}$) if $\varphi(\dot\bigcup_i A_i) = \sum_i \varphi(A_i)$ whenever the $A_i \in \mathcal{R}$ are pairwise disjoint and $\bigcup_i A_i \in \mathcal{R}$.

Remark 1.14

If φ is additive on $\mathcal{R}$, then by induction it is finitely additive, $\varphi(A_1 \dot\cup \cdots \dot\cup A_n) = \sum_i \varphi(A_i)$, and for all $A, B \in \mathcal{R}$:

$$\varphi(A \cup B) + \varphi(A \cap B) = \varphi(A) + \varphi(B), \tag{1.2}$$

$$A \subset B \Rightarrow \varphi(A) \leq \varphi(B) \quad \text{(monotonicity)}, \tag{1.3}$$

$$A \subset B,\ \varphi(A) < \infty \Rightarrow \varphi(B \setminus A) = \varphi(B) - \varphi(A),$$

$$\varphi\Big(\bigcup_{i \leq n} A_i\Big) \leq \sum_{i \leq n} \varphi(A_i) \quad \text{(finite subadditivity)}.$$

(1.2): decompose $A \cup B = (A \setminus B) \dot\cup (A \cap B) \dot\cup (B \setminus A)$, $A = (A \setminus B) \dot\cup (A \cap B)$, $B = (B \setminus A) \dot\cup (A \cap B)$. Monotonicity: $\varphi(B) = \varphi(A) + \varphi(B \setminus A)$; subadditivity: (1.2) and induction.

Notation. $A_n \nearrow A$ means $A_n \subset A_{n+1}$ and $A = \bigcup_n A_n$; $A_n \searrow A$ means $A_n \supset A_{n+1}$ and $A = \bigcap_n A_n$.

Theorem 1.15 (Continuity of measures)

Let φ be additive on a ring $\mathcal{R}$ and consider

(i) φ is σ-additive on $\mathcal{R}$;

(ii) (continuity from below) $A_n \in \mathcal{R}$, $A_n \nearrow A \in \mathcal{R} \Rightarrow \varphi(A) = \lim_n \varphi(A_n)$;

(iii) (continuity from above) $A_n \in \mathcal{R}$, $A_n \searrow A \in \mathcal{R}$, $\varphi(A_{n_0}) < \infty$ for some $n_0 \Rightarrow \varphi(A) = \lim_n \varphi(A_n)$.

Then (i) $\Leftrightarrow$ (ii) $\Rightarrow$ (iii).

Proof.

Step (i)⇒(ii). The disjoint $B_1 := A_1$, $B_n := A_n \setminus A_{n-1}$ give $A_n = B_1 \dot\cup \cdots \dot\cup B_n$, $\bigcup_n B_n = A$, so $\lim_n \varphi(A_n) = \lim_n \sum_{i\le n} \varphi(B_i) = \sum_i \varphi(B_i) = \varphi(A)$.

Step (ii)⇒(i). For disjoint $A_n \in \mathcal{R}$ with $A := \bigcup_n A_n \in \mathcal{R}$, $B_N := \bigcup_{n\le N} A_n \nearrow A$, so $\varphi(A) = \lim_N \varphi(B_N) = \lim_N \sum_{n\le N} \varphi(A_n)$.

Step (ii)⇒(iii) [Supplement]. $C_n := A_{n_0} \setminus A_n \nearrow A_{n_0} \setminus A$ $(n \ge n_0)$, all values finite, so by Remark 1.14 and (ii)

$$\varphi(A_{n_0}) - \varphi(A) = \varphi(A_{n_0} \setminus A) = \lim_n \varphi(C_n) = \varphi(A_{n_0}) - \lim_n \varphi(A_n).$$

■

Definition 1.16 (Intervals and elementary sets)

For $a, b \in \mathbb{R}^d$ with $a_i \le b_i$, an **interval** (bounded box) is any set of points x with $a_i \lhd_i x_i \lhd_i' b_i$, $\lhd_i, \lhd_i' \in \{\le, <\}$. It is open (closed) if all inequalities are strict (non-strict); $a_i = b_i$ is allowed, so an interval may be empty or a point. An **elementary set** is a finite union of intervals; $\mathcal{E}$ denotes the family of elementary sets. For an interval I put $\lambda(I) := \prod_{i=1}^d (b_i - a_i)$, and for $A = I_1 \dot\cup \cdots \dot\cup I_n$ with disjoint intervals

$$\lambda(A) := \sum_{i=1}^n \lambda(I_i). \tag{1.4}$$

$\mathcal{E}$ is a ring but not a σ-ring ($\mathbb{Q} \cap [0,1]$). Every $A \in \mathcal{E}$ is a finite *disjoint* union of intervals (cells of the grid of all endpoints), and (1.4) is independent of the decomposition (common grid refinement). So λ is finite, additive and nonnegative on $\mathcal{E}$.

Definition 1.17 (Regularity)

An additive $\varphi : \mathcal{E} \to [0, \infty]$ is **regular** if for every $A \in \mathcal{E}$ and $\varepsilon > 0$ there are $F, G \in \mathcal{E}$, F closed, G open, with $F \subset A \subset G$ and $\varphi(G) - \varepsilon \le \varphi(A) \le \varphi(F) + \varepsilon$.

Example 1.18

λ is regular and finite on $\mathcal{E}$: shrink or enlarge each interval to a closed or open one, changing the volume by less than ε.

From now on μ is a finite, regular, additive set function on $\mathcal{E}$ (e.g. λ).

Definition 1.19 (Outer measure)

For $E \subset \mathbb{R}^d$,

$$\mu^*(E) := \inf\Big\{ \sum_{n=1}^{\infty} \mu(A_n) \;\Big|\; A_n \in \mathcal{E} \text{ open},\ E \subset \bigcup_n A_n \Big\}.$$

Clearly $\mu^*(E) \geq 0$, $\mu^*(\emptyset) = 0$, and $E_1 \subset E_2 \Rightarrow \mu^*(E_1) \leq \mu^*(E_2)$.

Theorem 1.20

(i) $\mu^*(A) = \mu(A)$ for $A \in \mathcal{E}$, so μ^* extends μ to $\mathcal{P}(\mathbb{R}^d)$.

(ii) (σ-subadditivity) If $E = \bigcup_n E_n$ with arbitrary $E_n \subset \mathbb{R}^d$, then

$$\mu^*(E) \leq \sum_{n=1}^{\infty} \mu^*(E_n). \tag{1.5}$$

Proof. (i) By regularity an open $G \supset A$ in $\mathcal{E}$ has $\mu(G) \leq \mu(A) + \varepsilon$, so $\mu^*(A) \leq \mu(A)$.

Conversely, take an open cover $A_n \in \mathcal{E}$ with $\sum_n \mu(A_n) \leq \mu^*(A) + \varepsilon$ and closed $F \subset A$ with $\mu(A) \leq \mu(F) + \varepsilon$. F is compact, so $F \subset A_1 \cup \cdots \cup A_N$ and

$$\mu(A) \leq \mu(F) + \varepsilon \leq \mu(A_1 \cup \cdots \cup A_N) + \varepsilon \leq \sum_{k \leq N} \mu(A_k) + \varepsilon \leq \mu^*(A) + 2\varepsilon.$$

(ii) Let all $\mu^*(E_n) < \infty$ and choose open covers $(A_{nk})_k \subset \mathcal{E}$ of E_n with $\sum_k \mu(A_{nk}) \leq \mu^*(E_n) + 2^{-n}\varepsilon$; together they cover E, so $\mu^*(E) \leq \sum_n \sum_k \mu(A_{nk}) \leq \sum_n \mu^*(E_n) + \varepsilon$. ∎

Definition 1.21 (Approximation by elementary sets)

For $A, B \subset \mathbb{R}^d$ let

$$S(A,B) := (A \setminus B) \cup (B \setminus A) \quad \text{(symmetric difference)}, \qquad d(A,B) := \mu^*(S(A,B)) \in [0,\infty].$$

Write $A_n \to A$ if $d(A, A_n) \to 0$, and define

$$A \in \mathcal{M}_F(\mu) \text{ (finitely } \mu\text{-measurable)} :\Longleftrightarrow \exists A_n \in \mathcal{E} :\ A_n \to A,$$
$$A \in \mathcal{M}(\mu) \text{ } (\mu\text{-measurable)} :\Longleftrightarrow A = \textstyle\bigcup_n A_n \text{ with } A_n \in \mathcal{M}_F(\mu).$$

$\mathcal{E} \subset \mathcal{M}_F(\mu) \subset \mathcal{M}(\mu)$, and $A \in \mathcal{M}_F(\mu)$ has $\mu^*(A) < \infty$: $A \subset S(A, A_n) \cup A_n$, so $\mu^*(A) \leq d(A, A_n) + \mu(A_n) < \infty$.

Theorem 1.22

$\mathcal{M}(\mu)$ is a σ-ring and μ^* is σ-additive on $\mathcal{M}(\mu)$.

Lemma 1.23

For $A, B, C \subset \mathbb{R}^d$:

(1) $S(A,B) = S(B,A)$;

(2) $S(A,B) \subset S(A,C) \cup S(C,B)$;

(3) $S(A,B) = S(A^c, B^c)$;

(4) for A_1, A_2, B_1, B_2, each of $S(A_1 \cup A_2, B_1 \cup B_2)$, $S(A_1 \cap A_2, B_1 \cap B_2)$, $S(A_1 \setminus A_2, B_1 \setminus B_2)$

is contained in $S(A_1, B_1) \cup S(A_2, B_2)$.

Consequently, by monotonicity and subadditivity of μ^*: (i) $d(A, B) = d(B, A)$;
(ii) $d(A, B) \le d(A, C) + d(C, B)$;
(iii) $A = B \Rightarrow d(A, B) = 0$ (not conversely);
(iv) $d(A_1 \star A_2, B_1 \star B_2) \le d(A_1, B_1) + d(A_2, B_2)$ for $\star \in \{\cup, \cap, \setminus\}$.

Proof. (1) is clear.

(2) $x \in A \setminus B$ lies in $C \setminus B$ or $A \setminus C$, so $A \setminus B \subset (A \setminus C) \cup (C \setminus B)$; symmetrically $B \setminus A \subset (B \setminus C) \cup (C \setminus A)$.

(3) $S(A, B) = (A \cap B^c) \cup (B \cap A^c) = (B^c \setminus A^c) \cup (A^c \setminus B^c) = S(A^c, B^c)$.

(4) $(A_1 \cup A_2) \setminus (B_1 \cup B_2) \subset (A_1 \setminus B_1) \cup (A_2 \setminus B_2)$ and symmetrically, so

$$S(A_1 \cup A_2, B_1 \cup B_2) \subset S(A_1, B_1) \cup S(A_2, B_2). \tag{1.6}$$

Intersections, by (3) and (1.6): $S(A_1 \cap A_2, B_1 \cap B_2) = S(A_1^c \cup A_2^c, B_1^c \cup B_2^c) \subset S(A_1^c, B_1^c) \cup S(A_2^c, B_2^c) = S(A_1, B_1) \cup S(A_2, B_2)$. Differences: $A_1 \setminus A_2 = A_1 \cap A_2^c$ and (3). ∎

Lemma 1.24

If $\mu^*(A) < \infty$ or $\mu^*(B) < \infty$, then

$$|\mu^*(A) - \mu^*(B)| \le d(A, B). \tag{1.7}$$

Proof. Say $\mu^*(B) \le \mu^*(A)$, so $\mu^*(B) < \infty$; by Lemma 1.23(ii),

$$\mu^*(A) = d(A, \emptyset) \le d(A, B) + d(B, \emptyset) = d(A, B) + \mu^*(B).$$

The other case is symmetric. ∎

Remark 1.25

(i) For $\mu = \lambda$, $d = 1$, a countable $A \subset \mathbb{R}$ and $B = \emptyset$: $d(A, B) = \lambda^*(A) \le \sum_{p \in A} \lambda([p, p]) = 0$, although $A \ne B$: d is only a pseudo-metric.

(ii) On $\mathcal{F} := \{A \subset \mathbb{R}^d \mid \mu^*(A) < \infty\} \supset \mathcal{M}_F(\mu)$, modulo $d(A, B) = 0$, $(\mathcal{F}, d)$ is a metric space and $\mathcal{M}_F(\mu) = \overline{\mathcal{E}}$. [Supplement] *Reason:* d is finite on $\mathcal{F}$ (subadditivity), symmetric with triangle inequality (Lemma 1.23); $\mathcal{M}_F(\mu) \subset \overline{\mathcal{E}}$ by definition, and conversely by a diagonal choice.

Proof of Theorem 1.22.

Step 1 ($\mathcal{M}_F(\mu)$ is a ring). For elementary $A_n \to A$, $B_n \to B$, by Lemmas 1.23(iv) and 1.24,

$$\begin{aligned} &d(A_n \cup B_n, A \cup B) + d(A_n \cap B_n, A \cap B) + d(A_n \setminus B_n, A \setminus B) + |\mu^*(A_n) - \mu^*(A)| + |\mu^*(B_n) - \mu^*(B)| \\ &\quad + |\mu^*(A_n \cup B_n) - \mu^*(A \cup B)| + |\mu^*(A_n \cap B_n) - \mu^*(A \cap B)| \le 6\big(d(A_n, A) + d(B_n, B)\big) \to 0. \end{aligned}$$

Hence $A \cup B$, $A \setminus B \in \mathcal{M}_F(\mu)$ and

$$\mu^*(A_n) \to \mu^*(A), \quad \mu^*(A_n \cup B_n) \to \mu^*(A \cup B), \quad \mu^*(A_n \cap B_n) \to \mu^*(A \cap B). \tag{1.8}$$

Step 2 (μ^* is additive on $\mathcal{M}_F(\mu)$). Pass to the limit in (1.2) for A_n, B_n using (1.8): $\mu^*(A) + \mu^*(B) = \mu^*(A \cup B) + \mu^*(A \cap B)$, and $\mu^*(A \cap B) = 0$ if $A \cap B = \emptyset$.

Step 3 (disjoint representation). Disjointify $A = \bigcup_n A'_n$, $A'_n \in \mathcal{M}_F(\mu)$: $B'_n := A'_1 \cup \cdots \cup A'_n$, $A_1 := B'_1$, $A_n := B'_n \setminus B'_{n-1}$. Then $\mu^*(A) \le \sum_n \mu^*(A_n)$ (σ-subadditivity) and $\sum_{k\le n} \mu^*(A_k) = \mu^*(A_1 \dot\cup \cdots \dot\cup A_n) \le \mu^*(A)$, so

$$\mu^*(A) = \sum_{n=1}^{\infty} \mu^*(A_n). \tag{1.9}$$

Step 4 (finite measure implies $\mathcal{M}_F$). For $\mu^*(A) < \infty$ and $B_n := A_1 \cup \cdots \cup A_n$, $d(A, B_n) = \mu^*(\bigcup_{k>n} A_k) = \sum_{k>n} \mu^*(A_k) \to 0$ by (1.9) (Step 3 applied to $\bigcup_{k>n} A_k$). Elementary B_{n,k_n} with $d(B_{n,k_n}, B_n) < 1/n$ then converge to A:

$$A \in \mathcal{M}(\mu),\ \mu^*(A) < \infty \implies A \in \mathcal{M}_F(\mu). \tag{1.10}$$

Step 5 (σ-additivity on $\mathcal{M}(\mu)$). For disjoint $A_n \in \mathcal{M}(\mu)$ with union A: if $\mu^*(A) = \infty$, $\sum_n \mu^*(A_n) = \infty$ by σ-subadditivity; otherwise $A_n \in \mathcal{M}_F(\mu)$ by (1.10) and (1.9) applies.

Step 6 ($\mathcal{M}(\mu)$ is a σ-ring). $\bigcup_n A_n = \bigcup_{(n,k)\in\mathbb{N}^2} A_{nk} \in \mathcal{M}(\mu)$. For $A = \bigcup_n A_n$, $B = \bigcup_l B_l$ ($A_n, B_l \in \mathcal{M}_F(\mu)$), $A_n \cap B = \bigcup_l (A_n \cap B_l) \in \mathcal{M}(\mu)$ has finite measure, so lies in $\mathcal{M}_F(\mu)$ by (1.10); then $A_n \setminus B = A_n \setminus (A_n \cap B) \in \mathcal{M}_F(\mu)$ and $A \setminus B = \bigcup_n (A_n \setminus B) \in \mathcal{M}(\mu)$. ∎

Definition 1.26 (Lebesgue measure)

$\mu(A) := \mu^*(A)$ on $\mathcal{M}(\mu)$ extends μ to a measure on the σ-ring $\mathcal{M}(\mu)$ (Definition 1.10(ii)); for $\mu = \lambda$ it is the **Lebesgue measure** on $\mathbb{R}^d$.

Remark 1.27

(i) $\mathcal{M}(\mu)$ contains the open sets (countable unions of open intervals, via rational balls), hence $\mathbb{R}^d$: it is a σ-algebra containing the closed sets.

(ii) (*Regularity*) For $A \in \mathcal{M}(\mu)$, $\varepsilon > 0$ there are closed $F \subset A \subset G$ open with $\mu(G \setminus A) < \varepsilon$, $\mu(A \setminus F) < \varepsilon$: G by Proposition 1.28, and $F := \tilde G^c$ for an open $\tilde G \supset A^c$ with $\mu(\tilde G \setminus A^c) < \varepsilon$, since $A \setminus F = \tilde G \setminus A^c$.

(iii) With $G_n \supset A \supset F_n$ for $\varepsilon = 1/n$, $\mu(\bigcap_n G_n \setminus A) = 0 = \mu(A \setminus \bigcup_n F_n)$: $A = B \dot\cup (A \setminus B)$, $B := \bigcup_n F_n \in \mathcal{B}(\mathbb{R}^d)$, $A \setminus B$ null. Measurable = Borel + null, and $\mathcal{B}(\mathbb{R}^d) \subset \mathcal{M}(\mu)$.

(iv) The μ-null sets form a σ-ring, and $\mu^*(N) = 0$ makes N null. [Supplement] *Reason:* σ-subadditivity and monotonicity; $d(N, \emptyset) = 0$ gives $N \in \mathcal{M}_F(\mu)$.

(v) For λ every countable set is a null set, but there are uncountable null sets, e.g. the Cantor set.

Proposition 1.28

μ is regular on $\mathcal{M}(\mu)$: for $A \in \mathcal{M}(\mu)$ and $\varepsilon > 0$ there is an open $G \supset A$ with $\mu(G \setminus A) < \varepsilon$ (and hence Remark 1.27(ii) holds).

Proof.

Step 1 ($\mu(A) < \infty$). $A \in \mathcal{M}_F(\mu)$ by (1.10); open $O_n \in \mathcal{E}$ cover A with $\sum_n \mu(O_n) < \mu(A) + \varepsilon$, so $G := \bigcup_n O_n$ has $\mu(G \setminus A) = \mu(G) - \mu(A) < \varepsilon$.

Step 2 ($\mu(A) = \infty$) [Supplement]. $A = \bigcup_n A_n$, $A_n \in \mathcal{M}_F(\mu)$; Step 1 gives open $G_n \supset A_n$, $\mu(G_n \setminus A_n) < \varepsilon 2^{-n}$, and $G := \bigcup_n G_n$ has $G \setminus A \subset \bigcup_n (G_n \setminus A_n)$. ∎

1.3 Measure spaces and measurable functions

Motivation. On an arbitrary $(X, \mathcal{M}, \mu)$, measurable functions are stable under limits, sums, products and continuous maps.

Example 1.29

(i) $(\mathbb{R}^d, \mathcal{M}(\lambda), \lambda)$ is the **Lebesgue measure space**.
(ii) $\lambda|_{\mathcal{B}(\mathbb{R}^d)}$ is the **Lebesgue–Borel measure**, and $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda)$ the Lebesgue–Borel measure space (also just "Lebesgue measure").

Definition 1.30 (Measurable function)

An extended real-valued $f : X \to \overline{\mathbb{R}} = [-\infty, \infty]$ is **measurable** if $\{f > \alpha\} := \{x \in X \mid f(x) \in (\alpha, \infty]\} \in \mathcal{M}$ for every $\alpha \in \mathbb{R}$.

For $\mathcal{M}(\lambda)$: *Lebesgue measurable*; for $\mathcal{B}(\mathbb{R}^d)$: *Borel measurable.*

Example 1.31

(i) Borel measurable functions are Lebesgue measurable. The converse fails: $\mathcal{B}(\mathbb{R}^d)$ has the cardinality $\mathfrak{c}$ of the continuum while $\mathcal{M}(\lambda)$ has cardinality $2^{\mathfrak{c}}$ (Dudley, *Real Analysis and Probability*, Prop. 3.4.1), so there is a Lebesgue measurable, non-Borel set L, and $\mathbb{1}_L$ is Lebesgue but not Borel measurable. (ii) A continuous $f : \mathbb{R}^d \to \mathbb{R}$ is Borel measurable, since $\{f > \alpha\} = f^{-1}((\alpha, \infty))$ is open.

Remark 1.32

For measurable spaces $(X, \mathcal{M})$, $(\tilde{X}, \tilde{\mathcal{M}})$, a map $T : X \to \tilde{X}$ is $\mathcal{M}/\tilde{\mathcal{M}}$**-measurable** if $T^{-1}(\tilde{A}) \in \mathcal{M}$ for all $\tilde{A} \in \tilde{\mathcal{M}}$. If $\tilde{\mathcal{M}} = \sigma(\tilde{\mathcal{M}}_0)$, checking $\tilde{A} \in \tilde{\mathcal{M}}_0$ suffices. [Supplement] *Reason:* $\{\tilde{A} \mid T^{-1}(\tilde{A}) \in \mathcal{M}\}$ is a σ-algebra containing $\tilde{\mathcal{M}}_0$. In particular, with $\mathcal{B}(\overline{\mathbb{R}}) := \{A \subset \overline{\mathbb{R}} \mid A \cap \mathbb{R} \in \mathcal{B}(\mathbb{R})\} = \sigma(\{(\alpha, \infty] \mid \alpha \in \mathbb{R}\})$, Definition 1.30 is exactly $\mathcal{M}/\mathcal{B}(\overline{\mathbb{R}})$-measurability.

Theorem 1.33

For $f : X \to \overline{\mathbb{R}}$ the following are equivalent: (i) $\{f > \alpha\} \in \mathcal{M}\ \forall \alpha \in \mathbb{R}$;
(ii) $\{f \geq \alpha\} \in \mathcal{M}\ \forall \alpha$;
(iii) $\{f < \alpha\} \in \mathcal{M}\ \forall \alpha$;
(iv) $\{f \leq \alpha\} \in \mathcal{M}\ \forall \alpha$.

Proof. For $\alpha \in \mathbb{R}$:

(i)⇒(ii): $\{f \geq \alpha\} = \bigcap_n \{f > \alpha - \frac{1}{n}\}$.

(ii)⇒(iii): $\{f < \alpha\} = X \setminus \{f \geq \alpha\}$.

(iii)⇒(iv): $\{f \leq \alpha\} = \bigcap_n \{f < \alpha + \frac{1}{n}\}$.

(iv)⇒(i): $\{f > \alpha\} = X \setminus \{f \leq \alpha\}$. ∎

Theorem 1.34

f measurable $\Rightarrow |f|$ measurable.

Proof. $\{|f| < \alpha\} = \{f < \alpha\} \cap \{f > -\alpha\} \in \mathcal{M}$; apply Theorem 1.33. ■

Theorem 1.35

If f_n, $n \in \mathbb{N}$, are measurable, so are $\inf_n f_n$, $\sup_n f_n$, $\liminf_n f_n$ and $\limsup_n f_n$.

Proof. $\{\sup_n f_n > \alpha\} = \bigcup_n \{f_n > \alpha\}$ and $\{\inf_n f_n < \alpha\} = \bigcup_n \{f_n < \alpha\}$. Then $\limsup_n f_n = \inf_n \sup_{m \geq n} f_m$ and $\liminf_n f_n = \sup_n \inf_{m \geq n} f_m$ are measurable by two applications. ■

Corollary 1.36

(i) f, g measurable $\Rightarrow$ $\max(f, g)$, $\min(f, g)$ measurable; in particular $f^+ := \max(f, 0)$ and $f^- := -\min(f, 0)$ are measurable. (ii) A pointwise limit of measurable functions is measurable.

Proof. (i) is Theorem 1.35 for the sequence $f, g, g, \ldots$.

(ii) $f = \limsup_n f_n$. ■

Theorem 1.37

Let $f, g : X \to \mathbb{R}$ be measurable and $F : \mathbb{R}^2 \to \mathbb{R}$ continuous. Then $h(x) := F(f(x), g(x))$ is measurable; in particular $f + g$ and fg are measurable.

Proof. $G_\alpha := \{F > \alpha\} \subset \mathbb{R}^2$ is open, so (Remark 1.27(i)) a countable union of open rectangles $I_n = (\alpha_n, \beta_n) \times (\gamma_n, \delta_n)$. Each $\{(f, g) \in I_n\} = \{\alpha_n < f < \beta_n\} \cap \{\gamma_n < g < \delta_n\} \in \mathcal{M}$, so $\{h > \alpha\} = \bigcup_n \{(f, g) \in I_n\} \in \mathcal{M}$. ■

Remark 1.38

Compositions of Lebesgue measurable functions need not be Lebesgue measurable. There is a continuous increasing $f : [0, 1] \to [0, 1]$ and a Lebesgue measurable L with $f^{-1}(L)$ not Lebesgue measurable (Dudley, Prop. 4.2.1); then $g = \mathbb{1}_L$ gives $(g \circ f)^{-1}(\{1\}) = f^{-1}(L) \notin \mathcal{M}(\lambda)$. But $g \circ f$ is Lebesgue measurable for Borel g and Lebesgue measurable f (Borel if both are), and every Lebesgue measurable f is a.e. (Definition 1.49) Borel.

Definition 1.39 (Simple functions)

$f : X \to \mathbb{R}$ is **simple** if $\mathrm{im}(f) = \{f(x) \mid x \in X\}$ is finite. If $\mathrm{im}(f) = \{\alpha_1, \ldots, \alpha_n\}$ with distinct α_i and $E_i := \{f = \alpha_i\}$, then

$$f = \sum_{i=1}^{n} \alpha_i \mathbb{1}_{E_i}, \qquad \mathbb{1}_E(x) := \begin{cases} 1 & x \in E \\ 0 & x \notin E \end{cases} \tag{1.11}$$

($\mathbb{1}_E$ is the **indicator** or characteristic function of E). Since $\{\mathbb{1}_E \leq \alpha\}$ is $\emptyset$, E^c or X according as $\alpha < 0$, $0 \leq \alpha < 1$, $\alpha \geq 1$, the function $\mathbb{1}_E$ is measurable iff $E \in \mathcal{M}$; and f in (1.11) is measurable iff $E_1, \ldots, E_n \in \mathcal{M}$.

Theorem 1.40

For every $f : X \to \overline{\mathbb{R}}$ there are simple f_n with $f_n(x) \to f(x)$ for all x. If f is measurable the f_n can be chosen measurable, and if $f \geq 0$ they can be chosen pointwise increasing.

Proof. For $f \geq 0$ let

$$f_n := \sum_{i=1}^{n2^n} \frac{i-1}{2^n} \mathbb{1}_{\{f \in [\frac{i-1}{2^n}, \frac{i}{2^n})\}} + n \mathbb{1}_{\{f \geq n\}}. \tag{1.12}$$

Step 1 (increasing). $n \to n+1$ halves each dyadic cell without lowering values; on $\{f \geq n\}$ the new value is $\geq n$.

Step 2 (convergence). For $n > f(x)$, $0 \leq f(x) - f_n(x) < 2^{-n}$; if $f(x) = \infty$, $f_n(x) = n \to \infty$.

Step 3. The sets are measurable with f; in general take $f_n^+ - f_n^-$. ∎

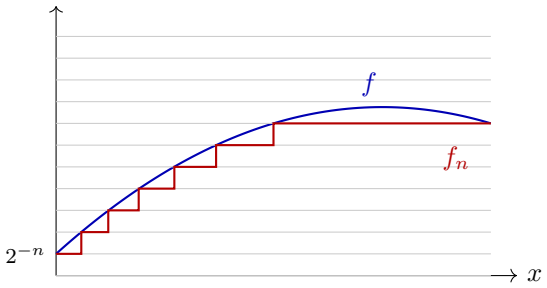


Figure. (1.12): *f_n rounds f down to $2^{-n}\mathbb{Z}$, capped at n.*

Note 1.41

For bounded f, (1.12) converges *uniformly* (error $< 2^{-n}$ once $n > \sup|f|$).

1.4 Integration

Motivation. Approximate from below by simple functions; the payoff is the three limit theorems (MCT, Fatou, DCT).

$(X, \mathcal{M}, \mu)$ is a measure space.

Definition 1.42 (Integral of simple functions)

For $s = \sum_{i=1}^n \alpha_i \mathbb{1}_{E_i}$ with $\alpha_i > 0$, pairwise disjoint $E_i \in \mathcal{M}$, and $E \in \mathcal{M}$, put

$$I_E(s) := \sum_{i=1}^n \alpha_i \mu(E \cap E_i). \tag{1.13}$$

Lemma 1.43

If $s, \tilde{s}$ are measurable simple functions as in Definition 1.42 with $0 \leq s \leq \tilde{s}$, then $I_E(s) \leq I_E(\tilde{s})$. In particular $I_E(s)$ does not depend on the representation of s.

Proof [Supplement]. For $s = \sum_i \alpha_i \mathbb{1}_{E_i}$, $\tilde{s} = \sum_j \beta_j \mathbb{1}_{F_j}$: $E_i \subset \bigcup_j F_j$ and $\alpha_i \leq \beta_j$ on nonempty $E_i \cap F_j$, so

$$I_E(s) = \sum_{i,j} \alpha_i \mu(E \cap E_i \cap F_j) \leq \sum_{i,j} \beta_j \mu(E \cap E_i \cap F_j) \leq \sum_j \beta_j \mu(E \cap F_j) = I_E(\tilde{s}).$$

Two representations: apply both ways. ■

For measurable $f \geq 0$ define

$$\int_E f\,d\mu := \sup\{I_E(s) \mid s \text{ simple, measurable, } 0 \leq s \leq f\}. \tag{1.14}$$

Remark 1.44

$\int_E f\,d\mu \in [0,\infty]$, and $I_E(s) = I_E(s\mathbb{1}_E)$, so

$$\int_E f\,d\mu = \sup\{I_E(s) \mid s \text{ simple, measurable, } 0 \leq s \leq f\mathbb{1}_E\}. \tag{1.15}$$

Hence for measurable $g \geq 0$:

$$f = g \text{ on } E \Rightarrow \int_E f\,d\mu = \int_E g\,d\mu, \qquad f \leq g \text{ on } E \Rightarrow \int_E f\,d\mu \leq \int_E g\,d\mu, \tag{1.16}$$

and by Lemma 1.43, $\int_E s\,d\mu = I_E(s)$ for every measurable simple $s \geq 0$.

Definition 1.45 (Integral and integrability)

For measurable f with $\min\left(\int_E f^+ d\mu, \int_E f^- d\mu\right) < \infty$ put

$$\int_E f\,d\mu := \int_E f^+ d\mu - \int_E f^- d\mu \in [-\infty,\infty]. \tag{1.17}$$

If both integrals are finite, f is **(Lebesgue) integrable** on E w.r.t. μ, written $f \in \mathcal{L}(E,\mu)$ ($f \in \mathcal{L}(E)$ if $\mu = \lambda$).

Remark 1.46

For $E \in \mathcal{M}$:

(i) f measurable and bounded on E, $\mu(E) < \infty \Rightarrow f \in \mathcal{L}(E,\mu)$.
(ii) f measurable, $\alpha \leq f \leq \beta$ on E, $\mu(E) < \infty \Rightarrow \alpha\mu(E) \leq \int_E f\,d\mu \leq \beta\mu(E)$.
(iii) $f, g \in \mathcal{L}(E,\mu)$, $f \leq g$ on $E \Rightarrow \int_E f\,d\mu \leq \int_E g\,d\mu$.
(iv) $f \in \mathcal{L}(E,\mu) \Rightarrow \alpha f \in \mathcal{L}(E,\mu)$ and $\int_E \alpha f\,d\mu = \alpha \int_E f\,d\mu$ for $\alpha \in \mathbb{R}$.
(v) $\mu(E) = 0$, f measurable $\Rightarrow \int_E f\,d\mu = 0$.
(vi) $f \in \mathcal{L}(E,\mu)$, $A \subset E$, $A \in \mathcal{M} \Rightarrow f \in \mathcal{L}(A,\mu)$.

[Supplement] *Proofs.*
(ii) $0 \leq s \leq c\mathbb{1}_E$ gives $I_E(s) \leq c\mu(E)$, attained by $c\mathbb{1}_E$: $\int_E c\,d\mu = c\mu(E)$. With $f^+ \leq \beta^+$, $f^- \geq \beta^-$, $f^+ \geq \alpha^+$, $f^- \leq \alpha^-$ on E and (1.16), $\alpha\mu(E) \leq \int_E f \leq \beta^+\mu(E) - \beta^-\mu(E) = \beta\mu(E)$.
(i) follows from the bounds in (ii).
(iii) $f^+ \leq g^+$ and $f^- \geq g^-$; use (1.16).
(iv) For $\alpha > 0$, $s \mapsto \alpha s$ matches simple functions below h and αh, $I_E(\alpha s) = \alpha I_E(s)$, and $(\alpha f)^\pm = \alpha f^\pm$; for $\alpha < 0$, $(\alpha f)^\pm = |\alpha| f^\mp$.
(v) Every $I_E(s) = \sum \alpha_i \mu(E \cap E_i) = 0$.
(vi) $I_A(s) \leq I_E(s)$ for $A \subset E$, so $\int_A f^\pm \leq \int_E f^\pm < \infty$.

Theorem 1.47

(i) If $f \geq 0$ is measurable, then $\varphi(A) := \int_A f\,d\mu$, $A \in \mathcal{M}$, is σ-additive, i.e. a measure on $(X, \mathcal{M})$. (ii) The same holds for $f \in \mathcal{L}(X, \mu)$, except that φ is then a difference of two finite measures.

Proof. (ii) is (i) for $f^{\pm}$. For (i) let $A = \dot{\bigcup}_i A_i$.

Step 1 (simple f). $f = \mathbb{1}_B$: $\varphi(A) = \mu(A \cap B) = \sum_i \mu(A_i \cap B)$; sum over $f = \sum_j \beta_j \mathbb{1}_{B_j}$, $\beta_j \geq 0$ (nonnegative series rearrange).

Step 2 ($\leq$). For simple $0 \leq s \leq f$, $\int_A s = \sum_i \int_{A_i} s \leq \sum_i \varphi(A_i)$; take $\sup_s$.

Step 3 ($\geq$). φ is monotone, so assume all $\varphi(A_n) < \infty$. Choose simple $0 \leq s_k \leq f$ with $\int_{A_k} s_k \geq \varphi(A_k) - \varepsilon$; $s := \mathbb{1}_{A_1} s_1 + \mathbb{1}_{A_2} s_2 \leq f$ and by Step 1

$$\varphi(A_1 \cup A_2) \geq \int_{A_1 \cup A_2} s\,d\mu = \int_{A_1} s_1\,d\mu + \int_{A_2} s_2\,d\mu \geq \varphi(A_1) + \varphi(A_2) - 2\varepsilon.$$

Induction and monotonicity: $\varphi(A) \geq \sum_{i \leq n} \varphi(A_i)$ for all n. ∎

Corollary 1.48

Let $A, E \in \mathcal{M}$, $A \subset E$, $\mu(E \setminus A) = 0$, and let $\int_E f\,d\mu$ be defined (e.g. $f \geq 0$ measurable or $f \in \mathcal{L}(E, \mu)$). Then $\int_E f\,d\mu = \int_A f\,d\mu$.

Proof. For $g = f^{\pm}$: $\int_E g = \int_A g + \int_{E \setminus A} g = \int_A g$ by Theorem 1.47 and Remark 1.46(v). ∎

Definition 1.49 (Almost everywhere)

A property P of points of $E \in \mathcal{M}$ holds **μ-almost everywhere** (μ-a.e.) on E if there is $N \in \mathcal{M}$ with $\mu(N) = 0$ such that P holds on $E \setminus N$ ("a.e." for Lebesgue measure).

Example 1.50

For measurable f, g: $f = g$ μ-a.e. on $E \iff \mu(\{f \neq g\} \cap E) = 0$. Here $\{f \neq g\} \in \mathcal{M}$ because $\{f < g\} = \bigcup_{q \in \mathbb{Q}} \{f < q\} \cap \{q < g\}$ and similarly for $\{f > g\}$. [Supplement]

Remark 1.51

"$f \sim g$ on $E :\iff f = g$ μ-a.e. on E" is an equivalence relation (unions of null sets are null). If $f = g$ off a null N and $\int_E f$ is defined, Corollary 1.48 and (1.16) give

$$\int_E f\,d\mu = \int_{E \setminus N} f\,d\mu = \int_{E \setminus N} g\,d\mu = \int_E g\,d\mu.$$

Theorem 1.52

Let $E \in \mathcal{M}$.

(i) If f is measurable, $g \in \mathcal{L}(E, \mu)$ and $|f| \leq g$ μ-a.e. on E, then $f \in \mathcal{L}(E, \mu)$.

(ii) $|f| \in \mathcal{L}(E, \mu)$ (i.e. $\int_E |f|\,d\mu < \infty$) $\iff$ $f \in \mathcal{L}(E, \mu)$.

(iii) $f \in \mathcal{L}(E, \mu) \Rightarrow |f| < \infty$ μ-a.e. on E.

(iv) $f \in \mathcal{L}(E, \mu) \Rightarrow \left|\int_E f\,d\mu\right| \leq \int_E |f|\,d\mu$.

Proof. (i) With $|f| \le g$ off a null N: $\int_E |f| = \int_{E\setminus N} |f| \le \int_{E\setminus N} g = \int_E g < \infty$. Since $f^{\pm} \le |f|$, (1.16) gives $f^{\pm} \in \mathcal{L}(E,\mu)$.

(ii) "$\Rightarrow$" by (i). "$\Leftarrow$": with $A := E \cap \{f \ge 0\}$, $B := E \cap \{f < 0\}$, $\int_E |f| = \int_A f^+ + \int_B f^- < \infty$ by Theorem 1.47 and Remark 1.46(vi).

(iii) For every n, $0 \le n\mathbb{1}_{\{|f|=\infty\}} \le |f|$, so $n\,\mu(\{|f| = \infty\} \cap E) \le \int_E |f| < \infty$; let $n \to \infty$.

(iv) By Remark 1.46(iii),(iv): $\pm \int_E f = \int_E (\pm f) \le \int_E |f|$. ∎

Theorem 1.53 (Monotone convergence theorem, Beppo Levi)

Let $0 \le f_1 \le f_2 \le \dots$ be measurable and $f := \lim_n f_n$. Then $\int_E f\,d\mu = \lim_n \int_E f_n\,d\mu$ for all $E \in \mathcal{M}$.

Proof. $\int_E f_n \nearrow \alpha \le \int_E f$. Conversely, for simple $0 \le s \le f$, $c \in (0,1)$, $E_n := E \cap \{f_n \ge cs\} \nearrow E$ (if $s(x) > 0$, $f_n(x) \to f(x) > cs(x)$), and

$$\int_E f_n\,d\mu \ge \int_{E_n} f_n\,d\mu \ge c \int_{E_n} s\,d\mu.$$

Continuity from below for the measure $A \mapsto \int_A s$ (Theorem 1.47) gives $\alpha \ge c \int_E s$; let $c \uparrow 1$, take $\sup_s$. ∎

Theorem 1.54 (Linearity)

Let $f_1, f_2 \in \mathcal{L}(E,\mu)$ with $f_1 + f_2$ well defined (no $\infty - \infty$). Then:

(i) $f_1 + f_2$ is measurable;

(ii) if f is measurable and $f = f_1 + f_2$ μ-a.e. on E, then $f \in \mathcal{L}(E,\mu)$ and

$$\int_E f\,d\mu = \int_E f_1\,d\mu + \int_E f_2\,d\mu. \tag{1.18}$$

(1.18) also holds if $f_1, f_2 \ge 0$ are merely measurable and f is measurable with $f = f_1 + f_2$ μ-a.e. on E.

Remark 1.55

Measurability of f in Theorem 1.54(ii) is automatic if $(X, \mathcal{M}, \mu)$ is **complete** (subsets of null sets are measurable): then $h = g$ a.e. with g measurable makes h measurable. [Supplement] *Reason:* if $h = g$ off a null N, $\{h > \alpha\} = (\{g > \alpha\} \setminus N) \cup (\{h > \alpha\} \cap N)$, the second set lying in N.

Proof of Theorem 1.54. (i) $f_1^+ - f_1^- + f_2^+ - f_2^-$ is a pointwise limit of simple combinations (Theorem 1.40; Corollary 1.36(ii)).

(ii) By Remark 1.51 take $f = f_1 + f_2$.

Step 1 ($f_1, f_2 \ge 0$). Simple: (1.13). In general simple $s_n^i \nearrow f_i$ and the MCT give $\int_E (f_1 + f_2) = \lim_n \int_E (s_n^1 + s_n^2) = \int_E f_1 + \int_E f_2$.

Step 2 (general). Take f_1, f_2 finite (Theorem 1.52(iii), Remark 1.51). $|f_1 + f_2| \le |f_1| + |f_2|$ gives $f \in \mathcal{L}(E,\mu)$; integrate $f^+ + f_1^- + f_2^- = f_1^+ + f_2^+ + f^-$ by Step 1 and rearrange the finite terms. ∎

Theorem 1.56

For measurable $f_k \geq 0$ and $E \in \mathcal{M}$: $\int_E \sum_{k=1}^{\infty} f_k \, d\mu = \sum_{k=1}^{\infty} \int_E f_k \, d\mu$.

Proof. $\sum_{k \leq n} f_k \nearrow \sum_k f_k$; apply the MCT and Theorem 1.54 to the partial sums. ∎

Theorem 1.57 (Fatou's lemma)

For measurable $f_n \geq 0$ and $E \in \mathcal{M}$: $\int_E \liminf_n f_n \, d\mu \leq \liminf_n \int_E f_n \, d\mu$.

Proof. $g_n := \inf_{k \geq n} f_k$ is measurable, $g_n \nearrow \liminf_n f_n$, and $g_n \leq f_k$ for $k \geq n$. By the MCT and (1.16),

$$\int_E \liminf_n f_n \, d\mu = \lim_n \int_E g_n \, d\mu \leq \liminf_n \int_E f_k \, d\mu = \liminf_n \int_E f_n \, d\mu.$$

∎

Theorem 1.58 (Dominated convergence theorem)

Let f_n be measurable with $f_n(x) \to f(x)$ for all $x \in E$, and suppose there is $g \in \mathcal{L}(E, \mu)$ (an *integrable majorant*) with

$$|f_n(x)| \leq g(x) \qquad \forall n \geq 1, \ x \in E. \tag{1.19}$$

Then $f \in \mathcal{L}(E, \mu)$ and $\int_E f \, d\mu = \lim_n \int_E f_n \, d\mu$.

Remark 1.59

In the MCT and DCT, a.e. convergence suffices if the limit is measurable (automatic when complete, Remark 1.55); (1.19) and $f_n \geq 0$ in Fatou need only hold a.e. [Supplement] *Reason:* multiplying by $\mathbb{1}_{N^c}$, N the union of the exceptional null sets, changes no integral (Corollary 1.48).

Proof of Theorem 1.58. $f, f_n \in \mathcal{L}(E, \mu)$ by Theorem 1.52(i), since $|f| \leq g$ as well.

Step 1. $f_n + g \geq 0$ and $f + g = \liminf_n (f_n + g)$. Fatou and Theorem 1.54 give $\int_E f + \int_E g \leq \liminf_n \int_E f_n + \int_E g$, i.e. $\int_E f \leq \liminf_n \int_E f_n$.

Step 2. $g - f_n \geq 0$ and $g - f = \liminf_n (g - f_n)$, so $\int_E g - \int_E f \leq \int_E g + \liminf_n \int_E (-f_n)$, i.e. $\int_E f \geq \limsup_n \int_E f_n$.

Step 3. Hence $\liminf_n \int_E f_n = \limsup_n \int_E f_n = \int_E f$. ∎

1.4.1 Comparison of the Lebesgue and Riemann integrals

Motivation. Lebesgue agrees with Riemann, and Riemann integrability means a null discontinuity set.

$X = [a, b]$, $\mu = \lambda$ on $\mathcal{M} = \{A \cap [a, b] \mid A \in \mathcal{M}(\lambda)\}$ (complete); $\int_a^b f \, dx := \int_X f \, d\lambda$, $f \in \mathcal{R}([a, b])$ if f is Riemann integrable, $\mathcal{R}\int_a^b f \, dx$ for the Riemann integral and $\underline{\mathcal{R}}\int$, $\overline{\mathcal{R}}\int$ for the lower and upper Riemann integrals. For a partition $P = \{x_0, \ldots, x_n\}$, $\mathrm{mesh}(P) := \max_i |x_i - x_{i-1}|$.

Theorem 1.60

(i) $f \in \mathcal{R}([a,b]) \Rightarrow f \in \mathcal{L}([a,b])$ and $\int_a^b f\,dx = \mathcal{R}\int_a^b f\,dx$.

(ii) For bounded f: $f \in \mathcal{R}([a,b]) \iff f$ is continuous a.e. on $[a,b]$.

Proof.

Step 1 (step functions). Let f be bounded. Choose partitions $P_k \subset P_{k+1}$ with $\operatorname{mesh}(P_k) \le 1/k$ such that the lower and upper sums satisfy

$$L(P_k,f) \to \underline{\mathcal{R}}\int_a^b f\,dx, \qquad U(P_k,f) \to \overline{\mathcal{R}}\int_a^b f\,dx. \tag{1.20}$$

For $P_k = \{x_0,\dots,x_{n(k)}\}$ define step functions L_k, U_k by $L_k(a) = U_k(a) = f(a)$ and, on $(x_{i-1},x_i]$, $L_k := m_i = \inf_{[x_{i-1},x_i]} f$, $U_k := M_i = \sup_{[x_{i-1},x_i]} f$. Then

$$L(P_k,f) = \int_a^b L_k\,dx, \qquad U(P_k,f) = \int_a^b U_k\,dx, \qquad L_1 \le L_2 \le \cdots \le f \le \cdots \le U_2 \le U_1. \tag{1.21}$$

Step 2 (limits). $L := \lim_k L_k$, $U := \lim_k U_k$ are measurable; with the bounded majorant $\max(|L_1|,|U_1|)$, the DCT, (1.21), (1.20) give

$$\int_a^b L\,dx = \underline{\mathcal{R}}\int_a^b f\,dx, \qquad \int_a^b U\,dx = \overline{\mathcal{R}}\int_a^b f\,dx. \tag{1.22}$$

Step 3 (i). Since $L \le f \le U$,

$$f \in \mathcal{R}([a,b]) \iff \int_a^b L\,dx = \int_a^b U\,dx \iff L = U \text{ a.e.} \iff L = f = U \text{ a.e.}, \tag{1.23}$$

the middle step by Lemma 1.62(i) for $U - L \ge 0$. Then f is measurable (completeness, Remark 1.55) and $\int_a^b f\,dx = \int_a^b L\,dx = \mathcal{R}\int_a^b f\,dx$ by Remark 1.51 and (1.22).

Step 4 (ii). Off the countable (hence null) set $\bigcup_k P_k$,

$$U(x) = L(x) \iff f \text{ is continuous at } x. \tag{1.24}$$

[Supplement] Such x is interior to an interval of each P_k, of length $\le 1/k$. Continuity at x puts it, for large k, where $|f - f(x)| < \varepsilon$, so $U_k(x) - L_k(x) \le 2\varepsilon$; conversely $U_k(x) - L_k(x) < \varepsilon$ gives $|f(y) - f(x)| \le M_i - m_i < \varepsilon$ around x. With (1.23): $f \in \mathcal{R}([a,b]) \iff L = U$ a.e. $\iff f$ continuous a.e. ∎

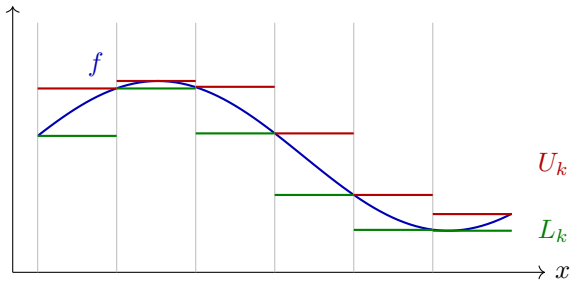


Figure. *Lower and upper step functions; refinement squeezes them to f at continuity points.*

1.5 Some fundamental inequalities and the L^p-spaces

Motivation. Markov, Jensen, Hölder and Minkowski control $\int |f|^p$, and L^p is complete.

Proposition 1.61 (Markov ($p=1$), Chebyshev ($p=2$))

For measurable f, $p \in (0,\infty)$ and $\varepsilon > 0$:

$$\mu(\{|f| \geq \varepsilon\}) \leq \frac{1}{\varepsilon^p} \int_X |f|^p \, d\mu.$$

Proof. $\mathbb{1}_{\{|f|\geq\varepsilon\}} = \mathbb{1}_{\{|f|^p\geq\varepsilon^p\}} \leq \varepsilon^{-p}|f|^p$. Integrate ((1.16); Remark 1.46(iv) for $h \geq 0$ needs no integrability). ■

Lemma 1.62

Let f be measurable. (i) If $f \geq 0$ μ-a.e. and $\int_X f \, d\mu = 0$, then $f = 0$ μ-a.e. (ii) If $\int_A f \, d\mu = 0$ for every $A \in \mathcal{M}$, then $f = 0$ μ-a.e.

Proof [Supplement]. (i) $|f| = f$ a.e., so $\int |f| \, d\mu = 0$ by Remark 1.51. Markov gives $\mu(\{|f| \geq \frac{1}{n}\}) \leq n \int |f| = 0$, and $\{f \neq 0\} = \bigcup_n \{|f| \geq \frac{1}{n}\}$.

(ii) $A := \{f > 0\}$: $\int f\mathbb{1}_A = 0$, so $f\mathbb{1}_A = 0$ a.e. by (i), and $f\mathbb{1}_A > 0$ on A forces $\mu(A) = 0$; likewise $\{f < 0\}$. ■

Theorem 1.63 (Jensen's inequality)

Let $(X, \mathcal{M}, \mu)$ be a probability space, $f \in \mathcal{L}(X,\mu)$ real-valued with $f(x) \in (a,b)$ for all x, and $\varphi : (a,b) \to \mathbb{R}$ convex. Then $(\varphi \circ f)^- \in \mathcal{L}(X,\mu)$ and

$$\varphi\Big(\int_X f \, d\mu\Big) \leq \int_X \varphi \circ f \, d\mu \in (-\infty, \infty].$$

Proof.

Step 1. φ is continuous, so $\varphi \circ f$ is measurable. $t := \int_X f \, d\mu \in (a,b)$, and with the right derivative β_+ at t, $\varphi(y) \geq \varphi(t) + \beta_+(y-t)$, so

$$\varphi \circ f \geq g := \varphi(t) + \beta_+(f - t) \in \mathcal{L}(X,\mu). \tag{1.25}$$

Step 2. $(\varphi \circ f)^- \leq g^-$, so $\int \varphi \circ f \in (-\infty, \infty]$ is defined.

Step 3. g finite a.e., so $(\varphi\circ f)^+ + g^- \geq (\varphi\circ f)^- + g^+ \geq 0$ a.e.; integrate (Theorem 1.54, nonnegative case) and rearrange the finite terms:

$$\varphi\Big(\int_X f \, d\mu\Big) = \int_X g \, d\mu \leq \int_X \varphi \circ f \, d\mu.$$

■

Example 1.64 (AM–GM)

Let $X = \{p_1, \ldots, p_n\}$, $\mu(\{p_i\}) = \alpha_i > 0$ with $\sum \alpha_i = 1$, $f(p_i) = x_i$, $\varphi = \exp$. Jensen gives $e^{\sum \alpha_i x_i} \leq \sum \alpha_i e^{x_i}$, i.e. with $y_i = e^{x_i}$,

$$\prod_{i=1}^n y_i^{\alpha_i} \leq \sum_{i=1}^n \alpha_i y_i \qquad (\alpha_i, y_i > 0,\ \textstyle\sum \alpha_i = 1).$$

Theorem 1.65 (Hölder and Minkowski inequalities)

Let $f, g : X \to [0,\infty]$ be measurable and $p, q \in (1,\infty)$ with $\frac{1}{p} + \frac{1}{q} = 1$. Then

$$\int_X fg\,d\mu \le \Big(\int_X f^p\,d\mu\Big)^{1/p}\Big(\int_X g^q\,d\mu\Big)^{1/q}, \tag{1.26}$$

$$\Big(\int_X (f+g)^p\,d\mu\Big)^{1/p} \le \Big(\int_X f^p\,d\mu\Big)^{1/p} + \Big(\int_X g^p\,d\mu\Big)^{1/p}. \tag{1.27}$$

Proof.

Step 1 (Hölder, trivial cases). A, B: the right factors of (1.26). $A = 0$ gives $f = 0$ a.e. (Lemma 1.62), so $fg = 0$ a.e.; same for B; A or $B = \infty$ is trivial. Else $F := f/A$, $G := g/B$ have $\int F^p = \int G^q = 1$.

Step 2 (Young). $F = e^{s/p}$, $G = e^{t/q}$: $FG \le \frac{1}{p}e^s + \frac{1}{q}e^t = \frac{1}{p}F^p + \frac{1}{q}G^q$ (also for $F, G \in \{0,\infty\}$); integrate: $\int FG \le 1$.

Step 3 (Minkowski). $(f+g)^p = f(f+g)^{p-1} + g(f+g)^{p-1}$, (1.26) and $(p-1)q = p$:

$$\int (f+g)^p \le \Big[\Big(\int f^p\Big)^{1/p} + \Big(\int g^p\Big)^{1/p}\Big]\Big(\int (f+g)^p\Big)^{1/q}. \tag{1.28}$$

Step 4. Assume $\int (f+g)^p > 0$, $\int f^p + \int g^p < \infty$; $(\frac{f+g}{2})^p \le \frac{1}{2}(f^p + g^p)$ gives $C := (\int (f+g)^p)^{1/q} \in (0,\infty)$; divide (1.28) by C. ∎

Definition 1.66 ($\mathcal{L}^p$)

For $p \in [1,\infty)$ let $\mathcal{L}^p(X,\mu) := \{f : X \to \mathbb{R} \text{ measurable} \mid \|f\|_p := (\int_X |f|^p\,d\mu)^{1/p} < \infty\}$.

Theorem 1.67

(i) For $p \in [1,\infty)$, $\mathcal{L}^p(X,\mu)$ is a vector space and $\|\cdot\|_p$ a semi-norm on it. If (f_n) is Cauchy in $\mathcal{L}^p$, there are $f \in \mathcal{L}^p(X,\mu)$ and a subsequence (f_{n_k}) with $\|f - f_n\|_p \to 0$ and $f_{n_k} \to f$ μ-a.e.

(ii) If $p, q \in (1,\infty)$, $\frac{1}{p} + \frac{1}{q} = 1$, $f \in \mathcal{L}^p$, $g \in \mathcal{L}^q$, then $fg \in \mathcal{L}^1$ and

$$\|fg\|_1 \le \|f\|_p \|g\|_q. \tag{1.29}$$

Proof.

Step 1 (semi-norm). $\|\alpha f\|_p = |\alpha|\|f\|_p$; $|f+g| \le |f| + |g|$ and Minkowski give the triangle inequality ($p = 1$: linearity). $\|f\|_p = 0$ gives only $f = 0$ a.e. (Lemma 1.62(i)).

Step 2 (a.e. convergent subsequence). $g_k := f_{n_{k+1}} - f_{n_k}$ with $\|g_k\|_p < 2^{-k}$; by the MCT and Minkowski,

$$\Big\|\sum_k |g_k|\Big\|_p = \lim_n \Big\|\sum_{k\le n} |g_k|\Big\|_p \le \limsup_n \sum_{k\le n} \|g_k\|_p < 1.$$

so (Theorem 1.52(iii)) off a null N, $f_{n_{k+1}} = f_{n_1} + \sum_{l\le k} g_l$ converges; $f := \lim_k f_{n_k} \mathbb{1}_{X\setminus N}$.

Step 3 (L^p convergence). If $\|f_m - f_n\|_p < \varepsilon$ for $m, n \ge N_0$ and $n_k \ge N_0$, Fatou gives $\|f - f_{n_k}\|_p \le \liminf_l \|f_{n_l} - f_{n_k}\|_p \le \varepsilon$. So $f \in \mathcal{L}^p$, $f_{n_k} \to f$, and the Cauchy sequence $f_n \to f$.

(ii) is (1.26) applied to $|f|, |g|$. ∎

Remark 1.68

(i) Identifying functions that agree μ-a.e. turns $\mathcal{L}^p$ into a normed space: $L^p(X,\mu) := \mathcal{L}^p(X,\mu)/\|\cdot\|_p^{-1}(0)$, where $\|\cdot\|_p^{-1}(0) = \{f \text{ measurable} \mid f = 0\ \mu\text{-a.e.}\}$. For the class $\tilde{f}$ of f put

$$\|\tilde{f}\|_p := \|f\|_p; \tag{1.30}$$

now $\|\tilde{f}\|_p = 0 \iff f = 0$ a.e. $\iff \tilde{f} = 0$. In particular $L^2(X,\mu)$ is an inner product space with

$$\langle \tilde{f}, \tilde{g}\rangle_2 := \int_X fg\, d\mu, \qquad \|\tilde{f}\|_2 = \sqrt{\langle \tilde{f}, \tilde{f}\rangle_2}, \tag{1.31}$$

for arbitrary representatives f, g.

(ii) We work with representatives modulo $f = g$ μ-a.e.

(iii) $\mathcal{L}^p$ is real-valued so sums are defined; $g : X \to [0,\infty]$ with $\int g^p < \infty$ is finite a.e. (Theorem 1.52(iii)) and tacitly replaced by a finite version.

Example 1.69

For $X = \mathbb{N}$, $\mathcal{M} = \mathcal{P}(\mathbb{N})$, $\mu = \sum_k \delta_k$ (counting measure), $\emptyset$ is the only null set and

$$L^p(X,\mu) = \mathcal{L}^p(X,\mu) = \ell^p := \Big\{x = (x_k) \in \mathbb{R}^{\mathbb{N}} \;\Big|\; \sum_k |x_k|^p < \infty\Big\}.$$

Theorem 1.70 (Riesz–Fischer)

For $p \in [1,\infty)$, $(L^p(X,\mu), \|\cdot\|_p)$ is a Banach space; in particular $L^2(X,\mu)$ is a Hilbert space with the inner product (1.31).

Proof. Normed by Remark 1.68(i). Cauchy $(\tilde{f}_n)$ have Cauchy representatives ((1.30)), which converge by Theorem 1.67(i). ∎

Chapter 2. Basic Probability Theory

2.1 Random variables and integration

Motivation. Probability is measure theory with total mass one: random variables, expectations, "a.s.".

$(\Omega, \mathcal{A})$ is a measurable space; $\mathbb{R}$ carries $\mathcal{B}(\mathbb{R})$, $\overline{\mathbb{R}} = \mathbb{R} \cup \{\pm\infty\}$ carries $\mathcal{B}(\overline{\mathbb{R}}) = \{B \subset \overline{\mathbb{R}} \mid B \cap \mathbb{R} \in \mathcal{B}(\mathbb{R})\}$.

Definition 2.1 (Random variable)

A **random variable** (r.v.) on $(\Omega, \mathcal{A})$ is an $\mathcal{A}/\mathcal{B}(\overline{\mathbb{R}})$-measurable map $X : \Omega \to \overline{\mathbb{R}}$. For real-valued X it suffices to check $\mathcal{A}/\mathcal{B}(\mathbb{R})$-measurability. We write $\{X \le c\} := \{\omega \mid X(\omega) \le c\}$, $\{X < c\}$, etc.

Facts 2.2

Mostly restatements of Section 1.3:

(i) X is a r.v. iff $\{X \le c\} \in \mathcal{A}$ for all $c \in \mathbb{R}$ ($\sigma(\{[-\infty, c]\}) = \mathcal{B}(\overline{\mathbb{R}})$), and c in a countable dense S (e.g. $\mathbb{Q}$) suffices since $\{X > c\} = \bigcup_{q \in S,\, q > c} \{X > q\}$.

(ii) If $\mathcal{A} = \mathcal{P}(\Omega)$, every function $\Omega \to \overline{\mathbb{R}}$ is a r.v.

(iii) If X is a r.v. and $h : \mathbb{R} \to \mathbb{R}$ (or $\overline{\mathbb{R}} \to \overline{\mathbb{R}}$) is Borel measurable, then $h(X)$ is a r.v. (preimages compose); e.g. $|X|$, X^2, $|X|^p$, e^X.

(iv) For r.v.'s $X_1, X_2, \ldots$ and $\alpha_i \in \mathbb{R}$, the following are r.v.'s: $\sum_{i \le n} \alpha_i X_i$ whenever no $\infty - \infty$ occurs; $\sum_{i=1}^{\infty} \alpha_i X_i$ if all $\alpha_i, X_i \ge 0$; $\sup_i X_i$, $\inf_i X_i$, $X_1 \wedge X_2 := \min$, $X_1 \vee X_2 := \max$; $\limsup_i X_i$, $\liminf_i X_i$, and $\lim_i X_i$ if it exists.

(v) $\mathbb{1}_A$ is a r.v. for $A \in \mathcal{A}$. An **elementary** (simple) r.v. is $X = \sum_{i=1}^{n} c_i \mathbb{1}_{A_i}$ with $c_i \in \mathbb{R}$ and disjoint $A_i \in \mathcal{A}$. Every r.v. is a limit of elementary ones (Theorem 1.40; $X = X^+ - X^-$, $X^+ := X \vee 0$, $X^- := -(X \wedge 0)$); for $X \ge 0$,

$$X_n := \sum_{i=0}^{n2^n - 1} \frac{i}{2^n} \mathbb{1}_{\{\frac{i}{2^n} \le X < \frac{i+1}{2^n}\}} + n \mathbb{1}_{\{X \ge n\}} \nearrow X.$$

On $(\Omega, \mathcal{A}, P)$: $\int \mathbb{1}_A \, dP = P(A)$; $\int \sum c_i \mathbb{1}_{A_i} \, dP = \sum c_i P(A_i)$; $\int X \, dP = \lim_n \int X_n \, dP$ for elementary $X_n \nearrow X \ge 0$ (MCT); and if $\min(\int X^+ dP, \int X^- dP) < \infty$,

$$\mathbb{E}[X] := \int_\Omega X \, dP := \int X^+ dP - \int X^- dP \in \overline{\mathbb{R}}$$

is the **expectation** (also $\int X(\omega) \, P(d\omega)$). Up to Theorem 2.8 everything translates Section 1.4.

Definition 2.3 ($\mathcal{L}^p$ and almost sure)

(i) $\mathcal{L}^1 := \mathcal{L}^1(\Omega, \mathcal{A}, P) := \{X : \Omega \to \mathbb{R} \text{ r.v.} \mid \mathbb{E}[|X|] < \infty\}$, and for $p \in [1, \infty)$, $\mathcal{L}^p := \{X \text{ real r.v.} \mid \mathbb{E}[|X|^p] < \infty\}$.

(ii) A property holds **P-almost surely** (P-a.s.) if it holds outside some $N \in \mathcal{A}$ with $P(N) = 0$.

With $\mathcal{N}_0 := \{X \mid X = 0 \ P\text{-a.s.}\}$, $L^p := \mathcal{L}^p / \mathcal{N}_0$ is a Banach space for $\|X\|_p := \mathbb{E}[|X|^p]^{1/p}$ (Theorem 1.70). The expectation is monotone ($0 \le X \le Y$ a.s. $\Rightarrow 0 \le \mathbb{E}[X] \le \mathbb{E}[Y]$) and

linear on $\mathcal{L}^1$ (Theorem 1.54).

2.2 Inequalities

Motivation. Jensen, Markov–Chebyshev and Hölder–Minkowski from Section 1.5, for $P(\Omega) = 1$.

Proposition 2.4 (Jensen's inequality)

Let φ be convex on an open interval $I \subset \mathbb{R}$ and $X \in \mathcal{L}^1$ with $X(\Omega) \subset I$. Then $\mathbb{E}[X] \in I$, $(\varphi \circ X)^- \in \mathcal{L}^1$ and $\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)]$.

Proof. Theorem 1.63 on $(\Omega, \mathcal{A}, P)$ (verbatim for infinite endpoints). ■

Example 2.5

For $X \in \mathcal{L}^1$: $\mathbb{E}[X]^2 \leq \mathbb{E}[X^2]$ ($\varphi(x) = x^2$). Moreover, for $1 \leq p \leq q$ and any r.v. X (integrability not needed),

$$\|X\|_p = \mathbb{E}[|X|^p]^{1/p} \leq \mathbb{E}[|X|^q]^{1/q} = \|X\|_q.$$

Proof. Jensen with convex $|x|^{q/p}$ on the bounded $(|X| \wedge n)^p$: $\mathbb{E}[(|X| \wedge n)^p]^{q/p} \leq \mathbb{E}[(|X| \wedge n)^q]$; take the $1/q$ power, $n \to \infty$ (MCT).

Corollary 2.6

Let X be a r.v.

(i) (Markov) $P(|X| \geq c) \leq \frac{1}{c}\mathbb{E}[|X|]$ for $c > 0$. In particular $\mathbb{E}[|X|] = 0 \Rightarrow X = 0$ a.s., and $\mathbb{E}[|X|] < \infty \Rightarrow |X| < \infty$ a.s.

(ii) (Chebyshev) For $X \in \mathcal{L}^1$, $P(|X - \mathbb{E}[X]| \geq c) \leq \frac{\mathrm{Var}(X)}{c^2}$, where $\mathrm{Var}(X) := \mathbb{E}[(X - \mathbb{E}[X])^2]$ is the **variance**. More generally $P(|X| \geq c) \leq c^{-p}\mathbb{E}[|X|^p]$ for $p > 0$.

Proof. Proposition 1.61 for X, resp. $X - \mathbb{E}[X]$; then Lemma 1.62(i) and Theorem 1.52(iii). ■

Theorem 2.7

(i) (Hölder) For $p, q \in (1, \infty)$, $\frac{1}{p} + \frac{1}{q} = 1$, $X \in \mathcal{L}^p$, $Y \in \mathcal{L}^q$: $\|XY\|_1 \leq \|X\|_p \|Y\|_q$. For $p = q = 2$ this is the Cauchy–Schwarz inequality in the Hilbert space L^2. (ii) (Minkowski) For $p \geq 1$ and $X, Y \in \mathcal{L}^p$: $\|X + Y\|_p \leq \|X\|_p + \|Y\|_p$.

Proof. Theorem 1.67(ii) and Step 1 of the proof of Theorem 1.67(i). ■

2.3 Convergence of random variables

Motivation. Almost sure, in probability, and L^p convergence differ; Borel–Cantelli links them.

Theorem 2.8 (Limit theorems)

(i) (MCT) $0 \le X_1 \le X_2 \le \dots \Rightarrow \lim_n \mathbb{E}[X_n] = \mathbb{E}[\lim_n X_n]$.

(ii) (Fatou) $X_n \ge 0 \Rightarrow \mathbb{E}[\liminf_n X_n] \le \liminf_n \mathbb{E}[X_n]$.

(iii) (DCT) If $|X_n| \le Y \in \mathcal{L}^1$ a.s. for all n and $\lim_n X_n$ exists a.s., then $\mathbb{E}[\lim_n X_n] = \lim_n \mathbb{E}[X_n]$. Here $\lim_n X_n$ means any r.v. X with $X = \lim_n X_n$ a.s., e.g. $X = \lim_n X_n \mathbb{1}_{\Omega\setminus N}$ with $P(N) = 0$.

Proof. Theorems 1.53, 1.57, 1.58 together with Remark 1.59; the a.s. limit $\lim_n X_n \mathbb{1}_{\Omega\setminus N}$ is measurable by Corollary 1.36(ii). ∎

Events. For $A_n \in \mathcal{A}$: $\bigcup_i A_i$ = "at least one A_i occurs", $\bigcap_i A_i$ = "all occur",

$$\limsup_n A_n := \bigcap_n \bigcup_{m\ge n} A_m = \text{"infinitely many } A_n \text{ occur" } (A_n \text{ i.o.}),$$
$$\liminf_n A_n := \bigcup_n \bigcap_{m\ge n} A_m = \text{"all but finitely many } A_n \text{ occur"},$$

and $\liminf_n A_n \subset \limsup_n A_n$.

Lemma 2.9 (Borel–Cantelli, part 1)

$\sum_n P(A_n) < \infty \;\Rightarrow\; P(\limsup_n A_n) = 0$.

Proof. $\bigcup_{m\ge n} A_m \searrow \limsup_n A_n$, so by continuity from above and subadditivity $P(\limsup_n A_n) = \lim_n P(\bigcup_{m\ge n} A_m) \le \lim_n \sum_{m\ge n} P(A_m) = 0$. ∎

Lemma 2.10 (Fast convergence in probability)

If $\sum_n P(|X_n| \ge \varepsilon) < \infty$ for every $\varepsilon > 0$, then $P(\lim_n X_n = 0) = 1$.

Proof.

Step 1. For $\varepsilon > 0$ let $N_\varepsilon := \limsup_n \{|X_n| \ge \varepsilon\}$; Borel–Cantelli gives $P(N_\varepsilon) = 0$. For $\omega \notin N_{1/k}$,

$$|X_m(\omega)| < \tfrac{1}{k} \quad \text{for all but finitely many } m. \tag{2.1}$$

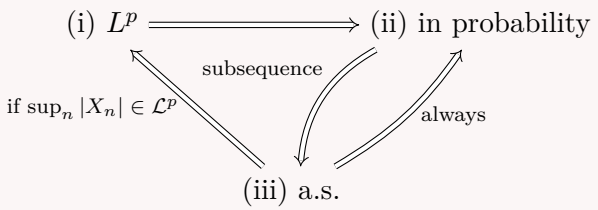
Step 2. $N := \bigcup_k N_{1/k}$ is null, and for $\omega \notin N$, (2.1) holds for every k, i.e. $X_n(\omega) \to 0$. ∎

Definition 2.11 (Modes of convergence)

For real r.v.'s $X, X_1, X_2, \dots$: (i) **L^p-convergence** ($p \ge 1$): $\|X_n - X\|_p \to 0$; (ii) **convergence in probability**: $P(|X_n - X| \ge \varepsilon) \to 0$ for every $\varepsilon > 0$; (iii) **P-a.s. convergence**: $P(\lim_n X_n = X) = 1$.

Theorem 2.12 (Comparison of the modes)

(i)⇒(ii); (iii)⇒(ii); (ii)⇒(iii) along a subsequence; (iii)⇒(i) if $\sup_n |X_n| \in \mathcal{L}^p$.



Proof.

Step (i)⇒(ii). Chebyshev: $P(|X_n - X| \geq \varepsilon) \leq \varepsilon^{-p}\mathbb{E}[|X_n - X|^p]$.

Step (iii)⇒(ii). $\{\lim_n X_n = X\} = \bigcap_k A_k$ with $A_k := \bigcup_m \bigcap_{n\geq m}\{|X_n - X| \leq \frac{1}{k}\}$. So $P(A_k) = 1$, and continuity from below gives

$$1 = P(A_k) = \lim_m P\Big(\bigcap_{n\geq m}\{|X_n - X| \leq \tfrac{1}{k}\}\Big) \leq \liminf_m P(|X_m - X| \leq \tfrac{1}{k}) \leq 1.$$

Hence $P(|X_m - X| > \frac{1}{k}) \to 0$ for every k.

Step (iii)⇒(i). With $Y := \sup_n |X_n|$ and $|X| \leq Y$ a.s., $|X_n - X|^p \leq 2^p Y^p \in \mathcal{L}^1$; apply the DCT.

Step (ii)⇒(iii) along a subsequence. Take $P(|X_{n_k} - X| \geq \frac{1}{k}) < 2^{-k}$; for $\frac{1}{k} < \varepsilon$, $P(|X_{n_k} - X| \geq \varepsilon) \leq 2^{-k}$, so $\sum_k P(|X_{n_k} - X| \geq \varepsilon) < \infty$ and Lemma 2.10 applies. ■

2.4 Image measure and distribution of random variables

Motivation. A random variable matters through its distribution P_X (or F_X) on $\mathbb{R}$.

$(\Omega, \mathcal{A})$, $(\tilde{\Omega}, \tilde{\mathcal{A}})$ are measurable spaces.

Proposition 2.13 (Image measure)

If $T : \Omega \to \tilde{\Omega}$ is $\mathcal{A}/\tilde{\mathcal{A}}$-measurable and P a probability measure on $(\Omega, \mathcal{A})$, then $\tilde{P}(\tilde{A}) := P(T \in \tilde{A}) = P(T^{-1}(\tilde{A}))$, $\tilde{A} \in \tilde{\mathcal{A}}$, is a probability measure on $(\tilde{\Omega}, \tilde{\mathcal{A}})$, the **image** (push-forward, induced) measure, written $P \circ T^{-1}$, $T(P)$ or P_T.

Proof. $\tilde{P}(\emptyset) = 0$, $\tilde{P}(\tilde{\Omega}) = 1$; preimages of disjoint sets are disjoint and commute with unions, so $\tilde{P}(\bigcup_i \tilde{A}_i) = \sum_i \tilde{P}(\tilde{A}_i)$. ■

Definition 2.14 (Distribution, distribution function)

The **distribution** of a real r.v. X on $(\Omega, \mathcal{A}, P)$ is $P_X(B) := P(X \in B)$, $B \in \mathcal{B}(\mathbb{R})$. Its **distribution function** is

$$F_X(t) := P(X \leq t) = P_X((-\infty, t]), \qquad t \in \mathbb{R}. \tag{2.2}$$

For μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, $F(x) := \mu((-\infty, x])$. (Extended real X: distribution on $\mathcal{B}(\overline{\mathbb{R}})$, but no distribution function.)

Remark 2.15

(i) F is increasing, right continuous, $F(-\infty) = 0$, $F(+\infty) = 1$, by continuity of μ: $(-\infty, b_n] \searrow (-\infty, a]$ for $b_n \searrow a$, $(-\infty, a] \searrow \emptyset$, $(-\infty, b] \nearrow \mathbb{R}$.

Conversely each such F is the distribution function of a unique μ on $\mathcal{B}(\mathbb{R})$, the distribution of $X(x) = x$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu)$.

[Supplement] *Proof.* Uniqueness: $\{(-\infty, t]\}$ is a $\cap$-stable generator (Theorem 1.9). Existence: on $((0,1), \mathcal{B}, \lambda)$, $G(u) := \inf\{t \mid F(t) \geq u\} \in \mathbb{R}$ attains its infimum (right continuity), so $G(u) \leq t \iff u \leq F(t)$, $\lambda(G \leq t) = F(t)$, and $\mu := \lambda \circ G^{-1}$.

(ii) For Borel $\Phi : \mathbb{R} \to \mathbb{R}$ with $\Phi(X) \in \mathcal{L}^1$ or $\Phi \geq 0$,

$$\mathbb{E}[\Phi(X)] = \int_\Omega \Phi(X)\, dP = \int_{\mathbb{R}} \Phi\, dP_X = \int_{\mathbb{R}} \Phi(s)\, dF_X(s) = \int_{\mathbb{R}} \Phi(s) F_X'(s)\, ds.$$

The third equality is notation; the last needs $F_X \in C^1$ (or absolutely continuous).

[Supplement] *Second equality (transformation theorem).* Indicators, simple $\Phi \geq 0$, MCT, then $\Phi^+ - \Phi^-$.

Last equality. $F_X(x) - F_X(-n) = \int_{-n}^x F_X'\, dt$ (FTC, Theorem 1.60); $n \to \infty$ (MCT) gives density F_X' (Remark 2.18(ii)); then repeat the indicator–simple–MCT argument.

Remark 2.16

(i) The **jump** $F(x) - F(x-) = \lim_n \mu((x - \frac{1}{n}, x]) = \mu(\{x\})$; F is continuous iff μ has no atoms ("μ is continuous").

(ii) An increasing bounded F has at most countably many discontinuities. [Supplement] *Reason:* at most k jumps are $\geq 1/k$.

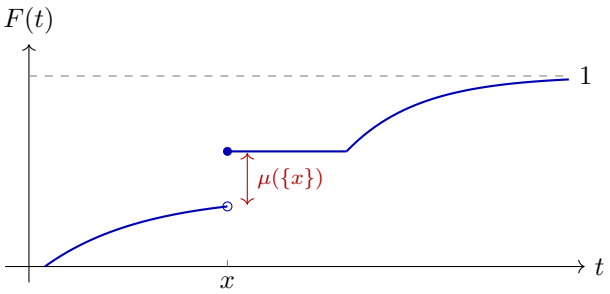


Figure. *A distribution function; its jump at x is the atom $\mu(\{x\})$.*

Definition 2.17 (Discrete and absolutely continuous distributions)

(i) F (or μ) is **discrete** if $\mu(S) = 1$ for a countable $S \subset \mathbb{R}$; then $F(x) = \sum_{y \in S,\, y \leq x} \mu(\{y\})$ is a step function.

(ii) F (or μ) is **absolutely continuous** if there is a measurable **density** $f \geq 0$ with

$$F(x) = \int_{-\infty}^x f(t)\, dt, \tag{2.3}$$

respectively

$$\mu(A) = \int_A f(t)\, dt = \int_{-\infty}^{\infty} \mathbb{1}_A f\, dt \quad \text{for all } A \in \mathcal{B}(\mathbb{R}). \tag{2.4}$$

In particular $\int_{-\infty}^{\infty} f\, dt = 1$.

Remark 2.18

(i) Every measurable $f \geq 0$ with $\int f\,dt = 1$ defines a probability measure $A \mapsto \int_A f\,dt$ (Theorem 1.47). (ii) (2.3)⇒(2.4) by uniqueness on the generator $\{(-\infty, x]\}$ (Theorem 1.9); the converse is trivial.

Example 2.19

(i) (*Uniform on* $[a,b]$) $f = \frac{1}{b-a}\mathbb{1}_{[a,b]}$, $F(x) = 0$ for $x \leq a$, $\frac{x-a}{b-a}$ on $[a,b]$, 1 for $x \geq b$.

(ii) (*Exponential with parameter* $\alpha > 0$) $f(x) = \alpha e^{-\alpha x}\mathbb{1}_{[0,\infty)}(x)$, $F(x) = (1-e^{-\alpha x})\mathbb{1}_{[0,\infty)}(x)$. Continuous geometric analogue: $\int_k^{k+1} f = e^{-\alpha k}(1-e^{-\alpha}) = (1-p)^k p$ with $p := 1-e^{-\alpha}$.

(iii) (*Normal* $N(m,\sigma^2)$) Density $f_{m,\sigma^2}(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-m)^2/(2\sigma^2)}$. $z = (y-m)/\sigma$ gives $F_{m,\sigma^2}(x) = F_{0,1}(\frac{x-m}{\sigma})$ (no closed form), $\mathbb{E}[X] = m$, $\mathrm{Var}(X) = \sigma^2$. [Supplement] *Reason:* $\mathbb{E}[X] - m = \sigma \int z f_{0,1}\,dz = 0$ (symmetry), $\mathrm{Var}(X) = \sigma^2 \int z^2 f_{0,1}\,dz = \sigma^2$ (parts: $\int z \cdot ze^{-z^2/2}dz = \int e^{-z^2/2}dz$). Central moments:

$$\mathbb{E}[|X-m|^p] = 2\int_0^\infty x^p \frac{e^{-x^2/(2\sigma^2)}}{\sqrt{2\pi\sigma^2}}\,dx = \frac{2^{p/2}\sigma^p}{\sqrt{\pi}}\,\Gamma\Big(\frac{p+1}{2}\Big) \qquad (y = x^2/(2\sigma^2)),$$

(iv) (*Binomial*) $\Omega = S^n$ uniform, success $x_i \in S_0 \subset S$, $p := |S_0|/|S|$; "exactly k successes" has $P(A_k) = \binom{n}{k}|S_0|^k|S\setminus S_0|^{n-k}/|S|^n = \binom{n}{k}p^k(1-p)^{n-k}$. The **binomial distribution** is $B(n,p) := \sum_{k=0}^n \binom{n}{k}p^k(1-p)^{n-k}\delta_k$.

(v) (*Poisson limit*) With $p_n = \lambda/n$,

$$\binom{n}{k}\Big(\frac{\lambda}{n}\Big)^k\Big(1-\frac{\lambda}{n}\Big)^{n-k} = \frac{\lambda^k}{k!}\underbrace{\frac{n(n-1)\cdots(n-k+1)}{n^k}}_{\to 1}\underbrace{\Big(1-\frac{\lambda}{n}\Big)^{n-k}}_{\to e^{-\lambda}} \to \frac{\lambda^k}{k!}e^{-\lambda}.$$

so $B(n,p) \approx$ the **Poisson distribution** $\pi_\lambda := e^{-\lambda}\sum_{j\geq 0}\frac{\lambda^j}{j!}\delta_j$ with $\lambda = np$.

2.5 Variance and covariance

Motivation. Variance is the risk of the best constant prediction; uncorrelatedness makes variances add.

Example 2.20

For the uniform distribution on $\{\omega_1,\ldots,\omega_n\}$, $\mathbb{E}[X] = \frac{1}{n}\sum_i X(\omega_i)$ is the arithmetic mean.

Definition 2.21 (Variance)

For $X \in \mathcal{L}^1$, $\mathrm{Var}(X) := \sigma^2(X) := \mathbb{E}[(X-\mathbb{E}[X])^2] \in [0,\infty]$ (the mean square prediction error).

Remark 2.22

For $X \in \mathcal{L}^1$: (i) $\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$; (ii) $\mathrm{Var}(X) = 0 \iff P(X = \mathbb{E}[X]) = 1$; (iii) $\mathrm{Var}(X) < \infty \iff X \in \mathcal{L}^2$.
[Supplement] *Proof.*

(i) Expand $(X-m)^2$, $m=\mathbb{E}[X]$ (both sides ∞ if $\mathbb{E}[X^2]=\infty$).
(ii) Corollary 2.6(i) applied to $(X-m)^2$.
(iii) Use (i) and $X^2 \le 2(X-m)^2 + 2m^2$.

Definition 2.23 (Covariance)

For $X, Y \in \mathcal{L}^2$ (so $XY \in \mathcal{L}^1$ by Cauchy–Schwarz)

$$\operatorname{Cov}(X,Y) := \mathbb{E}[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]. \tag{2.5}$$

Remark 2.24

(i) $\operatorname{Var}(aX+b) = a^2 \operatorname{Var}(X)$ for $X \in \mathcal{L}^1$. (ii) $\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X,Y)$ for $X, Y \in \mathcal{L}^2$. Both follow by expanding the square, using linearity.

Definition 2.25 (Uncorrelated)

$X, Y \in \mathcal{L}^2$ are **uncorrelated** if

$$\operatorname{Cov}(X,Y) = 0 \quad (\iff \operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y)). \tag{2.6}$$

Remark 2.26 (Bienaymé)

If $X_1, \dots, X_n \in \mathcal{L}^2$ are pairwise uncorrelated, then $\operatorname{Var}\left(\sum_i X_i\right) = \sum_i \operatorname{Var}(X_i)$, by expanding $\operatorname{Var}(\sum_i X_i) = \sum_{i,j} \operatorname{Cov}(X_i, X_j)$.

2.6 The monotone class theorem

Motivation. An identity on a multiplicative class that survives monotone limits holds for all bounded measurable functions.

Definition 2.27 (Monotone vector space)

A vector space $\mathcal{H}$ of bounded real functions on Ω is a **monotone vector space** (MVS) if (i) $1 = \mathbb{1}_\Omega \in \mathcal{H}$, and (ii) whenever $f_n \in \mathcal{H}$ with $0 \le f_1 \le f_2 \le \cdots \nearrow f$ and f bounded, then $f \in \mathcal{H}$.

Lemma 2.28

An MVS is closed under uniform convergence.

Proof. Let $f_n \in \mathcal{H}$ with $\|f - f_n\|_\infty := \sup_\omega |f(\omega) - f_n(\omega)| \to 0$.

Step 1. Assume $f \ge 0$ (add the constant $\|f\|_\infty \in \mathcal{H}$).

Step 2. Take summable $\varepsilon_k := \|f_{n_k} - f_{n_{k+1}}\|_\infty$, $a_k := \sum_{n \ge k} \varepsilon_n \searrow 0$, $g_k := f_{n_k} - a_k + 2a_1 \in \mathcal{H}$: $g_{k+1} - g_k = f_{n_{k+1}} - f_{n_k} + \varepsilon_k \ge 0$, $g_k \nearrow f + 2a_1$ (bounded), and $g_1 = f_{n_1} + a_1 \ge f \ge 0$ because $f - f_{n_1} = \sum_k (f_{n_{k+1}} - f_{n_k}) \le a_1$.

Step 3. By Definition 2.27(ii), $f + 2a_1 \in \mathcal{H}$, hence $f \in \mathcal{H}$. ∎

Notation. For a class $\mathcal{M}$ of functions on Ω, $\sigma(\mathcal{M})$ is the smallest σ-algebra making all $f \in \mathcal{M}$ measurable, and $\sigma(\mathcal{M})_b$ the bounded $\sigma(\mathcal{M})$-measurable functions.

Theorem 2.29 (Monotone class theorem, algebra form)

Let $\mathcal{M}$ be a multiplicative class of bounded functions ($f, g \in \mathcal{M} \Rightarrow fg \in \mathcal{M}$) and $\mathcal{H}$ an MVS with $\mathcal{M} \subset \mathcal{H}$. Then $\sigma(\mathcal{M})_b \subset \mathcal{H}$.

Proof.

Step 1 (algebra). $\mathcal{M}_0 := \operatorname{span}(\mathcal{M}, 1) \subset \mathcal{H}$ is an algebra; its uniform closure $\overline{\mathcal{M}_0}$ is an algebra in $\mathcal{H}$ (Lemma 2.28) with $\sigma(\overline{\mathcal{M}_0}) = \sigma(\mathcal{M})$.

Step 2 (truncation). $f \in \overline{\mathcal{M}_0} \Rightarrow f \wedge \alpha \in \overline{\mathcal{M}_0}$: Weierstrass polynomials P_ε with $|x \wedge \alpha - P_\varepsilon(x)| < \varepsilon$ on $[-\|f\|_\infty, \|f\|_\infty]$ give $\|f \wedge \alpha - P_\varepsilon(f)\|_\infty < \varepsilon$.

Step 3 (reduction to indicators). $f \in \sigma(\mathcal{M})_b$ is a uniform limit of elementary functions (Note 1.41), so it suffices that $\mathcal{S} := \{A \in \sigma(\overline{\mathcal{M}_0}) \mid \mathbb{1}_A \in \mathcal{H}\} = \sigma(\overline{\mathcal{M}_0})$.

Step 4 ($\mathcal{S}$ is a Dynkin system). $\Omega \in \mathcal{S}$ since $1 \in \mathcal{H}$; $\mathbb{1}_{A^c} = 1 - \mathbb{1}_A$; for disjoint A_i, $\sum_{i \le n} \mathbb{1}_{A_i} \nearrow \mathbb{1}_{\bigcup A_i}$, so MVS (ii) applies.

Step 5 (a $\cap$-stable generator). Let $\mathcal{E} := \{A \in \sigma(\overline{\mathcal{M}_0}) \mid \exists f_n \in \overline{\mathcal{M}_0},\ f_n \ge 0,\ f_n \nearrow \mathbb{1}_A\}$. $\mathcal{E}$ is $\cap$-stable (if $f_n \nearrow \mathbb{1}_A$, $g_n \nearrow \mathbb{1}_B$, then $f_n g_n \nearrow \mathbb{1}_{A \cap B}$), and $\mathcal{E} \subset \mathcal{S}$ by MVS (ii). Theorem 1.8 gives

$$\sigma(\mathcal{E}) = \mathcal{D}(\mathcal{E}) \subset \mathcal{S} \subset \sigma(\overline{\mathcal{M}_0}). \tag{2.7}$$

Step 6. $(n(f-\alpha)^+) \wedge 1 \in \overline{\mathcal{M}_0}$ (Step 2; $(f-\alpha)^+ = (f-\alpha) - ((f-\alpha) \wedge 0)$) increases to $\mathbb{1}_{\{f > \alpha\}}$, so the generating sets $\{f > \alpha\}$ lie in $\mathcal{E}$ and (2.7) is an equality. ∎

Example 2.30

Let μ_1, μ_2 be probability measures on a metric space (S, d) with $\int f\, d\mu_1 = \int f\, d\mu_2$ for all $f \in C_b(S)$ (bounded continuous). Then $\mu_1 = \mu_2$.
Proof. $\mathcal{H} := \{f \in \mathcal{B}(S)_b \mid \int f\, d\mu_1 = \int f\, d\mu_2\}$ is an MVS (MCT) containing $C_b(S)$, so $\mathcal{H} \supset \sigma(C_b(S))_b$. Closed $F = \{d(\cdot, F) \wedge 1 = 0\} \in \sigma(C_b(S))$, so $\sigma(C_b(S)) = \mathcal{B}(S)$ and $\mu_1 = \mu_2$.

2.7 Conditional expectation

Motivation. The best guess given $\mathcal{A}$: the L^2 projection, characterised by $\mathbb{E}[X\mathbb{1}_A] = \mathbb{E}[Y\mathbb{1}_A]$, which extends to $\mathcal{L}^1$.

$(\Omega, \mathcal{F}, P)$ is a probability space, $\mathcal{A} \subset \mathcal{F}$ a sub-σ-algebra.

Heuristics. For $X \in L^2(\Omega, \mathcal{F}, P)$ seek $Y \in L^2(\Omega, \mathcal{A}, P)$ with

$$\mathbb{E}[(X-Y)^2] \le \mathbb{E}[(X-Y_0)^2] \qquad \forall Y_0 \in L^2(\Omega, \mathcal{A}, P). \tag{2.8}$$

Then for $Z \in L^2(\mathcal{A})$, $\varepsilon \neq 0$, $\mathbb{E}[(X-Y)^2] \le \mathbb{E}[(X-Y-\varepsilon Z)^2]$ gives $2\varepsilon\mathbb{E}[(X-Y)Z] \le \varepsilon^2\mathbb{E}[Z^2]$, and $\varepsilon \to 0^\pm$ gives $\mathbb{E}[XZ] = \mathbb{E}[YZ]$. Y is the orthogonal projection onto the closed $L^2(\Omega, \mathcal{A}, P)$.

Definition 2.31 (Conditional expectation)

For $X \in \mathcal{L}^1(\Omega, \mathcal{F}, P)$, a **conditional expectation of X given $\mathcal{A}$** is any r.v. Y with (i) Y is $\mathcal{A}$-measurable ("$Y \in \mathcal{A}$"), and (ii) $\mathbb{E}[X\mathbb{1}_A] = \mathbb{E}[Y\mathbb{1}_A]$ for all $A \in \mathcal{A}$. Such Y exists, is

integrable and is a.s. unique; we write $Y = \mathbb{E}[X \mid \mathcal{A}]$.

Integrability. With $A = \{Y > 0\} \in \mathcal{A}$: $\mathbb{E}[Y^+] = \mathbb{E}[Y\mathbb{1}_A] = \mathbb{E}[X\mathbb{1}_A] \le \mathbb{E}[|X|]$ and $\mathbb{E}[Y^-] = \mathbb{E}[-X\mathbb{1}_{A^c}] \le \mathbb{E}[|X|]$.

Uniqueness. For another Y', $A_n := \{Y - Y' \ge \frac{1}{n}\} \in \mathcal{A}$ gives $0 = \mathbb{E}[(Y - Y')\mathbb{1}_{A_n}] \ge \frac{1}{n}P(A_n)$, so $P(Y > Y') \le \sum_n P(A_n) = 0$; symmetrically $P(Y' > Y) = 0$.

Existence follows from Radon–Nikodym ($\mathbb{E}[X \mid \mathcal{A}]$ as a density).

Definition 2.32 (Absolute continuity)

For finite measures ν, μ on $(\Omega, \mathcal{A})$, ν is **absolutely continuous** w.r.t. μ, $\nu \ll \mu$, if $\mu(N) = 0 \Rightarrow \nu(N) = 0$ for $N \in \mathcal{A}$.

Theorem 2.33 (Radon–Nikodym)

Let ν, μ be finite measures on $(\Omega, \mathcal{A})$ with $\nu \ll \mu$. Then there is $\varphi \in \mathcal{L}^1(\Omega, \mathcal{A}, \mu)$, $\varphi \ge 0$, with

$$\nu(A) = \int_A \varphi \, d\mu \qquad \text{for all } A \in \mathcal{A},$$

and φ is μ-a.e. unique. It is the **Radon–Nikodym derivative** $\frac{d\nu}{d\mu}$. (The theorem extends to σ-finite ν, μ, without integrability of the density.)

Proof [Supplement] (von Neumann).

Step 1 (a representing function). $\rho := \mu + \nu$; $\left|\int h \, d\nu\right| \le \int |h| \, d\rho \le \rho(\Omega)^{1/2}\|h\|_{L^2(\rho)}$, so by Riesz on the Hilbert space $L^2(\rho)$ (Theorem 1.70) some $g \in L^2(\rho)$ has

$$\int h \, d\nu = \int hg \, d\rho \qquad \forall h \in L^2(\rho). \tag{2.9}$$

Step 2 ($0 \le g \le 1$). With $h = \mathbb{1}_{\{g<0\}}$: $0 \le \nu(g < 0) = \int_{\{g<0\}} g \, d\rho \le 0$, so $\rho(g < 0) = 0$. With $h = \mathbb{1}_{\{g>1\}}$: $\nu(g > 1) = \int_{\{g>1\}} g \, d\rho \ge \rho(g > 1) \ge \nu(g > 1)$, so $\rho(g > 1) = 0$; modify g on a null set to get $0 \le g \le 1$.

Step 3. $\int h(1 - g) \, d\nu = \int hg \, d\mu$ for bounded, hence (MCT) all $h \ge 0$. $N := \{g = 1\}$: $\mu(N) = \int_N (1 - g) \, d\nu = 0$, so $\nu(N) = 0$.

Step 4. $h = \mathbb{1}_A(1 + g + \cdots + g^n)$: $\int_A (1 - g^{n+1}) \, d\nu = \int_A g(1 + \cdots + g^n) \, d\mu$; with $1 - g^{n+1} \nearrow \mathbb{1}_{N^c}$, $g(1 + \cdots + g^n) \nearrow \varphi := \frac{g}{1-g}\mathbb{1}_{N^c}$ and the MCT, $\nu(A) = \nu(A \cap N^c) = \int_A \varphi \, d\mu$; $A = \Omega$ gives $\varphi \in \mathcal{L}^1$.

Step 5 (uniqueness). $\int_A (\varphi - \varphi') \, d\mu = 0$ for all A; Lemma 1.62(ii). ∎

Existence of $\mathbb{E}[X \mid \mathcal{A}]$. $\nu^{\pm}(A) := \int_A X^{\pm} \, dP$ on $\mathcal{A}$ are finite, $\nu^{\pm} \ll P|_{\mathcal{A}}$; with $Y := \frac{d\nu^+}{dP|_{\mathcal{A}}} - \frac{d\nu^-}{dP|_{\mathcal{A}}}$ (Theorem 2.33), for $A \in \mathcal{A}$

$$\int_A X \, dP = \nu^+(A) - \nu^-(A) = \int_A \frac{d\nu^+}{dP|_{\mathcal{A}}} \, dP - \int_A \frac{d\nu^-}{dP|_{\mathcal{A}}} \, dP = \int_A Y \, dP,$$

so $Y = \mathbb{E}[X \mid \mathcal{A}]$. With $A = \Omega$: $\mathbb{E}[\mathbb{E}[X \mid \mathcal{A}]] = \mathbb{E}[X]$.

Interpretation. $\mathbb{E}[X \mid \mathcal{A}]$ is the best guess for X knowing which $A \in \mathcal{A}$ occurred.

Example 2.34

Let $X \in \mathcal{L}^1(\Omega, \mathcal{F}, P)$.

(i) X $\mathcal{A}$-measurable $\Rightarrow \mathbb{E}[X \mid \mathcal{A}] = X$.

(ii) $\mathcal{A} = \{\emptyset, \Omega\} \Rightarrow \mathbb{E}[X \mid \mathcal{A}] = \mathbb{E}[X]$.

(iii) $\mathcal{A} = \{\emptyset, A, A^c, \Omega\}$ with $0 < P(A) < 1 \Rightarrow \mathbb{E}[X \mid \mathcal{A}] = \frac{\mathbb{E}[X\mathbb{1}_A]}{P(A)}\mathbb{1}_A + \frac{\mathbb{E}[X\mathbb{1}_{A^c}]}{P(A^c)}\mathbb{1}_{A^c}$.

(iv) If X is independent of $\mathcal{A}$ (i.e. $\mathbb{1}_A$ and X are independent for all $A \in \mathcal{A}$, Section 3.4), then $\mathbb{E}[X\mathbb{1}_A] = \mathbb{E}[X]\mathbb{E}[\mathbb{1}_A] = \mathbb{E}[\mathbb{E}[X]\mathbb{1}_A]$, so $\mathbb{E}[X \mid \mathcal{A}] = \mathbb{E}[X]$.

Check Definition 2.31 directly.

Theorem 2.35 (Properties of conditional expectation)

Let $X, Y \in \mathcal{L}^1(\Omega, \mathcal{F}, P)$; (in)equalities hold P-a.s.

(i) (Linearity) $\mathbb{E}[\alpha X + \beta Y \mid \mathcal{A}] = \alpha\mathbb{E}[X \mid \mathcal{A}] + \beta\mathbb{E}[Y \mid \mathcal{A}]$.

(ii) (Monotonicity) $X \le Y \Rightarrow \mathbb{E}[X \mid \mathcal{A}] \le \mathbb{E}[Y \mid \mathcal{A}]$.

(iii) (Pulling out what is known) Y $\mathcal{A}$-measurable, $XY \in \mathcal{L}^1 \Rightarrow \mathbb{E}[XY \mid \mathcal{A}] = Y\mathbb{E}[X \mid \mathcal{A}]$.

(iv) (Tower property) For $\mathcal{A}_1 \subset \mathcal{A}_2 \subset \mathcal{F}$: $\mathbb{E}[\mathbb{E}[X \mid \mathcal{A}_1] \mid \mathcal{A}_2] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{A}_2] \mid \mathcal{A}_1] = \mathbb{E}[X \mid \mathcal{A}_1]$.

(v) (Contraction) $|\mathbb{E}[X \mid \mathcal{A}]| \le \mathbb{E}[|X| \mid \mathcal{A}]$.

(vi) (DCT) If $|X_n| \le Y \in \mathcal{L}^1$ and $X_n \to X$ a.s., then $\mathbb{E}[X_n \mid \mathcal{A}] \to \mathbb{E}[X \mid \mathcal{A}]$ in $L^1(\Omega, \mathcal{A}, P)$ and a.s.

(vii) (Jensen) $\varphi : \mathbb{R} \to \mathbb{R}$ convex, $\varphi(X) \in \mathcal{L}^1 \Rightarrow \varphi(\mathbb{E}[X \mid \mathcal{A}]) \le \mathbb{E}[\varphi(X) \mid \mathcal{A}]$.

Proof. (i) For $A \in \mathcal{A}$, $\mathbb{E}[(\alpha X + \beta Y)\mathbb{1}_A] = \alpha\mathbb{E}[\mathbb{E}[X \mid \mathcal{A}]\mathbb{1}_A] + \beta\mathbb{E}[\mathbb{E}[Y \mid \mathcal{A}]\mathbb{1}_A] = \mathbb{E}[(\alpha\mathbb{E}[X \mid \mathcal{A}] + \beta\mathbb{E}[Y \mid \mathcal{A}])\mathbb{1}_A]$, and the bracket is $\mathcal{A}$-measurable.

(ii) $A := \{\mathbb{E}[X \mid \mathcal{A}] > \mathbb{E}[Y \mid \mathcal{A}]\} \in \mathcal{A}$ and $0 \ge \mathbb{E}[(X - Y)\mathbb{1}_A] = \mathbb{E}[(\mathbb{E}[X \mid \mathcal{A}] - \mathbb{E}[Y \mid \mathcal{A}])\mathbb{1}_A]$ by (i); the integrand is > 0 on A, so $P(A) = 0$.

(iii) First X^+Y^+: take $\mathcal{A}$-elementary $Y_n^+ = \sum_i c_{i,n}\mathbb{1}_{A_{i,n}} \nearrow Y^+$. For $A \in \mathcal{A}$, the MCT twice gives

$$\mathbb{E}[X^+Y^+\mathbb{1}_A] = \lim_n \sum_i c_{i,n}\mathbb{E}[X^+\mathbb{1}_{A_{i,n}\cap A}] = \lim_n \sum_i c_{i,n}\mathbb{E}[\mathbb{E}[X^+ \mid \mathcal{A}]\mathbb{1}_{A_{i,n}\cap A}] = \mathbb{E}[\mathbb{E}[X^+ \mid \mathcal{A}]Y^+\mathbb{1}_A].$$

So $\mathbb{E}[X^+Y^+ \mid \mathcal{A}] = Y^+\mathbb{E}[X^+ \mid \mathcal{A}]$; the same holds for $X^\pm Y^\mp$, X^-Y^-, and linearity gives the claim.

(iv) $\mathbb{E}[X \mid \mathcal{A}_1]$ is $\mathcal{A}_2$-measurable, so the first expression equals $\mathbb{E}[X \mid \mathcal{A}_1]$ by Example 2.34(i). For $A \in \mathcal{A}_1 \subset \mathcal{A}_2$: $\mathbb{E}[\mathbb{1}_A\mathbb{E}[\mathbb{E}[X \mid \mathcal{A}_2] \mid \mathcal{A}_1]] = \mathbb{E}[\mathbb{1}_A\mathbb{E}[X \mid \mathcal{A}_2]] = \mathbb{E}[\mathbb{1}_A X] = \mathbb{E}[\mathbb{1}_A\mathbb{E}[X \mid \mathcal{A}_1]]$.

(v) $\pm X \le |X|$ with (i), (ii).

(vi) $X \in \mathcal{L}^1$ (DCT); by (v), (i), $\mathbb{E}|\mathbb{E}[X_n \mid \mathcal{A}] - \mathbb{E}[X \mid \mathcal{A}]| \le \mathbb{E}|X_n - X| \to 0$. [Supplement] *A.s.:* $Z_n := \sup_{k\ge n}|X_k - X| \le 2Y$ decreases to 0, $|\mathbb{E}[X_n \mid \mathcal{A}] - \mathbb{E}[X \mid \mathcal{A}]| \le \mathbb{E}[Z_n \mid \mathcal{A}] \searrow Z \ge 0$ (by (ii)), and $\mathbb{E}[Z] \le \mathbb{E}[Z_n] \to 0$ (DCT).

(vii) [Supplement] $\varphi = \sup_k \ell_k$ for countably many affine $\ell_k(x) = a_k x + b_k$ (supporting lines at rationals); by linearity and monotonicity, $\ell_k(\mathbb{E}[X \mid \mathcal{A}]) = \mathbb{E}[\ell_k(X) \mid \mathcal{A}] \le \mathbb{E}[\varphi(X) \mid \mathcal{A}]$ off a null N_k; take $\sup_k$ off $\bigcup_k N_k$. ∎

Definition 2.36 (Conditional expectation of nonnegative r.v.'s)

For $\mathcal{F}$-measurable $X : \Omega \to [0,\infty]$, $\mathbb{E}[X \mid \mathcal{A}]$ is any Y with (i) Y $\mathcal{A}$-measurable and (ii) $\mathbb{E}[X\mathbb{1}_A] = \mathbb{E}[Y\mathbb{1}_A]$ for all $A \in \mathcal{A}$. Such Y exists, is a.s. unique and a.s. $[0,\infty]$-valued.

Existence. $\mathbb{E}[X \wedge n \mid \mathcal{A}]$ increases a.s. (Theorem 2.35(ii)); $Y := \lim_n \max_{k\le n} \mathbb{E}[X_k \mid \mathcal{A}]$ ($X_k := X \wedge k$) is $\mathcal{A}$-measurable and, by the MCT, for $A \in \mathcal{A}$

$$\mathbb{E}[X\mathbb{1}_A] = \lim_n \mathbb{E}[X_n\mathbb{1}_A] = \lim_n \mathbb{E}[\mathbb{E}[X_n \mid \mathcal{A}]\mathbb{1}_A] = \lim_n \mathbb{E}\big[\max_{k\le n} \mathbb{E}[X_k \mid \mathcal{A}]\mathbb{1}_A\big] = \mathbb{E}[Y\mathbb{1}_A].$$

Uniqueness and positivity [Supplement]. For rational $a < b$, $A := \{Y' \le a < b \le Y\}$ has $bP(A) \le \mathbb{E}[Y\mathbb{1}_A] = \mathbb{E}[Y'\mathbb{1}_A] \le aP(A) < \infty$, so $P(A) = 0$; hence $Y = Y'$ a.s. $A_n := \{Y < -\frac{1}{n}\}$ has $0 \le \mathbb{E}[Y\mathbb{1}_{A_n}] \le -\frac{1}{n}P(A_n)$, so $Y \ge 0$ a.s.

Likewise, for $X, Y \ge 0$: $\mathbb{E}[\mathbb{E}[X \mid \mathcal{A}]] = \mathbb{E}[X]$; $X \in \mathcal{A} \Rightarrow \mathbb{E}[X \mid \mathcal{A}] = X$; $X = Y$ a.s. $\Rightarrow$ equal conditional expectations; linearity with coefficients $\alpha, \beta \ge 0$; and $X \le Y$ a.s. $\Rightarrow \mathbb{E}[X \mid \mathcal{A}] \le \mathbb{E}[Y \mid \mathcal{A}]$.

Theorem 2.37

Let $X_n : \Omega \to [0,\infty]$ be $\mathcal{F}$-measurable; (in)equalities hold a.s.

(1) (Beppo Levi) $X_n \nearrow \Rightarrow \mathbb{E}[\sup_n X_n \mid \mathcal{A}] = \sup_n \mathbb{E}[X_n \mid \mathcal{A}]$.

(2) (Fatou) $\mathbb{E}[\liminf_n X_n \mid \mathcal{A}] \le \liminf_n \mathbb{E}[X_n \mid \mathcal{A}]$.

Proof [Supplement]. (1) $Y_n := \mathbb{E}[X_n \mid \mathcal{A}] \nearrow Y := \sup_n Y_n$ a.s.; for $A \in \mathcal{A}$, the MCT twice: $\mathbb{E}[\sup_n X_n\mathbb{1}_A] = \lim_n \mathbb{E}[X_n\mathbb{1}_A] = \lim_n \mathbb{E}[Y_n\mathbb{1}_A] = \mathbb{E}[Y\mathbb{1}_A]$, so $Y = \mathbb{E}[\sup_n X_n \mid \mathcal{A}]$.

(2) $g_n := \inf_{k\ge n} X_k \nearrow \liminf_n X_n$, $\mathbb{E}[g_n \mid \mathcal{A}] \le \mathbb{E}[X_k \mid \mathcal{A}]$ ($k \ge n$); by (1) $\mathbb{E}[\liminf_n X_n \mid \mathcal{A}] = \lim_n \mathbb{E}[g_n \mid \mathcal{A}] \le \lim_n \inf_{k\ge n} \mathbb{E}[X_k \mid \mathcal{A}]$. ■

Chapter 3. Sequences of Random Variables

3.1 Infinitely many fair coin tosses

Motivation. Limits need all tosses on one space, built from Lebesgue measure via binary expansions.

Ω is the set of outcomes of a **random experiment**; probability attaches a number to every event.

One toss. $\Omega = \{0,1\}$, $\mathcal{A} = \mathcal{P}(\Omega)$, $P(\{0\}) = P(\{1\}) = \frac{1}{2}$.

n **tosses.** $\Omega = \{0,1\}^n$, $\mathcal{A} = \mathcal{P}(\Omega)$ and $P(A) = |A|/2^n$: the **uniform distribution**, each binary word having probability 2^{-n}.

Infinitely many tosses. $\Omega = \{0,1\}^{\mathbb{N}} = \{(x_i)_{i\in\mathbb{N}} \mid x_i \in \{0,1\}\}$ is uncountable; $(x_i) \mapsto \sum_i x_i 2^{-i}$ maps it onto $[0,1]$ (not injectively). Let $\mathcal{A} := \sigma(\mathcal{A}_0)$ with the *cylinder sets*

$$\mathcal{A}_0 := \{B \subset \Omega \mid B = B_0 \times \{0,1\} \times \{0,1\} \times \cdots \text{ for some } n \text{ and } B_0 \subset \{0,1\}^n\},$$

(events depending on finitely many tosses); we need P on $\mathcal{A}$ with $P(x_1 = \bar{x}_1, \dots, x_n = \bar{x}_n) = 2^{-n}$.

Existence via Lebesgue measure. Let $(\tilde{\Omega}, \tilde{\mathcal{A}}, \tilde{P}) = ([0,1], \mathcal{B}([0,1]), \lambda)$. On $\Omega = \{0,1\}^{\mathbb{N}}$ let $X_i((x_n)_n) := x_i$ (the i-th toss) and $\mathcal{A} := \sigma(\{X_i = 1\} \mid i \in \mathbb{N})$, which equals $\sigma(\mathcal{A}_0)$. The non-terminating binary expansion of $\tilde{\omega} \in [0,1]$ is

$$\tilde{\omega} = \sum_{i=1}^{\infty} T_i(\tilde{\omega}) 2^{-i}, \qquad T_i := \sum_{k=1}^{2^{i-1}} \mathbb{1}_{\left(\frac{2k-1}{2^i}, \frac{2k}{2^i}\right]},$$

and defines $T : \tilde{\Omega} \to \Omega$, $T(\tilde{\omega}) := (T_1(\tilde{\omega}), T_2(\tilde{\omega}), \dots)$. T misses the sequences ending in zeros (other than $(0,0,\dots)$), i.e. the countably many dyadic rationals.

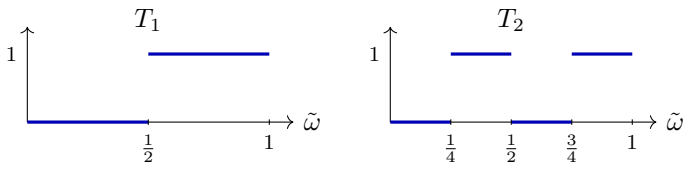


Figure. *Digit maps T_1, T_2; fixing n digits gives an interval of length 2^{-n}.*

Since $T_i = X_i \circ T$, $T^{-1}(\{X_i = 1\}) = \{T_i = 1\}$ is a finite union of intervals, so T is $\tilde{\mathcal{A}}/\mathcal{A}$-measurable (Remark 1.32). $P := \tilde{P} \circ T^{-1}$ is a probability measure on $(\Omega, \mathcal{A})$ (Proposition 2.13), and for $(x_1, \dots, x_n) \in \{0,1\}^n$

$$P(X_1 = x_1, \dots, X_n = x_n) = \tilde{P}\Big(\bigcap_{i \le n} \{T_i = x_i\}\Big) = \tilde{P}(\text{an interval of length } 2^{-n}) = 2^{-n}.$$

So P extends the model for n tosses; uniqueness follows from Theorem 1.9, since the cylinder sets form a $\cap$-stable generator.

Some expectations.

- The i-th toss: $\mathbb{E}[X_i] = 1 \cdot P(X_i = 1) + 0 \cdot P(X_i = 0) = \frac{1}{2}$.
- Number of successes $S_n := X_1 + \cdots + X_n$: $P(S_n = k) = \binom{n}{k} 2^{-n}$, so $\mathbb{E}[S_n] = \sum_k k\binom{n}{k} 2^{-n} = \frac{n}{2}$; more simply, $\mathbb{E}[S_n] = \sum_i \mathbb{E}[X_i] = \frac{n}{2}$ by linearity.

- Waiting time $S := \min\{n \mid X_n = 1\}$ ($\min\emptyset := \infty$), measurable since $\{S = k\} = \{X_1 = \cdots = X_{k-1} = 0, X_k = 1\}$: $P(S = k) = 2^{-k}$, $P(S = \infty) = 0$, and

$$\mathbb{E}[S] = \sum_{k\geq 1} k2^{-k} = \frac{1}{2}\sum_{k\geq 1} k\Big(\frac{1}{2}\Big)^{k-1} = 2, \qquad \text{using } \sum_{k\geq 1} kq^{k-1} = \frac{d}{dq}\frac{q}{1-q} = \frac{1}{(1-q)^2}.$$

- (*Doubling strategy*) Doubling a unit stake until the first success gives stake $I_n = 2^{n-1}\mathbb{1}_{\{T>n-1\}}$ (T the waiting time): $\mathbb{E}[I_n] = 2^{n-1}2^{-(n-1)} = 1$, yet $I_n \to 0$ a.s.

3.2 Infinitely many biased coin tosses

Motivation. Success probability p: moments from the joint law.

With $\Omega = \{0,1\}^{\mathbb{N}}$, X_i and $\mathcal{A}$ as before, there is a unique probability measure $P = P_p$ on $(\Omega, \mathcal{A})$ with

$$P_p(X_{i_1} = x_1, \ldots, X_{i_n} = x_n) = p^{\sum_i x_i}(1-p)^{n-\sum_i x_i} \qquad (1 \leq i_1 < \cdots < i_n).$$

(Existence for $p = \frac{1}{2}$ was shown above; the general case and uniqueness come in Section 3.6 and Example 3.13.) Then $P_p(X_i = 1) = p$ and $P_p(X_i = X_j = 1) = p^2$ for $i \neq j$, so

$$\mathbb{E}_p[X_i] = p, \qquad \mathrm{Var}(X_i) = p - p^2 = p(1-p), \qquad \mathrm{Cov}(X_i, X_j) = \mathbb{E}_p[X_iX_j] - p^2 = 0 \ (i \neq j).$$

The X_i are pairwise uncorrelated (in fact independent, Example 3.13). For $S_n = X_1 + \cdots + X_n$: $\mathbb{E}_p[S_n] = np$, $\mathrm{Var}(S_n) = np(1-p)$ (Bienaymé).

- $X := \sum_n 2^{-n}X_n$ has $\mathbb{E}_p[X] = \sum_n 2^{-n}p = p$ by Theorem 1.56. The partial sums are Cauchy in L^2, and uncorrelatedness gives $\mathrm{Var}(X) = \sum_n 4^{-n}p(1-p) = \frac{1}{3}p(1-p)$.
- The waiting time S for the first success is **geometric**: $P_p(S = n) = (1-p)^{n-1}p$. Hence $\mathbb{E}_p[S] = \sum_n n(1-p)^{n-1}p = \frac{1}{p}$ and $\mathrm{Var}(S) = \frac{1-p}{p^2}$. [Supplement] For the variance, $\mathbb{E}_p[S(S-1)] = p(1-p)\sum_n n(n-1)(1-p)^{n-2} = \frac{2(1-p)}{p^2}$ (second derivative of the geometric series), so $\mathrm{Var}(S) = \frac{2(1-p)}{p^2} + \frac{1}{p} - \frac{1}{p^2} = \frac{1-p}{p^2}$.

3.3 The (strong and the weak) law of large numbers

Motivation. Empirical means of uncorrelated, bounded-variance variables converge in L^2, in probability and a.s.

$X_1, X_2, \cdots \in \mathcal{L}^2$ are uncorrelated with $\sup_i \sigma_i^2 < \infty$, $\sigma_i^2 := \mathrm{Var}(X_i)$, and $S_n := X_1 + \cdots + X_n$ (S_n/n: *empirical mean*).

Remark 3.1

We may assume $\mathbb{E}[X_i] = 0$: the centred $\tilde{X}_i := X_i - \mathbb{E}[X_i]$ are in $\mathcal{L}^2$, uncorrelated, have the same variances, and $\frac{\tilde{S}_n}{n} - \frac{\mathbb{E}[\tilde{S}_n]}{n} = \frac{S_n}{n} - \frac{\mathbb{E}[S_n]}{n}$.

Theorem 3.2 (L^2 law of large numbers)

$\lim_n \mathbb{E}\Big[\Big(\frac{S_n}{n} - \frac{\mathbb{E}[S_n]}{n}\Big)^2\Big] = 0$ (resp. $\mathbb{E}[(\frac{S_n}{n} - m)^2] \to 0$ if $\mathbb{E}[X_i] \equiv m$).

Proof. By Bienaymé (Remark 2.26),

$$\mathbb{E}\Big[\Big(\frac{S_n}{n}-\frac{\mathbb{E}[S_n]}{n}\Big)^2\Big]=\operatorname{Var}\Big(\frac{S_n}{n}\Big)=\frac{1}{n^2}\sum_{i=1}^{n}\sigma_i^2\le\frac{1}{n}\sup_i\sigma_i^2\to 0.$$

■

Remark 3.3

This is a Hilbert space fact: in L^2 with $(X,Y)=\mathbb{E}[XY]$, the arithmetic mean of uniformly bounded orthogonal vectors tends to 0, since $\|S_n/n\|^2=n^{-2}\sum_{i\le n}\|X_i\|^2\to 0$.

Theorem 3.4 (Weak law of large numbers)

If $X_i\in\mathcal{L}^2$ are uncorrelated with uniformly bounded variances and $\mathbb{E}[X_i]=m$, then for every $\varepsilon>0$, $\lim_n P(|\frac{S_n}{n}-m|\ge\varepsilon)=0$, i.e. $S_n/n\to m$ in probability.

Proof. Chebyshev: $P(|\frac{S_n}{n}-m|\ge\varepsilon)\le\varepsilon^{-2}\operatorname{Var}(\frac{S_n}{n})\to 0$ by Theorem 3.2. ■

Example 3.5 (Bernoulli)

For Bernoulli experiments with parameter p, $\mathbb{E}_p[X_i]=p$ and $\operatorname{Var}(X_i)=p(1-p)\le\frac{1}{4}$, so

$$P_p\Big(\Big|\frac{S_n}{n}-p\Big|\ge\varepsilon\Big)\to 0. \tag{3.1}$$

(J. Bernoulli, *Ars Conjectandi*): p is a long-run relative frequency.

Example 3.6 (Bernstein polynomials)

Every $f\in C([0,1])$ is a uniform limit of polynomials. For $p\in[0,1]$ the transformation theorem (Remark 2.15) gives

$$B_n(p):=\sum_{k=0}^{n}f\Big(\frac{k}{n}\Big)\binom{n}{k}p^k(1-p)^{n-k}=\sum_k f\Big(\frac{k}{n}\Big)P_p(S_n=k)=\mathbb{E}_p\Big[f\Big(\frac{S_n}{n}\Big)\Big].$$

With δ from uniform continuity, splitting on $\{|\frac{S_n}{n}-p|\le\delta\}$ and Chebyshev:

$$|B_n(p)-f(p)|\le\mathbb{E}_p\Big|f\Big(\frac{S_n}{n}\Big)-f(p)\Big|\le\varepsilon+2\|f\|_\infty P_p\Big(\Big|\frac{S_n}{n}-p\Big|>\delta\Big)\le\varepsilon+\frac{2\|f\|_\infty}{4\delta^2 n}.$$

Hence $\limsup_n\|B_n-f\|_\infty\le\varepsilon$ for every ε, i.e. $\|B_n-f\|_\infty\to 0$.

Theorem 3.7 (Strong law of large numbers)

Let $X_1,X_2,\dots\in\mathcal{L}^2$ be uncorrelated with $\sup_i\sigma^2(X_i)=c<\infty$. Then $\frac{S_n}{n}-\frac{\mathbb{E}[S_n]}{n}\to 0$ P-a.s. (resp. $\frac{S_n}{n}\to m$ a.s. if $\mathbb{E}[X_i]\equiv m$).

Proof. By Remark 3.1 let $\mathbb{E}[X_i]=0$.

Step 1 (the subsequence $n_k=k^2$). For $\varepsilon>0$, Chebyshev gives $P(|S_{k^2}|/k^2>\varepsilon)\le\frac{\operatorname{Var}(S_{k^2})}{\varepsilon^2k^4}\le\frac{c}{\varepsilon^2k^2}$, which is summable. By Lemma 2.10, $S_{k^2}/k^2\to 0$ outside a null set N_1.

Step 2 (fluctuations between squares). Let $D_k := \max_{k^2 \le l < (k+1)^2} |S_l - S_{k^2}|$. Since $\operatorname{Var}(S_l - S_{k^2}) = \sum_{k^2 < i \le l} \sigma_i^2 \le 2kc$ and there are $2k$ indices l,

$$P\Big(\frac{D_k}{k^2} \ge \varepsilon\Big) \le \sum_{l=k^2+1}^{k^2+2k} P(|S_l - S_{k^2}| \ge \varepsilon k^2) \le 2k \cdot \frac{2kc}{\varepsilon^2 k^4} = \frac{4c}{\varepsilon^2 k^2}.$$

Lemma 2.10 gives $D_k/k^2 \to 0$ outside a null set N_2.

Step 3. For $k^2 \le n < (k+1)^2$: $\frac{|S_n|}{n} \le \frac{|S_{k^2}| + D_k}{k^2} \to 0$ on $(N_1 \cup N_2)^c$. ∎

Example 3.8

For Bernoulli experiments, $\frac{1}{n}\sum_{i \le n} X_i \to p$ P_p-a.s. (É. Borel 1909, strengthening Bernoulli's (3.1)). For a fair coin, $S_n = Y_1 + \cdots + Y_n$, $Y_i := 2X_i - 1$, is a **simple random walk** (rescaled: Brownian motion, Section 4.1); $S_n/n \to 0$ a.s., and the exact order is the *law of the iterated logarithm*:

$$\limsup_n \frac{S_n}{\sqrt{2n \log\log n}} = +1, \qquad \liminf_n \frac{S_n}{\sqrt{2n \log\log n}} = -1 \qquad P\text{-a.s.}$$

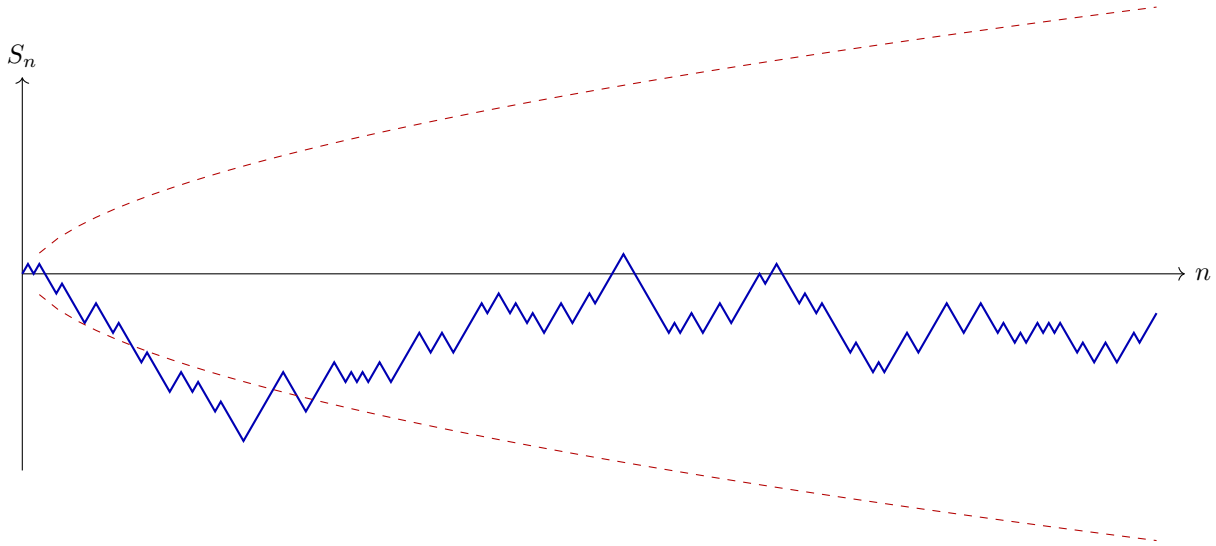


Figure. *A 200-step random walk with $\pm\sqrt{2n \log\log n}$ (dashed).*

3.4 Independence

Motivation. Independence passes from generators to σ-algebras, giving the 0–1 law and Borel–Cantelli 2.

3.4.1 Independent events

Definition 3.9 (Independence)

Events $A_i \in \mathcal{A}$, $i \in I$, are **independent** if $P(\bigcap_{j \in J} A_j) = \prod_{j \in J} P(A_j)$ for every finite $J \subset I$. Families $\mathcal{B}_i \subset \mathcal{A}$, $i \in I$, are **independent** if this holds for all finite $J \subset I$ and all choices $A_j \in \mathcal{B}_j$.

Theorem 3.10

Let $\mathcal{B}_i$, $i \in I$, be independent $\cap$-stable families. Then (i) $\sigma(\mathcal{B}_i)$, $i \in I$, are independent; (ii) for a partition $I = \bigcup_{k\in K} J_k$, the σ-algebras $\sigma\big(\bigcup_{i\in J_k} \mathcal{B}_i\big)$, $k \in K$, are independent.

Proof. (i) Let $J = \{j_1, \dots, j_n\}$ and $A_{j_\nu} \in \sigma(\mathcal{B}_{j_\nu})$; we show

$$P(A_{j_1} \cap \cdots \cap A_{j_n}) = P(A_{j_1}) \cdots P(A_{j_n}). \tag{3.2}$$

Step 1. Fix $A_{j_2} \in \mathcal{B}_{j_2}, \dots, A_{j_n} \in \mathcal{B}_{j_n}$. Then $\mathcal{D}_{j_1} := \{A \in \sigma(\mathcal{B}_{j_1}) \mid P(A \cap A_{j_2} \cap \cdots \cap A_{j_n}) = P(A)P(A_{j_2}) \cdots P(A_{j_n})\}$ is a Dynkin system containing $\mathcal{B}_{j_1}$, so $\mathcal{D}_{j_1} = \sigma(\mathcal{B}_{j_1})$ by Theorem 1.8.

Step 2. Now $\sigma(\mathcal{B}_{j_1}), \mathcal{B}_{j_2}, \dots, \mathcal{B}_{j_n}$ are independent $\cap$-stable families; repeat Step 1 for j_2, then j_3, and so on, to get (3.2).

(ii) $\mathcal{C}_k := \{\bigcap_{j\in J} A_j \mid J \subset J_k \text{ finite}, \ A_j \in \mathcal{B}_j\}$ are $\cap$-stable and independent: $P\big(\bigcap_\nu \bigcap_{i\in J_\nu} A_i\big) = \prod_\nu P\big(\bigcap_{i\in J_\nu} A_i\big)$. By (i) the $\sigma(\mathcal{C}_k)$ are independent, and $\sigma(\mathcal{C}_k) = \sigma\big(\bigcup_{i\in J_k} \mathcal{B}_i\big)$. ∎

Example 3.11

If A_i, $i \in I$, are independent, then the σ-algebras $\sigma(\{A_i\}) = \{\emptyset, A_i, A_i^c, \Omega\}$ are independent (Theorem 3.10(i) with $\mathcal{B}_i = \{A_i\}$). But $\{A_i, A_i^c \mid i \in I\}$ is not independent unless $P(A_i) \in \{0, 1\}$.

Remark 3.12

Pairwise independence is weaker: for two fair tosses, A = "first 1", B = "second 1", C = "equal" have probability $\frac{1}{2}$, pairwise intersections $\frac{1}{4}$, but $P(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{8} = P(A)P(B)P(C)$.

Example 3.13 (Independent 0-1 experiments)

On $\Omega = \{0, 1\}^{\mathbb{N}}$ with $\mathcal{A} = \sigma(\{X_i = 1\}, i \in \mathbb{N})$ let P_p satisfy (i) $P_p(X_i = 1) = p$ and (ii) the events $\{X_i = 1\}$, $i \in \mathbb{N}$, are independent. By (ii) and Example 3.11, for $i_1 < \cdots < i_n$ and $x_j \in \{0, 1\}$

$$P_p(X_{i_1} = x_1, \dots, X_{i_n} = x_n) = \prod_j P_p(X_{i_j} = x_j) = p^k(1-p)^{n-k}, \qquad k = \textstyle\sum_j x_j.$$

These cylinder probabilities form a $\cap$-stable generator, so P_p is uniquely determined by (i) and (ii) (Theorem 1.9).

Theorem 3.14 (Kolmogorov's 0–1 law)

Let $\mathcal{B}_n$, $n \in \mathbb{N}$, be independent σ-algebras and $\mathcal{B}_\infty := \bigcap_{n=1}^\infty \sigma\big(\bigcup_{m\geq n} \mathcal{B}_m\big)$ the **tail σ-algebra**. Then $P(A) \in \{0, 1\}$ for all $A \in \mathcal{B}_\infty$.

Illustration. For $\mathcal{B}_i = \sigma(\{X_i = 1\})$, $\mathcal{B}_\infty \ni \limsup_i \{X_i = 1\}$, $\{\lim_n \frac{1}{n} \sum_{i\leq n} X_i \text{ exists}\}$.

Proof.

Step 1. By Theorem 3.10(ii), $\mathcal{B}_1, \dots, \mathcal{B}_{n-1}, \sigma(\bigcup_{m\geq n} \mathcal{B}_m) \supset \mathcal{B}_\infty$ are independent for all n, so $\mathcal{B}_\infty, \mathcal{B}_1, \mathcal{B}_2, \dots$ are.

Step 2. Again by Theorem 3.10(ii), $\mathcal{B}_\infty \subset \sigma(\bigcup_n \mathcal{B}_n)$ is independent of $\sigma(\bigcup_n \mathcal{B}_n)$, hence of itself; apply Lemma 3.15. ∎

Lemma 3.15

If a σ-algebra $\mathcal{B} \subset \mathcal{A}$ is independent of itself, then $P(A) \in \{0,1\}$ for all $A \in \mathcal{B}$.

Proof. $P(A) = P(A \cap A) = P(A)^2$. ■

For independent A_n, $P(\limsup_n A_n) \in \{0,1\}$ ($\mathcal{B}_n := \sigma(\{A_n\})$, Example 3.11); which value occurs:

Lemma 3.16 (Borel–Cantelli, part 2)

If $A_i \in \mathcal{A}$ are independent and $\sum_i P(A_i) = \infty$, then $P(\limsup_i A_i) = 1$.

Proof. Show $P(\bigcap_{m \geq n} A_m^c) = 0$: by continuity, independence of complements (Example 3.11) and $1 - \alpha \leq e^{-\alpha}$,

$$P\Big(\bigcap_{m \geq n} A_m^c\Big) = \lim_k P\Big(\bigcap_{m=n}^{n+k} A_m^c\Big) = \lim_k \prod_{m=n}^{n+k} (1 - P(A_m)) \leq \lim_k \exp\Big(-\sum_{m=n}^{n+k} P(A_m)\Big) = 0.$$

■

Example 3.17 (A text appears infinitely often)

Fix a word $(x_1, \ldots, x_N)$ with K ones, $p \in (0,1)$; $A_i =$ "the i-th length-N block is the word" are independent (Theorem 3.10(ii)) with probability $\alpha := p^K(1-p)^{N-K} > 0$. Borel–Cantelli 2: infinitely many occurrences a.s.; the strong law: $\frac{1}{n}\sum_{i \leq n} \mathbb{1}_{A_i} \to \alpha$ a.s.

3.4.2 Independent random variables

Definition 3.18 (Independent random variables)

(i) For a r.v. $X : (\Omega, \mathcal{A}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, $\sigma(X) := X^{-1}(\mathcal{B}(\mathbb{R})) = \{\{X \in A\} \mid A \in \mathcal{B}(\mathbb{R})\} \subset \mathcal{A}$ is the σ-algebra **generated by** X.

(ii) R.v.'s X_i, $i \in I$, are **independent** if the $\sigma(X_i)$ are independent, i.e. $P(\bigcap_{j \in J}\{X_j \in A_j\}) = \prod_{j \in J} P(X_j \in A_j)$ for finite J and Borel A_j.

(iii) X_i, $i \in I$, are **independent of** a sub-σ-algebra $\tilde{\mathcal{A}}$ if $\sigma(X_i \mid i \in I)$ (the smallest σ-algebra making all X_i measurable) is independent of $\tilde{\mathcal{A}}$.

Remark 3.19

If X_i, $i \in I$, are independent and $h_i : \mathbb{R} \to \mathbb{R}$ are Borel, then $Y_i := h_i(X_i)$ are independent, because $\sigma(Y_i) \subset \sigma(X_i)$.

Theorem 3.20 (Product rule)

If $X_1, \ldots, X_n \geq 0$ are independent, then $\mathbb{E}[X_1 \cdots X_n] = \mathbb{E}[X_1] \cdots \mathbb{E}[X_n]$.

Proof.

Step 1 (reduction to $n = 2$). $X_1 \cdots X_{n-1}$ is $\sigma(\sigma(X_1) \cup \cdots \cup \sigma(X_{n-1}))$-measurable, independent of $\sigma(X_n)$ (Theorem 3.10(ii)); induct. For independent $X, Y \geq 0$ show

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]. \tag{3.3}$$

Step 2 (simple functions). For $X_n = \sum_i \alpha_i \mathbb{1}_{A_i}$, $Y_n = \sum_j \beta_j \mathbb{1}_{B_j}$ with $\alpha_i, \beta_j \geq 0$, $A_i \in \sigma(X)$, $B_j \in \sigma(Y)$:

$$\mathbb{E}[X_n Y_n] = \sum_{i,j} \alpha_i \beta_j P(A_i \cap B_j) = \sum_{i,j} \alpha_i \beta_j P(A_i) P(B_j) = \mathbb{E}[X_n]\mathbb{E}[Y_n].$$

Step 3. Choose $\sigma(X)$- resp. $\sigma(Y)$-measurable simple $X_n \nearrow X$, $Y_n \nearrow Y$ (Facts 2.2(v)); then $X_n Y_n \nearrow XY$, and the MCT turns Step 2 into (3.3). ■

Corollary 3.21

X, Y independent and in $\mathcal{L}^1 \Rightarrow XY \in \mathcal{L}^1$ and $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

Proof. $\mathbb{E}[X^{\varepsilon_1} Y^{\varepsilon_2}] = \mathbb{E}[X^{\varepsilon_1}]\mathbb{E}[Y^{\varepsilon_2}] < \infty$ for $\varepsilon_i \in \{+, -\}$ (Remark 3.19); expand $XY = X^+Y^+ + X^-Y^- - X^+Y^- - X^-Y^+$. ■

Remark 3.22

The converse fails. For $X \sim N(0,1)$ and $Y = X^2$, X, Y are dependent but $\mathbb{E}[XY] = \mathbb{E}[X^3] = 0 = \mathbb{E}[X]\mathbb{E}[Y]$. Uncorrelatedness is also not preserved under functions: $\mathbb{E}[X^+Y^+] \neq \mathbb{E}[X^+]\mathbb{E}[Y^+]$. [Supplement] Indeed $\mathbb{E}[X^+Y^+] = \int_0^\infty x^3 \varphi(x)\, dx = \frac{2}{\sqrt{2\pi}}$ (since $\int_0^\infty x^3 e^{-x^2/2} dx = 2$), while $\mathbb{E}[X^+]\mathbb{E}[Y^+] = \frac{1}{\sqrt{2\pi}} \cdot 1$.
Independent $X, Y \in \mathcal{L}^2$ are uncorrelated.

Corollary 3.23 (Strong law for independent variables)

If $X_1, X_2, \cdots \in \mathcal{L}^2$ are independent with $\sup_i \operatorname{Var}(X_i) < \infty$, then $\frac{1}{n}\sum_{i \leq n}(X_i - \mathbb{E}[X_i]) \to 0$ a.s.; if $\mathbb{E}[X_i] \equiv m$, then $\frac{1}{n}\sum_{i \leq n} X_i \to m$ a.s.

Proof. Independent $\mathcal{L}^2$ variables are uncorrelated (Remark 3.22); apply Theorem 3.7. ■

3.5 Kolmogorov's law of large numbers

Motivation. For identically distributed, pairwise independent variables a first moment suffices: truncate, use geometric subsequences, interpolate.

(X_i) is **identically distributed** (i.d.) if $P \circ X_i^{-1} = P \circ X_1^{-1}$ for all i; i.i.d. means independent and identically distributed.

Theorem 3.24 (Kolmogorov 1930)

If $X_1, X_2, \cdots \in \mathcal{L}^1$ are i.i.d. with $m = \mathbb{E}[X_1]$, then $\frac{1}{n}\sum_{i=1}^n X_i \to m$ P-a.s.

Proof. Independent implies pairwise independent; apply Theorem 3.25. ■

Theorem 3.25 (Etemadi 1981)

If $X_1, X_2, \cdots \in \mathcal{L}^1$ are pairwise independent and identically distributed with $m = \mathbb{E}[X_1]$, then $\frac{1}{n}\sum_{i=1}^n X_i \to m$ P-a.s.

Proof. Assume $X_i \geq 0$ (else treat $X_i^{\pm}$).

Step 1 (truncation). $\tilde{X}_i := X_i \mathbb{1}_{\{X_i<i\}}$ are pairwise independent (Remark 3.19); it suffices that $\tilde{S}_n/n \to m$ a.s., $\tilde{S}_n := \sum_{i\leq n} \tilde{X}_i$, since

$$\begin{aligned}\sum_n P(X_n \neq \tilde{X}_n) &= \sum_n P(X_1 \geq n) = \sum_n \sum_{k\geq n} P(X_1 \in [k, k+1)) \\ &= \sum_k k\, P(X_1 \in [k, k+1)) \leq \mathbb{E}[X_1] < \infty,\end{aligned}$$

and by Borel–Cantelli $X_n = \tilde{X}_n$ eventually, a.s.

Step 2 (geometric subsequence). $\alpha > 1$, $k_n := \lfloor \alpha^n \rfloor$; assume (Step 3)

$$\frac{\tilde{S}_{k_n} - \mathbb{E}[\tilde{S}_{k_n}]}{k_n} \to 0 \quad \text{a.s.} \tag{3.4}$$

$\mathbb{E}[\tilde{X}_i] \nearrow m$, so by Cesàro $\tilde{S}_{k_n}/k_n \to m$ off a null N_α. For $k_n \leq l < k_{n+1}$, by monotonicity

$$\frac{k_n}{k_{n+1}} \cdot \frac{\tilde{S}_{k_n}}{k_n} \leq \frac{\tilde{S}_l}{l} \leq \frac{\tilde{S}_{k_{n+1}}}{k_{n+1}} \cdot \frac{k_{n+1}}{k_n},$$

and $k_{n+1}/k_n \to \alpha$, so $\frac{m}{\alpha} \leq \liminf_l \frac{\tilde{S}_l}{l} \leq \limsup_l \frac{\tilde{S}_l}{l} \leq \alpha m$ off N_α; let $\alpha_j \searrow 1$.

Step 3 (proof of (3.4)). By Lemma 2.10 and Chebyshev (pairwise uncorrelated), it is enough that

$$s := \sum_n \frac{1}{k_n^2} \sum_{i\leq k_n} \mathbb{E}[\tilde{X}_i^2] = \sum_i \Big(\sum_{n:\, k_n \geq i} \frac{1}{k_n^2} \Big) \mathbb{E}[\tilde{X}_i^2] < \infty.$$

Step 4 (a geometric bound). For some c,

$$\sum_{n:\, k_n \geq i} \frac{1}{k_n^2} \leq \frac{c}{i^2}. \tag{3.5}$$

since $k_n > \alpha^n - 1 \geq c_\alpha \alpha^n$, $c_\alpha = \frac{\alpha-1}{\alpha}$, and the least n_i with $k_n \geq i$ has $\alpha^{n_i} \geq i$:

$$\sum_{n\geq n_i} \frac{1}{k_n^2} \leq c_\alpha^{-2} \sum_{n\geq n_i} \alpha^{-2n} = \frac{c_\alpha^{-2} \alpha^{-2n_i}}{1-\alpha^{-2}} \leq \frac{c_\alpha^{-2}}{1-\alpha^{-2}} \cdot \frac{1}{i^2}.$$

Step 5. With (3.5), $\mathbb{E}[\tilde{X}_i^2] = \mathbb{E}[X_1^2 \mathbb{1}_{\{X_1<i\}}] \leq \sum_{l\leq i} l^2 P(X_1 \in [l-1, l))$ and $\sum_{i\geq l} \frac{1}{i^2} \leq \frac{1}{l^2} + \sum_{i>l} \big(\frac{1}{i-1} - \frac{1}{i}\big) \leq \frac{2}{l}$:

$$\begin{aligned}s &\leq c \sum_i \frac{1}{i^2} \sum_{l\leq i} l^2 P(X_1 \in [l-1, l)) = c \sum_l l^2 \Big(\sum_{i\geq l} \frac{1}{i^2} \Big) P(X_1 \in [l-1, l)) \\ &\leq 2c \sum_l \mathbb{E}\big[l \mathbb{1}_{\{X_1 \in [l-1, l)\}}\big] \leq 2c(\mathbb{E}[X_1] + 1).\end{aligned}$$

■

Corollary 3.26

If $X_1, X_2, \cdots \geq 0$ are pairwise independent and identically distributed, then $\frac{1}{n} \sum_{i\leq n} X_i \to \mathbb{E}[X_1] \in [0, \infty]$ a.s.

Proof. If $\mathbb{E}[X_1] < \infty$ this is Theorem 3.25. If $\mathbb{E}[X_1] = \infty$, apply Theorem 3.25 to $X_i \wedge N$:

$$\frac{1}{n} \sum_{i\leq n} X_i \geq \frac{1}{n} \sum_{i\leq n} (X_i \wedge N) \to \mathbb{E}[X_1 \wedge N] \quad \text{a.s.},$$

and $\mathbb{E}[X_1 \wedge N] \nearrow \infty$ (MCT). ■

3.6 Finite products of probability measures

Motivation. The product measure, built by integrating sections, comes with Fubini's theorem.

For probability spaces $(S_1, \mathcal{S}_1, \mu_1)$, $(S_2, \mathcal{S}_2, \mu_2)$, $\Omega := S_1 \times S_2$, $X_i(x_1, x_2) = x_i$, and

$$\mathcal{A} := \mathcal{S}_1 \otimes \mathcal{S}_2 := \sigma(X_1, X_2) = \sigma(\{A_1 \times A_2 \mid A_i \in \mathcal{S}_i\}).$$

we seek the **product measure** $P =: \mu_1 \otimes \mu_2$ with

$$P(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2) \qquad (A_i \in \mathcal{S}_i). \tag{3.6}$$

unique by Theorem 1.9 (rectangles are a $\cap$-stable generator).

Theorem 3.27 (Fubini for probability measures)

(i) For every $\mathcal{A}$-measurable $f : \Omega \to \overline{\mathbb{R}}$ and $x_1 \in S_1$, $x_2 \in S_2$, the sections $f_{x_1} := f(x_1, \cdot)$ and $f_{x_2} := f(\cdot, x_2)$ are measurable. If moreover $f \geq 0$, then $x_1 \mapsto \int_{S_2} f_{x_1}\, d\mu_2$ and $x_2 \mapsto \int_{S_1} f_{x_2}\, d\mu_1$ are $\mathcal{S}_1$- resp. $\mathcal{S}_2$-measurable.

(ii) There is a unique probability measure $P = \mu_1 \otimes \mu_2$ on $(\Omega, \mathcal{A})$ such that for all $\mathcal{A}$-measurable $f \geq 0$

$$\int_\Omega f\, dP = \int_{S_1} \Big(\int_{S_2} f(x_1, x_2)\, \mu_2(dx_2) \Big) \mu_1(dx_1), \tag{3.7}$$

and the order may be interchanged:

$$\int_\Omega f\, dP = \int_{S_2} \Big(\int_{S_1} f(x_1, x_2)\, \mu_1(dx_1) \Big) \mu_2(dx_2). \tag{3.8}$$

In particular, with the sections $A_{x_1} := \{x_2 \mid (x_1, x_2) \in A\}$, $A_{x_2} := \{x_1 \mid (x_1, x_2) \in A\}$,

$$P(A) = \int_{S_1} \mu_2(A_{x_1})\, \mu_1(dx_1) = \int_{S_2} \mu_1(A_{x_2})\, \mu_2(dx_2) \qquad (A \in \mathcal{A}). \tag{3.9}$$

Proof.

Step 1 (sections are measurable). $\phi_{x_1}(x_2) := (x_1, x_2)$ is measurable, because $\phi_{x_1}^{-1}(A_1 \times A_2)$ is A_2 or $\emptyset$ and rectangles generate $\mathcal{A}$ (Remark 1.32). So $f_{x_1} = f \circ \phi_{x_1}$ is measurable; likewise f_{x_2}.

Step 2 (indicators). For $f \geq 0$, the map

$$x_1 \mapsto \int_{S_2} f(x_1, x_2)\, \mu_2(dx_2) \in [0, \infty] \tag{3.10}$$

is defined; for $f = \mathbb{1}_A$ it is $\mu_2(A_{x_1})$. $\mathcal{D} := \{A \in \mathcal{A} \mid x_1 \mapsto \mu_2(A_{x_1})$ measurable$\}$ is a Dynkin system ($\Omega_{x_1} = S_2$; complements $1 - \mu_2(A_{x_1})$; disjoint unions give series) containing rectangles $(\mathbb{1}_{A_1}(x_1)\mu_2(A_2))$, so $\mathcal{D} = \mathcal{A}$ (Theorem 1.8).

Step 3 (measure-theoretic induction). Linearity for simple $f \geq 0$, then simple $f_n \nearrow f$ (Facts 2.2(v)) and the MCT; likewise in the other order.

Step 4 (existence). $P(A) := \int_{S_1} \mu_2(A_{x_1})\, \mu_1(dx_1)$ has $P(\Omega) = 1$, is σ-additive (Theorem 1.56) and satisfies (3.6). (3.7) holds for indicators, hence for $f \geq 0$ (linearity, MCT).

Step 5 (interchanging the order). $Q(A) := \int_{S_2} \mu_1(A_{x_2})\, \mu_2(dx_2)$ agrees with P on rectangles, so $Q = P$ (Theorem 1.9); extend (3.8) as before; (3.9) is $f = \mathbb{1}_A$. ∎

Remark 3.28

If $f \in \mathcal{L}^1(\mu_1 \otimes \mu_2)$, Theorem 3.27 applies to f^+ and f^-, hence by linearity to f.

Definition 3.29 (Finite products of measurable spaces)

For measurable spaces $(S_i, \mathcal{S}_i)$, $1 \le i \le n$, $\Omega := S_1 \times \cdots \times S_n$ carries the **product σ-algebra**

$$\mathcal{A} := \mathcal{S}_1 \otimes \cdots \otimes \mathcal{S}_n = \bigotimes_{i=1}^{n} \mathcal{S}_i := \sigma(\{A_1 \times \cdots \times A_n \mid A_i \in \mathcal{S}_i\}).$$

Equivalently $\mathcal{A} = \sigma(X_1, \dots, X_n)$ with the coordinate projections X_i. [Supplement] *Reason:* $A_1 \times \cdots \times A_n = \bigcap_i X_i^{-1}(A_i)$, and $X_i^{-1}(A_i)$ is a rectangle.

Theorem 3.30 (Finite products of probability measures)

For probability spaces $(S_i, \mathcal{S}_i, \mu_i)$, $1 \le i \le n$, there is a unique probability measure $P = \mu_1 \otimes \cdots \otimes \mu_n = \bigotimes_{i=1}^n \mu_i$ on $(S_1 \times \cdots \times S_n, \mathcal{S}_1 \otimes \cdots \otimes \mathcal{S}_n)$ with $P(A_1 \times \cdots \times A_n) = \prod_i \mu_i(A_i)$.

Proof. Uniqueness: Theorem 1.9. Existence by induction: $P := P_{n-1} \otimes \mu_n$ (Theorem 3.27), as $\mathcal{A}_{n-1} \otimes \mathcal{S}_n = \mathcal{S}_1 \otimes \cdots \otimes \mathcal{S}_n$ (same generating rectangles), and $P(A_1 \times \cdots \times A_n) = P_{n-1}(A_1 \times \cdots \times A_{n-1})\mu_n(A_n) = \prod_i \mu_i(A_i)$. ∎

Theorem 3.31 (Fubini for finite products)

With $P = \mu_1 \otimes \cdots \otimes \mu_n$ and $\mathcal{S}_1 \otimes \cdots \otimes \mathcal{S}_n$-measurable $f \ge 0$,

$$\int f \, dP = \int_{S_1} \cdots \int_{S_n} f(x_1, \dots, x_n) \, \mu_n(dx_n) \cdots \mu_1(dx_1), \tag{3.11}$$

and for every permutation π of $\{1, \dots, n\}$

$$\int f \, dP = \int_{S_{\pi(1)}} \cdots \int_{S_{\pi(n)}} f(x_1, \dots, x_n) \, \mu_{\pi(n)}(dx_{\pi(n)}) \cdots \mu_{\pi(1)}(dx_{\pi(1)}). \tag{3.12}$$

Proof.

Step 1 ((3.11)). Induction: $\int f \, dP = \int_{\Omega_{n-1}} \left(\int_{S_n} f \, d\mu_n \right) dP_{n-1}$ (Theorem 3.27), and the inner integral is $\mathcal{A}_{n-1}$-measurable and ≥ 0.

Step 2 ((3.12)) [Supplement]. Regrouping $\Omega = \left(\prod_{i \ne \pi(n)} S_i \right) \times S_{\pi(n)}$ keeps the product σ-algebra, so the iterated integrals are defined (Theorem 3.27(i), induction). For $f = \mathbb{1}_A$ the right side is a probability measure in A agreeing with P on rectangles, hence $= P$ (Theorem 1.9); extend by linearity and the MCT. ∎

Remark 3.32

If $f \in \mathcal{L}^1(\mu_1 \otimes \cdots \otimes \mu_n)$, Theorem 3.31 applies to $f^\pm$ and hence to f.

3.7 Weak convergence of probability measures

Motivation. Distributions converge when integrals of bounded continuous functions do ($\delta_{1/n} \not\to \delta_0$ setwise).

Definition 3.33 (Weak convergence)

Let Ω be a metric space with Borel σ-algebra $\mathcal{B}(\Omega)$, and μ, μ_n probability measures on it. $\mu_n \to \mu$ **weakly** if $\int f\,d\mu_n \to \int f\,d\mu$ for all $f \in C_b(\Omega)$ (bounded continuous functions).

Example 3.34

(i) $x_n \to x$ implies $\delta_{x_n} \to \delta_x$ weakly, since $f(x_n) \to f(x)$ for continuous f. (ii) $\mu_n := N(0, \frac{1}{n}) \to \delta_0$ weakly: substituting $x = y/\sqrt{n}$ and using the DCT,

$$\int f\,d\mu_n = \int f(x)\frac{e^{-nx^2/2}}{\sqrt{2\pi/n}}\,dx = \int f\Big(\frac{y}{\sqrt{n}}\Big)\frac{e^{-y^2/2}}{\sqrt{2\pi}}\,dy \to f(0) = \int f\,d\delta_0.$$

(Also for finite measures.)

Theorem 3.35 (Portmanteau theorem)

For probability measures μ, μ_n on a metric space (Ω, d) the following are equivalent:

(i) $\mu_n \to \mu$ weakly;

(ii) $\int f\,d\mu_n \to \int f\,d\mu$ for all bounded uniformly continuous f;

(iii) $\limsup_n \mu_n(F) \le \mu(F)$ for all closed F;

(iv) $\liminf_n \mu_n(G) \ge \mu(G)$ for all open G;

(v) $\mu_n(A) \to \mu(A)$ for all μ-continuity sets A, i.e. $A \in \mathcal{B}(\Omega)$ with $\mu(\partial A) = 0$;

(vi) $\int f\,d\mu_n \to \int f\,d\mu$ for all bounded measurable f that are continuous μ-a.e.

Proof.

Step Easy implications. (i)⇒(ii) and (vi)⇒(i) are special cases; (iii)⇔(iv) by taking complements.

Step (ii)⇒(iii). Open $G_m := \{d(\cdot, F) < \frac{1}{m}\} \searrow F$, so $\mu(G_m) < \mu(F) + \varepsilon$ for some m. With $\phi = 1$ on $(-\infty, 0]$, $1 - x$ on $[0, 1]$, 0 on $[1, \infty)$, $f := \phi(m\,d(\cdot, F))$ is uniformly continuous, $\mathbb{1}_F \le f \le \mathbb{1}_{G_m}$, so $\limsup_n \mu_n(F) \le \lim_n \int f\,d\mu_n = \int f\,d\mu \le \mu(G_m) < \mu(F) + \varepsilon$.

Step (iii)⇒(v). With interior $\mathring{A}$ and closure $\bar{A}$, $\mu(\partial A) = 0$ gives $\mu(\mathring{A}) = \mu(A) = \mu(\bar{A})$, and

$$\mu(A) = \mu(\mathring{A}) \overset{\text{(iv)}}{\le} \liminf_n \mu_n(\mathring{A}) \le \liminf_n \mu_n(A) \le \limsup_n \mu_n(A) \le \limsup_n \mu_n(\bar{A}) \overset{\text{(iii)}}{\le} \mu(\bar{A}) = \mu(A).$$

Step (v)⇒(vi). $D := \{x \mid \mu(f = x) \ne 0\}$ is countable (Remark 2.16); pick $c_0 < \cdots < c_m$ off D with $c_{k+1} - c_k \le \varepsilon$, $c_0 \le f < c_m$. $A_k := f^{-1}([c_k, c_{k+1}))$ are μ-continuity sets, since (below)

$$\partial A_k \subset \{f = c_k\} \cup \{f = c_{k+1}\} \cup D_f, \qquad D_f := \{\omega \mid f \text{ is discontinuous at } \omega\}, \tag{3.13}$$

and all three sets are μ-null. With $g := \sum_k c_k \mathbb{1}_{A_k}$, $\|f - g\|_\infty \le \varepsilon$ and

$$\begin{aligned}\Big|\int f\,d\mu - \int f\,d\mu_n\Big| &\le \int |f - g|\,d\mu + \Big|\int g\,d\mu - \int g\,d\mu_n\Big| + \int |g - f|\,d\mu_n \\ &\le 2\varepsilon + \sum_k |c_k||\mu(A_k) - \mu_n(A_k)| \to 2\varepsilon.\end{aligned}$$

Step Proof of (3.13). For $\omega \in \partial A_k \setminus D_f$ and $\varepsilon' > 0$, some $B(\omega, \delta)$ maps into $B(f(\omega), \varepsilon')$ and meets A_k and A_k^c, so $B(f(\omega), \varepsilon')$ meets $[c_k, c_{k+1})$ and its complement: $f(\omega) \in \{c_k, c_{k+1}\}$.

[Supplement] $D_f = \bigcup_k \{\mathrm{osc}_f \ge \frac{1}{k}\}$, $\mathrm{osc}_f(\omega) := \lim_{\delta \to 0} \sup\{|f(x) - f(y)| \mid x, y \in B(\omega, \delta)\}$, is measurable: each set is closed (a small ball with oscillation $< \frac{1}{k}$ serves nearby points). ■

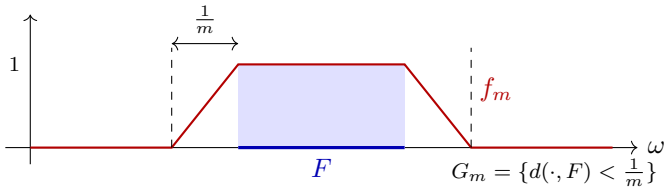


Figure. *The test function of (ii)⇒(iii):* $\mathbb{1}_F \le f_m \le \mathbb{1}_{G_m}$.

Definition 3.36 (Support, C_0)

For continuous f on a metric space Ω, $\operatorname{supp} f := \overline{\{f \neq 0\}}$, and $C_0(\Omega) := \{f \in C_b(\Omega) \mid \operatorname{supp} f \text{ compact}\}$.

Corollary 3.37

For probability measures μ, μ_n on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ with distribution functions F, F_n, the following are equivalent:

(i) $\mu_n \to \mu$ **vaguely**: $\int f\, d\mu_n \to \int f\, d\mu$ for all $f \in C_0(\mathbb{R})$;

(ii) $\mu_n \to \mu$ weakly;

(iii) $F_n(x) \to F(x)$ at every continuity point x of F;

(iv) $\mu_n((a,b]) \to \mu((a,b])$ whenever $\mu(\{a\}) = \mu(\{b\}) = 0$.

Proof.

Step (i)⇒(ii) [Supplement]. For $f \in C_b(\mathbb{R})$ take $\mu([-R,R]) > 1-\varepsilon$ and $g \in C_0(\mathbb{R})$, $0 \le g \le 1$, $g = 1$ on $[-R,R]$; then $\int (1-g)\, d\mu_n < \varepsilon$ eventually, and with $fg \in C_0(\mathbb{R})$

$$\Big|\int f\, d\mu_n - \int f\, d\mu\Big| \le \Big|\int fg\, d\mu_n - \int fg\, d\mu\Big| + \|f\|_\infty \Big(\int (1-g)\, d\mu_n + \int (1-g)\, d\mu\Big),$$

whose lim sup is at most $2\|f\|_\infty \varepsilon$.

Step (ii)⇒(iii). If F is continuous at x, then $\mu(\{x\}) = 0 = \mu(\partial(-\infty, x])$, and Portmanteau (v) gives $F_n(x) = \mu_n((-\infty, x]) \to F(x)$.

Step (iii)⇒(iv). F is continuous at a and b, so $\mu((a,b]) = F(b) - F(a) = \lim_n (F_n(b) - F_n(a))$.

Step (iv)⇒(i). $D := \{x \mid \mu(\{x\}) \neq 0\}$ is countable. For $f \in C_0(\mathbb{R})$ (uniformly continuous, δ for ε) pick $c_0 < \cdots < c_m$ off D, $\operatorname{supp} f \subset (c_0, c_m]$, $c_k - c_{k-1} \le \delta$; $g := \sum_{k=1}^m f(c_{k-1}) \mathbb{1}_{(c_{k-1}, c_k]}$ has $\|f - g\|_\infty \le \varepsilon$ and

$$\Big|\int f\, d\mu - \int f\, d\mu_n\Big| \le 2\varepsilon + \sum_k |f(c_{k-1})| |\mu((c_{k-1}, c_k]) - \mu_n((c_{k-1}, c_k])| \to 2\varepsilon.$$

■

3.8 Joint distribution and convolution

Motivation. $S_n/n \to \delta_{\mathbb{E}[X_1]}$ weakly; the interesting scale $S_n/\sqrt{n}$ needs laws of sums, i.e. convolutions.

Definition 3.38 (Joint distribution)

For real r.v.'s $X_1, \dots, X_n$ on $(\Omega, \mathcal{A}, P)$, the law $P_{\bar{X}} := P \circ \bar{X}^{-1}$ of $\bar{X} := (X_1, \dots, X_n) : \Omega \to \mathbb{R}^n$ is the **joint distribution** of $X_1, \dots, X_n$.

Remark 3.39

(i) $P_{\bar{X}}(A) = P(\bar{X} \in A)$, $A \in \mathcal{B}(\mathbb{R}^n)$, is a probability measure (Proposition 2.13), because $\bar{X}$ is $\mathcal{A}/\mathcal{B}(\mathbb{R}^n)$-measurable. Indeed

$$\mathcal{B}(\mathbb{R}^n) = \bigotimes_{i=1}^{n} \mathcal{B}(\mathbb{R}) = \sigma(\{A_1 \times \cdots \times A_n \mid A_i = (-\infty, x_i]\}),$$

and $\bar{X}^{-1}(A_1 \times \cdots \times A_n) = \bigcap_i \{X_i \in A_i\} \in \mathcal{A}$ (Remark 1.32). [Supplement] *The equalities:* open sets are countable unions of open boxes (Remark 1.27(i)), and half-open boxes are finite differences of orthants.

(ii) By Theorem 1.9, $P_{\bar{X}}$ is determined by $P_{\bar{X}}(A_1 \times \cdots \times A_n) = P(X_1 \in A_1, \dots, X_n \in A_n)$, even with $A_i = (-\infty, x_i]$.

Remark 3.40 (Transformation theorem)

For Borel $\Phi : \mathbb{R}^n \to \mathbb{R}$ with $\Phi(\bar{X}) \in \mathcal{L}^1$ or $\Phi \geq 0$: $\mathbb{E}[\Phi(\bar{X})] = \int_\Omega \Phi(\bar{X})\, dP = \int_{\mathbb{R}^n} \Phi\, dP_{\bar{X}}$ (same proof as in Remark 2.15).

Example 3.41

(i) X, Y uniform on $[0,1]$: independent gives uniform on $[0,1]^2$ (Theorem 3.42); $X = Y$ gives the image of $\lambda|_{[0,1]}$ under $x \mapsto (x, x)$. (ii) Independent $X, Y \sim N(m, \sigma^2)$: joint density $f(x,y) = \frac{1}{2\pi\sigma^2} \exp\big(-\frac{(x-m)^2+(y-m)^2}{2\sigma^2}\big)$.

Theorem 3.42

Let $X_1, \dots, X_n$ have laws $\mu_1, \dots, \mu_n$ and joint law $P_{\bar{X}}$. Then $X_1, \dots, X_n$ are independent $\iff$ $P_{\bar{X}} = \bigotimes_i \mu_i$, i.e. $P_{\bar{X}}(A_1 \times \cdots \times A_n) = \prod_i \mu_i(A_i)$. In this case:

(i) $P_{\bar{X}}$ is determined by $\mu_1, \dots, \mu_n$;

(ii) for every permutation π and $\phi \geq 0$ (or $P_{\bar{X}}$-integrable), $\int \phi\, dP_{\bar{X}}$ equals the iterated integral in the order π;

(iii) if $\mu_i(dx_i) = f_i(x_i)\, dx_i$, then $P_{\bar{X}}(d\bar{x}) = \bar{f}(\bar{x})\, d\bar{x}$ with $\bar{f}(\bar{x}) = \prod_i f_i(x_i)$.

Proof. The equivalence is Definition 3.18 rewritten, since $\mu_i(A_i) = P(X_i \in A_i)$.

(i) Theorem 3.30.

(ii) Theorem 3.31 and Remark 3.32.

(iii) $\int \bar{f}\, d\bar{x} = \prod_i \int f_i = 1$ (Fubini; Lebesgue measure on $\mathbb{R}^n$ is the product measure by Theorem 1.9), and $\bar{\mu}(A) := \int_A \bar{f}\, d\bar{x}$ agrees with $P_{\bar{X}}$ on rectangles, so $\bar{\mu} = P_{\bar{X}}$ (Theorem 1.9). ■

Law of S_n for independent summands, with $T_x(y) := x + y$:

Theorem 3.43 (Convolution)

Let X_1, X_2 be independent with laws μ_1, μ_2.

(i) $P_{X_1+X_2} = \mu_1 * \mu_2 := \int \mu_2 \circ T_{x_1}^{-1}(\cdot)\, \mu_1(dx_1)$, i.e. for $A \in \mathcal{B}(\mathbb{R})$

$$\mu_1 * \mu_2(A) = \iint \mathbb{1}_A(x_1 + x_2)\, \mu_1(dx_1)\, \mu_2(dx_2) = \int \mu_2(A - x_1)\, \mu_1(dx_1),$$

where $A - x_1 := \{x - x_1 \mid x \in A\}$.

(ii) If μ_2 has density f_2, then $\mu_1 * \mu_2$ has density $f(x) = \int f_2(x - x_1)\, \mu_1(dx_1)$, which equals $(f_1 * f_2)(x) := \int f_1(x_1) f_2(x - x_1)\, dx_1$ if $\mu_1(dx_1) = f_1(x_1)\, dx_1$.

Proof. (i) With $\bar{A} := \{(x_1, x_2) \mid x_1 + x_2 \in A\} \in \mathcal{B}(\mathbb{R}^2)$ and Theorem 3.42,

$$P(X_1 + X_2 \in A) = (\mu_1 \otimes \mu_2)(\bar{A}) = \int \Big(\int \mathbb{1}_{A - x_1}(x_2)\, \mu_2(dx_2) \Big) \mu_1(dx_1) = \int \mu_2(A - x_1)\, \mu_1(dx_1).$$

(ii) Substituting $x = x_1 + x_2$ and using Fubini (Theorem 3.42(ii)):

$$\begin{aligned}(\mu_1 * \mu_2)(A) &= \int \int_{A - x_1} f_2(x_2)\, dx_2\, \mu_1(dx_1) = \int \int_A f_2(x - x_1)\, dx\, \mu_1(dx_1) \\ &= \int_A \Big(\int f_2(x - x_1)\, \mu_1(dx_1) \Big) dx.\end{aligned}$$

■

Example 3.44

(i) If $X_i \sim N(m_i, \sigma_i^2)$ are independent, then $X_1 + X_2 \sim N(m_1 + m_2, \sigma_1^2 + \sigma_2^2)$, i.e. $f_{m_1,\sigma_1^2} * f_{m_2,\sigma_2^2} = f_{m_1+m_2,\sigma_1^2+\sigma_2^2}$. [Supplement] After shifting, let $m_i = 0$, $a = \sigma_1^2$, $b = \sigma_2^2$. Completing the square in y,

$$\int e^{-\frac{y^2}{2a} - \frac{(x-y)^2}{2b}}\, dy = e^{-\frac{x^2}{2(a+b)}} \int e^{-\frac{a+b}{2ab}(y - \frac{ax}{a+b})^2}\, dy = e^{-\frac{x^2}{2(a+b)}} \sqrt{\frac{2\pi ab}{a+b}},$$

and multiplying by $\frac{1}{2\pi\sqrt{ab}}$ gives $f_{0,a+b}(x)$.

(ii) The **Gamma distribution** $\Gamma_{\alpha,p}$ has density $\gamma_{\alpha,p}(x) = \frac{\alpha^p}{\Gamma(p)} x^{p-1} e^{-\alpha x}$ for $x > 0$ (0 otherwise). If $X_i \sim \Gamma_{\alpha,p_i}$ are independent, then $X_1 + X_2 \sim \Gamma_{\alpha,p_1+p_2}$. [Supplement] For $x > 0$, substituting $y = xt$,

$$\begin{aligned}(\gamma_{\alpha,p_1} * \gamma_{\alpha,p_2})(x) &= \frac{\alpha^{p_1+p_2} e^{-\alpha x}}{\Gamma(p_1)\Gamma(p_2)} \int_0^x y^{p_1-1}(x - y)^{p_2-1} dy \\ &= \frac{\alpha^{p_1+p_2} e^{-\alpha x} x^{p_1+p_2-1}}{\Gamma(p_1)\Gamma(p_2)} B(p_1, p_2) = \gamma_{\alpha,p_1+p_2}(x),\end{aligned}$$

using $B(p_1, p_2) = \Gamma(p_1)\Gamma(p_2)/\Gamma(p_1 + p_2)$. For $p_i = 1$: a sum $S_n = T_1 + \cdots + T_n$ of independent $\mathrm{Exp}(\alpha)$ variables has the Gamma density $\gamma_{\alpha,n}(x) = \frac{\alpha^n}{(n-1)!} x^{n-1} e^{-\alpha x}$, $x > 0$.

Example 3.45 (The waiting time paradox)

Independent $\mathrm{Exp}(\alpha)$ gaps T_i ($\mathbb{E}[T_i] = \frac{1}{\alpha}$, e.g. buses), $S_n := T_1 + \cdots + T_n$, $S_0 := 0$; for fixed $t > 0$, $X = t - S_n$, $Y = S_{n+1} - t$ on $\{S_n \le t < S_{n+1}\}$. Not $\mathbb{E}[Y] = \frac{1}{2\alpha}$, but:

(i) $Y \sim \mathrm{Exp}(\alpha)$, so $\mathbb{E}[Y] = \frac{1}{\alpha}$;

(ii) X is Exp(α) "compressed to $[0,t]$": $P(X \geq s) = e^{-\alpha s}$ for $0 \leq s \leq t$ and $P(X = t) = e^{-\alpha t}$, so $\mathbb{E}[X] = \int_0^t s\alpha e^{-\alpha s} ds + te^{-\alpha t} = \frac{1}{\alpha}(1 - e^{-\alpha t})$;

(iii) X and Y are independent.

So $[t - X, t + Y]$ has mean length $\frac{1}{\alpha}(2 - e^{-\alpha t}) \approx \frac{2}{\alpha}$: t tends to fall in long gaps.

Proof. Split by the number n of events in $[0,t]$ (independent $S_n \sim \gamma_{\alpha,n}$, T_{n+1}); for $0 \leq x \leq t$, $y \geq 0$:

$$\begin{aligned} P(X \geq x, Y \geq y) &= P(T_1 \geq t + y) + \sum_{n \geq 1} P(S_n \leq t - x,\ T_{n+1} \geq t + y - S_n) \\ &= e^{-\alpha(t+y)} + \sum_{n \geq 1} \int_0^{t-x} \gamma_{\alpha,n}(s) e^{-\alpha(t+y-s)}\, ds \\ &= e^{-\alpha(t+y)} \Big(1 + \int_0^{t-x} e^{\alpha s} \sum_{n \geq 1} \gamma_{\alpha,n}(s)\, ds\Big). \end{aligned}$$

Since $\sum_{n \geq 1} \gamma_{\alpha,n}(s) = \alpha e^{-\alpha s} \sum_{n \geq 1} \frac{(\alpha s)^{n-1}}{(n-1)!} = \alpha$, the bracket is $1 + (e^{\alpha(t-x)} - 1) = e^{\alpha(t-x)}$, and $P(X \geq x, Y \geq y) = e^{-\alpha y} e^{-\alpha x}$. Setting $x = 0$ gives (i), $y = 0$ gives (ii), and the product form gives (iii).

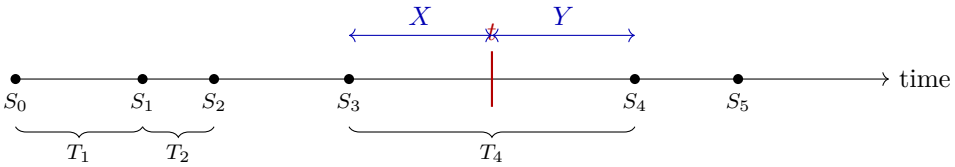


Figure. *The gap containing t (here T_4) is size-biased.*

3.9 Characteristic functions

Motivation. Fourier transforms of laws turn convolutions into products and weak convergence into pointwise convergence.

$\mathcal{M}_+^1(\mathbb{R}^n)$: probability measures on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$. The **characteristic function** of μ is

$$\hat{\mu}(u) := \int e^{i\langle u,y\rangle} \mu(dy) := \int \cos\langle u, y\rangle\, \mu(dy) + i \int \sin\langle u, y\rangle\, \mu(dy), \qquad u \in \mathbb{R}^n.$$

Theorem 3.46

For $\mu \in \mathcal{M}_+^1(\mathbb{R}^n)$: (i) $|\hat{\mu}(u)| \leq \hat{\mu}(0) = 1$; (ii) $\hat{\mu}(-u) = \overline{\hat{\mu}(u)}$; (iii) $\hat{\mu}$ is uniformly continuous; (iv) $\hat{\mu}$ is positive definite: $\sum_{j,k=1}^m c_j \bar{c}_k \hat{\mu}(u_j - u_k) \geq 0$ for $c_j \in \mathbb{C}$, $u_j \in \mathbb{R}^n$.

Proof [Supplement]. (i) $|\hat{\mu}(u)| \leq \int |e^{i\langle u,y\rangle}|\, d\mu = 1 = \hat{\mu}(0)$.

(ii) Conjugate the integrand.

(iii) $|\hat{\mu}(u + h) - \hat{\mu}(u)| \leq \int |e^{i\langle h,y\rangle} - 1|\, \mu(dy) \to 0$ (DCT), uniformly in u.

(iv) $\sum_{j,k} c_j \bar{c}_k \hat{\mu}(u_j - u_k) = \int \big|\sum_j c_j e^{i\langle u_j,y\rangle}\big|^2 \mu(dy) \geq 0$. ∎

Theorem 3.47 (Uniqueness theorem)

If $\mu_1, \mu_2 \in \mathcal{M}_+^1(\mathbb{R}^n)$ and $\hat{\mu}_1 = \hat{\mu}_2$, then $\mu_1 = \mu_2$.

Proof [Supplement] (Gaussian smoothing).

Step 1 (a Fourier formula for the Gaussian). For $p_\sigma(x) := (2\pi\sigma^2)^{-n/2} e^{-\|x\|^2/(2\sigma^2)}$, Theorem 3.52 (independent of this section) with $y = \sigma u$, $u \to x/\sigma$ gives

$$p_\sigma(x) = (2\pi)^{-n} \int e^{i\langle x,u\rangle} e^{-\sigma^2\|u\|^2/2}\, du.$$

Step 2 (smoothed measure is determined by $\hat{\mu}$). $h_\sigma(x) := \int p_\sigma(x-y)\,\mu(dy)$; by Step 1 and Fubini (bound $e^{-\sigma^2\|u\|^2/2}$),

$$h_\sigma(x) = (2\pi)^{-n} \int e^{i\langle x,u\rangle} e^{-\sigma^2\|u\|^2/2} \Big(\int e^{-i\langle y,u\rangle} \mu(dy) \Big) du = (2\pi)^{-n} \int e^{i\langle x,u\rangle} e^{-\sigma^2\|u\|^2/2} \hat{\mu}(-u)\, du,$$

which depends on μ only through $\hat{\mu}$.

Step 3 (letting $\sigma \to 0$). For $f \in C_b(\mathbb{R}^n)$, with γ the standard Gaussian on $\mathbb{R}^n$ and the substitution $x = y + \sigma z$,

$$\int f(x) h_\sigma(x)\, dx = \iint f(y+\sigma z)\, \gamma(dz)\, \mu(dy) \xrightarrow[\sigma\to 0]{} \int f\, d\mu$$

(DCT); the left side is the same for μ_1, μ_2 (Step 2), so $\mu_1 = \mu_2$ (Example 2.30). ∎

Theorem 3.48 (Bochner)

If $\phi : \mathbb{R}^n \to \mathbb{C}$ is continuous, positive definite and $\phi(0) = 1$, there is exactly one $\mu \in \mathcal{M}_+^1(\mathbb{R}^n)$ with $\hat{\mu} = \phi$.

On the proof

Uniqueness is Theorem 3.47. Existence is not proved in the lectures; see Klenke, *Probability Theory*, Thm. 15.29. It is not used below.

Theorem 3.49 (Lévy's continuity theorem)

Let $\mu_m \in \mathcal{M}_+^1(\mathbb{R}^n)$.

(i) If $\mu_m \to \mu$ weakly, then $\hat{\mu}_m \to \hat{\mu}$ uniformly on compact sets.

(ii) If $\hat{\mu}_m \to \phi$ pointwise for some $\phi : \mathbb{R}^n \to \mathbb{C}$ continuous at 0, then there is a unique $\mu \in \mathcal{M}_+^1(\mathbb{R}^n)$ with $\hat{\mu} = \phi$, and $\mu_m \to \mu$ weakly.

Proof [Supplement] (part (ii) for $n = 1$, the case used in the CLT).

Step (i) Tightness. $\mu(\{\|y\| \ge R\}) < \varepsilon$ and Portmanteau (iii) give $\mu_m(\{\|y\| \ge R\}) < \varepsilon$ for $m \ge m_0$; enlarge R for $m < m_0$.

Step (i) Equicontinuity. $|\hat{\mu}_m(u) - \hat{\mu}_m(v)| \le \int_{\{\|y\|<R\}} |e^{i\langle u-v,y\rangle} - 1|\, d\mu_m + 2\varepsilon \le R\|u-v\| + 2\varepsilon$; with pointwise convergence ($\cos, \sin \in C_b$) and a finite cover, convergence is uniform on compacts.

Step (ii) Tightness from ϕ. For $\delta > 0$ and $\nu \in \mathcal{M}_+^1(\mathbb{R})$, Fubini gives

$$\frac{1}{\delta} \int_{-\delta}^{\delta} (1 - \hat{\nu}(u))\, du = 2 \int \Big(1 - \frac{\sin \delta x}{\delta x}\Big) \nu(dx) \ge \nu(\{|x| \ge 2/\delta\}),$$

as $1-\frac{\sin s}{s}\geq\frac{1}{2}$ for $|s|\geq 2$. Continuity of ϕ at 0, $\phi(0)=1$, gives δ with $\frac{1}{\delta}\int_{-\delta}^{\delta}|1-\phi|<\varepsilon$, and by the DCT $\mu_m(\{|x|\geq 2/\delta\})<2\varepsilon$ for large m: tightness.

Step (ii) Helly selection. A diagonal subsequence has $F_{m_k}(q)\to H(q)$ for rational q; $G(x):=\inf_{q>x}H(q)$ is increasing, right continuous, and at continuity points $H(q)\leq\liminf_k F_{m_k}(x)\leq\limsup_k F_{m_k}(x)\leq H(q')$ $(q<x<q')$ gives $F_{m_k}(x)\to G(x)$. Tightness gives $G(\mp\infty)=0,1$, so G is the distribution function of some ν (Remark 2.15) and $\mu_{m_k}\to\nu$ (Corollary 3.37).

Step (ii) Identification. $\hat{\nu}=\phi$ by (i). Every subsequence has a further one converging to some ν' with $\hat{\nu}'=\phi$, so $\nu'=\nu=:\mu$ (Theorem 3.47); a subsequence with $|\int f\,d\mu_m-\int f\,d\mu|\geq\varepsilon$ would contradict this.

($n\geq 2$: Helly with multivariate distribution functions; not needed.) ■

For an $\mathcal{A}/\mathcal{B}(\mathbb{R}^n)$-measurable $\bar{X}=(X_1,\dots,X_n)$ on $(\Omega,\mathcal{A},P)$,

$$\varphi_{\bar{X}}(u):=\hat{P}_{\bar{X}}(u)=\int e^{i\langle u,y\rangle}P_{\bar{X}}(dy)=\mathbb{E}\big[e^{i\langle u,\bar{X}\rangle}\big]$$

is the **characteristic function of** $\bar{X}$.

Remark 3.50

$X_1,\dots,X_n$ are independent iff $\varphi_{\bar{X}}(u_1,\dots,u_n)=\prod_j\varphi_{X_j}(u_j)$ for all $u\in\mathbb{R}^n$. "⇒": Corollary 3.21 on real and imaginary parts. "⇐": the right side is the characteristic function of $\bigotimes_j P_{X_j}$ (Fubini); apply Theorems 3.47, 3.42.

Theorem 3.51

If $X_1,\dots,X_n$ are independent, $\alpha\in\mathbb{R}$ and $S:=\alpha\sum_k X_k$, then $\varphi_S(u)=\prod_{k=1}^n\varphi_{X_k}(\alpha u)$.

Proof. $\varphi_S(u)=\mathbb{E}[\prod_k e^{i\alpha u X_k}]=\prod_k\mathbb{E}[e^{i\alpha u X_k}]$ by independence (Corollary 3.21, applied to the bounded complex factors via real and imaginary parts and induction). ■

Theorem 3.52 (Gaussian characteristic function)

For all $u\in\mathbb{R}^n$: $(2\pi)^{-n/2}\int_{\mathbb{R}^n}e^{i\langle u,y\rangle}e^{-\frac{1}{2}\|y\|^2}dy=e^{-\frac{1}{2}\|u\|^2}$.

Proof [Supplement]. By Fubini the integral factorises over coordinates, so let $n=1$ and $\varphi(u):=(2\pi)^{-1/2}\int e^{iuy}e^{-y^2/2}dy$. Differentiating under the integral (dominated by $|y|e^{-y^2/2}$) and integrating by parts,

$$\varphi'(u)=(2\pi)^{-1/2}\int iy\,e^{iuy}e^{-y^2/2}dy=(2\pi)^{-1/2}\int i\,e^{iuy}\,\frac{d}{dy}\big(-e^{-y^2/2}\big)dy=-u\varphi(u).$$

With $\varphi(0)=1$ this gives $\varphi(u)=e^{-u^2/2}$. ■

Example 3.53

(i) $\hat{\delta}_a(u)=e^{iua}$. (ii) For $\mu=\sum_i\alpha_i\delta_{a_i}$ $(\alpha_i\geq 0,\ \sum\alpha_i=1)$, $\hat{\mu}(u)=\sum_i\alpha_i e^{iua_i}$. In particular the binomial distribution $\beta_n^p=\sum_k\binom{n}{k}p^kq^{n-k}\delta_k$ $(q=1-p)$ has $\hat{\beta}_n^p(u)=(q+pe^{iu})^n$, and the Poisson distribution $\pi_\alpha=\sum_n e^{-\alpha}\frac{\alpha^n}{n!}\delta_n$ has $\hat{\pi}_\alpha(u)=e^{-\alpha}\sum_n\frac{(\alpha e^{iu})^n}{n!}=e^{\alpha(e^{iu}-1)}$.

3.10 Central limit theorem

Motivation. Standardised sums of many independent, individually negligible (Lindeberg) terms are approximately normal.

Definition 3.54 (Central limit property)

Let $X_1, X_2, \dots \in \mathcal{L}^2$ be independent with positive variances, $S_n := X_1 + \cdots + X_n$ and $S_n^* := \frac{S_n - \mathbb{E}[S_n]}{\sqrt{\operatorname{Var}(S_n)}}$ (so $\mathbb{E}[S_n^*] = 0$, $\operatorname{Var}(S_n^*) = 1$). (X_n) has the **central limit property** (CLP) if $P_{S_n^*} \to N(0,1)$ weakly, equivalently (Corollary 3.37)

$$\lim_n P(S_n^* \le b) = \Phi(b) := \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{b} e^{-x^2/2} dx \qquad \forall b \in \mathbb{R}.$$

Theorem 3.55 (Central limit theorem, Lindeberg)

Let $X_1, X_2, \dots \in \mathcal{L}^2$ be independent, $\sigma_n^2 := \operatorname{Var}(X_n) > 0$, $s_n := (\sum_{k \le n} \sigma_k^2)^{1/2}$, and assume **Lindeberg's condition**

$$L_n(\varepsilon) := \sum_{k=1}^{n} \int_{\{|X_k - \mathbb{E}[X_k]| \ge \varepsilon s_n\}} \Big(\frac{X_k - \mathbb{E}[X_k]}{s_n} \Big)^2 dP \xrightarrow[n\to\infty]{} 0 \qquad \forall \varepsilon > 0. \tag{L}$$

Then (X_n) has the CLP.

Remark 3.56

(i) i.i.d. sequences in $\mathcal{L}^2$ satisfy (L): with $m = \mathbb{E}[X_1]$, $\sigma^2 = \operatorname{Var}(X_1)$, $s_n^2 = n\sigma^2$ and $L_n(\varepsilon) = \sigma^{-2} \int_{\{|X_1 - m| \ge \varepsilon \sqrt{n}\sigma\}} (X_1 - m)^2 dP \to 0$ by the DCT.

(ii) **Lyapunov's condition** (often with $\delta = 1$) implies (L):

$$\exists \delta > 0 : \quad \lim_n \sum_{k=1}^{n} \frac{\mathbb{E}|X_k - \mathbb{E}[X_k]|^{2+\delta}}{s_n^{2+\delta}} = 0. \tag{Lya}$$

Indeed $|X_k - \mathbb{E}X_k| \ge \varepsilon s_n$ implies $|X_k - \mathbb{E}X_k|^2 \le |X_k - \mathbb{E}X_k|^{2+\delta} / (\varepsilon s_n)^{\delta}$, so $L_n(\varepsilon) \le \varepsilon^{-\delta} \sum_k \mathbb{E}|X_k - \mathbb{E}X_k|^{2+\delta} / s_n^{2+\delta}$.

(iii) Uniformly bounded X_n with $s_n \to \infty$ satisfy (Lya) for every $\delta > 0$: if $|X_k| \le \alpha/2$, then $|X_k - \mathbb{E}X_k| \le \alpha$ and

$$\sum_k \frac{\mathbb{E}|X_k - \mathbb{E}X_k|^{2+\delta}}{s_n^{2+\delta}} \le \sum_k \frac{\alpha^{\delta} \mathbb{E}|X_k - \mathbb{E}X_k|^2}{s_n^2 s_n^{\delta}} = \Big(\frac{\alpha}{s_n} \Big)^{\delta} \to 0.$$

Lemma 3.57

Lindeberg's condition implies

$$\lim_n \max_{1 \le k \le n} \frac{\sigma_k}{s_n} = 0. \tag{3.14}$$

Proof. For $k \le n$ and $\varepsilon > 0$, splitting the integral on $\{|X_k - \mathbb{E}X_k| \ge \varepsilon s_n\}$:

$$\Big(\frac{\sigma_k}{s_n} \Big)^2 = \int \Big(\frac{X_k - \mathbb{E}X_k}{s_n} \Big)^2 dP \le L_n(\varepsilon) + \varepsilon^2.$$

■

Lemma 3.58

For $t \in \mathbb{R}$, $n \in \mathbb{N}$: $\left|e^{it} - \sum_{k=0}^{n-1} \frac{(it)^k}{k!}\right| \le \frac{|t|^n}{n!}$.

Proof. Taylor's formula with integral remainder for e^{it} (real and imaginary parts) gives the remainder $R_n(t) = \frac{1}{(n-1)!}\int_0^t (t-s)^{n-1} i^n e^{is} ds$, and $|R_n(t)| \le \frac{1}{(n-1)!}\int_0^{|t|} s^{n-1} ds = \frac{|t|^n}{n!}$. ■

Theorem 3.59

For $X \in \mathcal{L}^2$, φ_X is twice continuously differentiable with $\varphi_X'(u) = i\int X e^{iuX} dP$, $\varphi_X''(u) = -\int X^2 e^{iuX} dP$; in particular $\varphi_X'(0) = i\mathbb{E}[X]$, $\varphi_X''(0) = -\mathbb{E}[X^2]$, $|\varphi_X''| \le \mathbb{E}[X^2]$. Moreover

$$\varphi_X(u) = 1 + iu\mathbb{E}[X] + \tfrac{1}{2}\theta(u)u^2\mathbb{E}[X^2] \qquad \text{with } \theta(u) \in \mathbb{C},\ |\theta(u)| \le 1.$$

Proof.

Step 1 (derivatives). Difference quotients of e^{iuX} and iXe^{iuX} are dominated by $|X|$, X^2 (Lemma 3.58, $n = 1$); the DCT gives differentiation under the integral and continuity.

Step 2 (expansion). Lemma 3.58, $n = 2$: $|\varphi_X(u) - 1 - iu\mathbb{E}[X]| \le \frac{1}{2}u^2\mathbb{E}[X^2]$; let $\theta(u)$ be the ratio (or 0). ■

From now on $X_1, X_2, \cdots \in \mathcal{L}^2$ are independent with $\mathbb{E}[X_n] = 0$, $\sigma_n^2 = \mathrm{Var}(X_n) > 0$, $s_n = (\sum_{k\le n}\sigma_k^2)^{1/2}$.

Theorem 3.60

Assume

$$\text{(F)}\quad \lim_n \max_{k\le n} \frac{\sigma_k}{s_n} = 0 \ \text{(Feller)}, \qquad \text{(b)}\quad \lim_n \sum_{k=1}^n \left(\varphi_{X_k}\Big(\frac{u}{s_n}\Big) - 1\right) = -\frac{u^2}{2} \quad \forall u \in \mathbb{R}.$$

Then (X_n) has the CLP.

Proof.

Step 1 (reduction). By Theorem 3.51, $\varphi_{S_n^*}(u) = \prod_{k\le n}\varphi_{X_k}(u/s_n)$. If

$$\lim_n \prod_{k=1}^n \varphi_{X_k}\Big(\frac{u}{s_n}\Big) = e^{-u^2/2} \qquad \forall u, \tag{3.15}$$

then $\varphi_{S_n^*} \to e^{-u^2/2} = \widehat{N(0,1)}$ (Theorem 3.52), which is continuous at 0. Lévy's theorem 3.49 and Theorem 3.47 give $P_{S_n^*} \to N(0,1)$ weakly.

Step 2 (comparison with exponentials). By (b), $\prod_k \exp(\varphi_{X_k}(u/s_n) - 1) \to e^{-u^2/2}$. All factors lie in the unit disc ($|\exp(\varphi_{X_k} - 1)| = \exp(\mathrm{Re}\,\varphi_{X_k} - 1) \le 1$), and $|\prod_k a_k - \prod_k b_k| \le \sum_k |a_k - b_k|$ there, so with $z_k := \varphi_{X_k}(u/s_n) - 1$

$$\Big|\prod_k \varphi_{X_k}\Big(\frac{u}{s_n}\Big) - \prod_k e^{z_k}\Big| \le D_n := \sum_{k=1}^n |z_k + 1 - e^{z_k}|.$$

Step 3. For $\varepsilon \in (0, \frac{1}{2}]$ and $|z| \le \varepsilon$, $|z + 1 - e^z| \le |z|^2 \le \varepsilon|z|$; $|z_k| \le \frac{1}{2}\frac{u^2}{s_n^2}\sigma_k^2 < \varepsilon$ for $n \ge n_0$ (Theorem 3.59, (F)), so

$$D_n \le \varepsilon \sum_k |z_k| \le \varepsilon \frac{u^2}{2} \sum_k \frac{\sigma_k^2}{s_n^2} = \varepsilon \frac{u^2}{2},$$

so $D_n \to 0$. ∎

Proof of Theorem 3.55.

Step 1. We may assume $\mathbb{E}[X_n] = 0$. By Lemma 3.57, (L)⇒(F), so by Theorem 3.60 it remains to derive (b) from (L).

Step 2 (pointwise bounds). Fix u and let

$$Y_k := \Big| e^{i\frac{u}{s_n}X_k} - 1 - i\frac{u}{s_n}X_k + \frac{1}{2}\frac{u^2}{s_n^2}X_k^2 \Big|.$$

Lemma 3.58 ($n = 3$) gives $Y_k \le \frac{1}{6}|\frac{u}{s_n}X_k|^3$, and the triangle inequality with Lemma 3.58 ($n = 2$) gives $Y_k \le \frac{u^2}{s_n^2}X_k^2$.

Step 3. Since $\mathbb{E}[X_k] = 0$ and $\sum_k \sigma_k^2/s_n^2 = 1$,

$$\Big| \sum_{k=1}^{n} \Big(\varphi_{X_k}\Big(\frac{u}{s_n}\Big) - 1 \Big) + \frac{u^2}{2} \Big| = \Big| \sum_k \mathbb{E}\Big[e^{i\frac{u}{s_n}X_k} - 1 - i\frac{u}{s_n}X_k + \frac{u^2}{2s_n^2}X_k^2 \Big] \Big| \le \sum_k \mathbb{E}[Y_k].$$

Step 4. For $\varepsilon > 0$, use the second bound on $\{|X_k| \ge \varepsilon s_n\}$ and the first on the complement, where $\frac{1}{s_n^3}\int_{\{|X_k|<\varepsilon s_n\}} |X_k|^3 dP \le \frac{\varepsilon}{s_n^2}\sigma_k^2$:

$$\sum_k \mathbb{E}[Y_k] \le u^2 \sum_k \int_{\{|X_k/s_n| \ge \varepsilon\}} \Big(\frac{X_k}{s_n}\Big)^2 dP + \frac{|u|^3}{6}\varepsilon \sum_k \frac{\sigma_k^2}{s_n^2} = u^2 L_n(\varepsilon) + \frac{|u|^3}{6}\varepsilon.$$

Hence $\limsup_n \sum_k \mathbb{E}[Y_k] \le \frac{|u|^3}{6}\varepsilon$ for every $\varepsilon > 0$, which proves (b). ∎

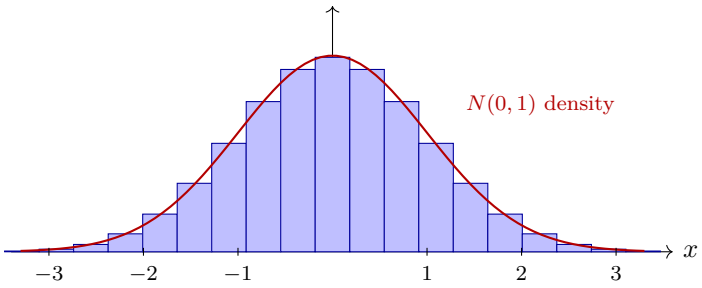


Figure. *Law of S_n^* for $n = 30$ fair tosses (bars of area $P(S_n = k)$) against the $N(0,1)$ density.*

Example 3.61 (Applications)

(i) Ruin probability. n i.i.d. claims $X_i \ge 0$ in $\mathcal{L}^2(m, \sigma^2)$, premium $\Pi := m + \lambda\sigma^2$ ($\lambda > 0$), capital K: ruin $R := \{S_n > K + n\Pi\}$, and the CLT gives

$$P(R) = P\Big(S_n^* > \frac{K + n\lambda\sigma^2}{\sqrt{n}\sigma}\Big) \approx 1 - \Phi(c_n), \qquad c_n := \frac{K + n\lambda\sigma^2}{\sqrt{n}\sigma} \to \infty.$$

c_n increases for $n > K/(\lambda\sigma^2)$: larger portfolios eventually lower the ruin probability.

Remark 3.62

At levels $c_n \to \infty$ the CLT gives no rate; if $\mathbb{E}|X_1 - m|^3 < \infty$, Berry–Esseen (Klenke,

Thm. 15.51) gives

$$\sup_x |P(S_n^* \le x) - \Phi(x)| \le \frac{C}{\sqrt{n}} \frac{\mathbb{E}|X_1 - m|^3}{\sigma^3},$$

which justifies the numbers: with $n = 2000$, $\sigma = 60$, $\lambda = 0.5‰$: (a) $K = 0$ gives $P(R) \approx 1 - \Phi(1.342) \approx 9\%$;
(b) $K = 1500$ gives $P(R) \approx 3\%$. For $P(R) \le 1‰$ with $K = 0$ one needs $\Phi(\sqrt{n}\lambda\sigma) \ge 0.999$, i.e. $n \ge 10\,611$.

(ii) Stirling's formula via the CLT (Weng 1977). Stirling (1730) proved

$$n! \approx \sqrt{2\pi}\, n^{n+\frac{1}{2}} e^{-n}, \tag{3.16}$$

used by De Moivre (1733); conversely the CLT (proved without Stirling) yields it. For i.i.d. Poisson(1) X_i, $S_n \sim \pi_n$ with $\mathbb{E}[S_n] = \operatorname{Var}(S_n) = n$ and $S_n^* = t_n(S_n)$, $t_n(x) = \frac{x-n}{\sqrt{n}}$. For $f_\infty(x) := x^- = (-x) \vee 0$, a telescoping sum gives

$$\int f_\infty \, dP_{S_n^*} = e^{-n} \sum_{k=0}^{n} \frac{n^k}{k!} \frac{n-k}{\sqrt{n}} = \frac{e^{-n}}{\sqrt{n}} \Big(n + \sum_{k=1}^{n} \Big(\frac{n^{k+1}}{k!} - \frac{n^k}{(k-1)!} \Big) \Big) = \frac{e^{-n} n^{n+\frac{1}{2}}}{n!},$$

while $\int f_\infty \, dN(0,1) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{0} (-x) e^{-x^2/2} dx = \frac{1}{\sqrt{2\pi}}$. So (3.16) follows once

$$\int f_\infty \, dP_{S_n^*} \to \int f_\infty \, dN(0,1). \tag{3.17}$$

f_∞ is unbounded, but $f_m := f_\infty \wedge m \in C_b$ converges by the CLT and $g_m := f_\infty - f_m$ is uniformly small:

$$\int g_m \, dP_{S_n^*} \le \int_{(-\infty, -m]} |x| \, dP_{S_n^*} \le \frac{1}{m} \int x^2 \, dP_{S_n^*} = \frac{1}{m} \operatorname{Var}(S_n^*) = \frac{1}{m},$$

and likewise $\int g_m \, dN(0,1) \le \frac{1}{m}$. A 3ε-argument gives (3.17).

Chapter 4. Brownian Motion

4.1 Definition and some properties

Motivation. The continuous limit of the random walk (Example 3.8): quadratic variation t, hence infinite length.

For $I \subset \mathbb{R}_+ := [0,\infty)$ (e.g. $\mathbb{N}_0$, $[0,T]$, dyadic rationals $\mathbb{Q}_2^+ := \{m2^{-n} \mid m,n \geq 0\}$, $\mathbb{R}_+$), a **stochastic process** is a family $(X_t)_{t\in I}$ of r.v.'s on $(\Omega,\mathcal{F},P)$, possibly with values in a metric **state space** S; $t \mapsto X_t(\omega)$ is a **sample path**.

Definition 4.1 (Brownian motion)

A real-valued process $(B_t)_{t\geq 0}$ is a (standard) **Brownian motion** (BM) if

(i) $P(B_0 = 0) = 1$;

(ii) for $0 \leq s < t$, $B_t - B_s \sim N(0, t-s)$, i.e. $P(B_t - B_s \leq b) = \frac{1}{\sqrt{2\pi(t-s)}} \int_{-\infty}^{b} e^{-x^2/(2(t-s))} dx$;

(iii) it has **independent increments**: for $0 \leq t_1 < \cdots < t_n$, the r.v.'s $B_{t_1} - B_0, B_{t_2} - B_{t_1}, \ldots, B_{t_n} - B_{t_{n-1}}$ are independent;

(iv) almost all paths are continuous: $P(t \mapsto B_t \text{ is continuous on } [0,\infty)) = 1$.

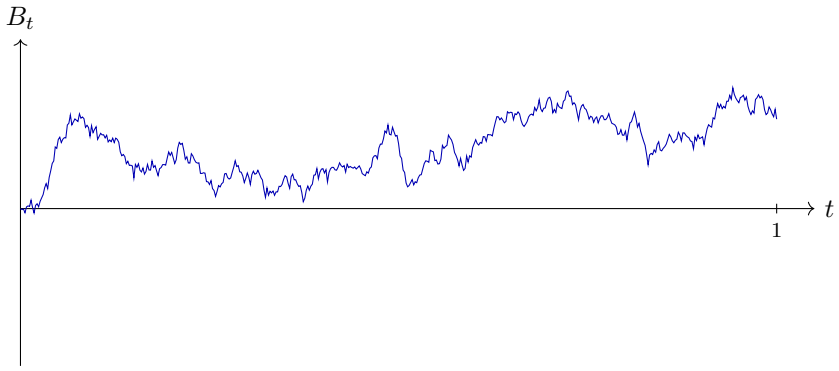


Figure. *A simulated Brownian path on $[0,1]$ (2000 steps); quadratic variation t (Theorem 4.3), infinite total variation (Proposition 4.6).*

Proposition 4.2

Let $(B_t)_{t\geq 0}$ be a BM. (i) (Translation invariance) For fixed $t_0 \geq 0$, $\tilde{B}_t := B_{t+t_0} - B_{t_0}$ is a BM. (ii) (Scaling invariance) For $\lambda > 0$, $\tilde{B}_t := \frac{1}{\sqrt{\lambda}} B_{\lambda t}$ is a BM.

Proof. (i) Definition 4.1(i),(iv) are clear, and $\tilde{B}_t - \tilde{B}_s = B_{t+t_0} - B_{s+t_0} \sim N(0, t-s)$. For $0 \leq t_1 < \cdots < t_n$, the increments of $\tilde{B}$ are increments of B along $t_0 \leq t_1 + t_0 < \cdots < t_n + t_0$ (if $t_1 = 0$ the first one is the constant 0, which is independent of everything), hence independent by (iii) for B.

(ii) (i), (iii),

(iv) carry over directly. For (ii), substituting $y = x/\sqrt{\lambda}$,

$$P(\tilde{B}_t - \tilde{B}_s \leq b) = P(B_{\lambda t} - B_{\lambda s} \leq \sqrt{\lambda} b) = \int_{-\infty}^{\sqrt{\lambda} b} \frac{e^{-x^2/(2\lambda(t-s))}}{\sqrt{2\pi\lambda(t-s)}} dx = \int_{-\infty}^{b} \frac{e^{-y^2/(2(t-s))}}{\sqrt{2\pi(t-s)}} dy.$$

■

Let $\tau_n = \{0 = t_1^{(n)} < t_2^{(n)} < \cdots < t_{N(\tau_n)}^{(n)} < \infty\}$ be partitions of $[0,\infty)$ with

$$|\tau_n| := \max_i \left(t_{i+1}^{(n)} - t_i^{(n)}\right) \to 0 \quad \text{(mesh)}, \qquad t_{N(\tau_n)}^{(n)} \to \infty. \tag{$\star$}$$

Theorem 4.3 (Quadratic variation)

Let (B_t) be a BM and (τ_n) as in $(\star)$.

(i) For every $t \geq 0$, $S_t^n := \sum_{t_i^{(n)} \in \tau_n,\ t_{i+1}^{(n)} \leq t} \left(B_{t_{i+1}^{(n)}} - B_{t_i^{(n)}}\right)^2 \to t$ in $L^2(\Omega, \mathcal{F}, P)$.

(ii) If $\tau_n \subset \tau_{n+1}$ for all n, or $\sum_n |\tau_n| < \infty$, then $P(\lim_n S_t^n = t \ \forall t \geq 0) = 1$.

Proof. (i) Write $\Delta_i := t_{i+1}^{(n)} - t_i^{(n)}$; the sums run over i with $t_{i+1}^{(n)} \leq t$. Then $\mathbb{E}[S_t^n] = \sum \Delta_i \to t$ in $\mathbb{R}$, by $(\star)$. By independence of increments and $\mathbb{E}[N^4] = 3\sigma^4$ for $N \sim N(0,\sigma^2)$,

$$\operatorname{Var}(S_t^n) = \sum \operatorname{Var}\left((B_{t_{i+1}} - B_{t_i})^2\right) = \sum (3\Delta_i^2 - \Delta_i^2) = 2\sum \Delta_i^2 \leq 2t|\tau_n| \to 0.$$

Hence $S_t^n - \mathbb{E}[S_t^n] \to 0$ in L^2, and with $\mathbb{E}[S_t^n] \to t$ we get $S_t^n \to t$ in L^2.

(ii) [Supplement] *Case* $\sum_n |\tau_n| < \infty$ (this includes the dyadic partitions $t_i^{(n)} = (i-1)2^{-n}$, the case treated in the lectures).

Step 1 (fixed t). By Chebyshev and the bound above, $\sum_n P(|S_t^n - \mathbb{E}S_t^n| \geq \varepsilon) \leq \varepsilon^{-2} \sum_n 2t|\tau_n| < \infty$. By Lemma 2.10, $S_t^n - \mathbb{E}[S_t^n] \to 0$ a.s., hence $S_t^n \to t$ on a set Ω_t with $P(\Omega_t) = 1$.

Step 2 (all t at once). Let $\Omega' := \bigcap_{t \in \mathbb{Q}_2^+} \Omega_t$, so $P(\Omega') = 1$. For $\omega \in \Omega'$ and arbitrary t, take dyadic $t_k^- \nearrow t$, $t_k^+ \searrow t$. Since S_t^n is increasing in t,

$$t_k^- = \lim_n S_{t_k^-}^n(\omega) \leq \liminf_n S_t^n(\omega) \leq \limsup_n S_t^n(\omega) \leq \lim_n S_{t_k^+}^n(\omega) = t_k^+,$$

and $k \to \infty$ gives the claim.

General nested partitions: see Revuz–Yor, *Continuous Martingales and Brownian Motion*, Ch. II, Prop. (2.12); not proved in the lectures. ∎

Remark 4.4

Let Ω'' with $P(\Omega'') = 1$ be such that $t \mapsto B_t(\omega)$ is continuous for $\omega \in \Omega''$, and $\tilde{\Omega} := \Omega' \cap \Omega''$ with Ω' from the proof above. For $\tilde{\omega} \in \tilde{\Omega}$ the path is continuous and, along (τ_n) as in Theorem 4.3(ii),

$$\langle B \rangle_t(\tilde{\omega}) := \lim_n \sum_{t_{i+1}^{(n)} \leq t} \left(B_{t_{i+1}^{(n)}}(\tilde{\omega}) - B_{t_i^{(n)}}(\tilde{\omega})\right)^2 = t \qquad \forall t \geq 0.$$

Such a path is a **typical path**, and $\langle B \rangle_t(\tilde{\omega})$ its **quadratic variation**.

Definition 4.5 (Finite variation)

$g : \mathbb{R}_+ \to \mathbb{R}$ has **finite variation** on $[0,t]$ if

$$\operatorname{Var}_t g := \sup_\tau \sum_{t_i \in \tau,\ t_{i+1} \leq t} |g(t_{i+1}) - g(t_i)| < \infty, \tag{4.1}$$

the supremum over all partitions $0 = t_1 < \cdots < t_{N(\tau)} < \infty$; g has finite variation if this holds for every t.

Proposition 4.6

A typical path of BM is continuous and of unbounded variation on every interval $[0,T]$, $T>0$.

Proof. Suppose $\mathrm{Var}_T B_\cdot(\tilde\omega)<\infty$ for some $T>0$. The path is uniformly continuous on $[0,T]$, so for $\varepsilon>0$ the increments are $<\varepsilon$ once $|\tau_n|<\delta$, and

$$\sum_{t_{i+1}\le T}\big(B_{t_{i+1}}-B_{t_i}\big)^2(\tilde\omega)\le \max_i|B_{t_{i+1}}-B_{t_i}|(\tilde\omega)\cdot\sum_i|B_{t_{i+1}}-B_{t_i}|(\tilde\omega)\le\varepsilon\,\mathrm{Var}_T B_\cdot(\tilde\omega).$$

Hence $\langle B\rangle_T(\tilde\omega)=0$, contradicting $\langle B\rangle_T(\tilde\omega)=T>0$. ∎

4.2 Construction (hence existence) of Brownian motion

Motivation. Build the process on dyadic times (Kolmogorov extension), prove continuity (Kolmogorov–Čentsov), and extend.

For a process $(X_t)_{t\in I}$ and pairwise distinct $t_1,\dots,t_n\in I$ let

$$\mu_{t_1\dots t_n}(A_1\times\cdots\times A_n):=P\big((X_{t_1},\dots,X_{t_n})\in A_1\times\cdots\times A_n\big),\qquad A_i\in\mathcal{B}(\mathbb{R}).\tag{4.2}$$

By Theorem 1.9 this determines the law $P((X_{t_1},\dots,X_{t_n})\in A)$, $A\in\mathcal{B}(\mathbb{R}^n)$, because rectangles are a $\cap$-stable generator of $\mathcal{B}(\mathbb{R}^n)$. These **finite-dimensional distributions** are **consistent**:

$$\mu_{t_1\dots t_n}(A_1\times\cdots\times A_n)=\mu_{t_{\pi(1)}\dots t_{\pi(n)}}(A_{\pi(1)}\times\cdots\times A_{\pi(n)})\quad\text{for permutations }\pi,\tag{4.3}$$

$$\mu_{t_1\dots t_n}(A_1\times\cdots\times A_{n-1}\times\mathbb{R})=\mu_{t_1\dots t_{n-1}}(A_1\times\cdots\times A_{n-1})\quad(n\ge 2).\tag{4.4}$$

Any family of probability measures $\mu_{t_1\dots t_n}$ on $\mathbb{R}^n$ satisfying (4.3), (4.4) is called consistent. Conversely, does every consistent family come from a process?

On $\mathbb{R}^I:=\{\omega: I\to\mathbb{R}\}$ let the **cylinder sets** be

$$\mathcal{C}_I:=\big\{\{\omega\mid(\omega(t_1),\dots,\omega(t_n))\in A_1\times\cdots\times A_n\}\;\big|\;t_1<\cdots<t_n\in I,\ A_i\in\mathcal{B}(\mathbb{R}),\ n\in\mathbb{N}\big\}.$$

$\sigma(\mathcal{C}_I)$ is the smallest σ-algebra making all finite-dimensional projections $f_{t_1\dots t_n}(\omega):=(\omega(t_1),\dots,\omega(t_n))$ measurable, and, since $f^{-1}_{t_1\dots t_n}(A_1\times\cdots\times A_n)=\bigcap_i f_{t_i}^{-1}(A_i)$, even the smallest one making all f_t, $t\in I$, measurable.

Theorem 4.7 (Kolmogorov's extension theorem)

For every consistent family $\mu_{t_1\dots t_n}$ there is a unique probability measure P on $(\mathbb{R}^I,\sigma(\mathcal{C}_I))$ such that, with $X_t(\omega):=\omega(t)$, (4.2) holds, i.e. $\mu_{t_1\dots t_n}=P\circ f^{-1}_{t_1\dots t_n}$.

Proof [Supplement].

Step 1 (inner regularity). Every probability measure μ on $(\mathbb{R}^n,\mathcal{B}(\mathbb{R}^n))$ satisfies $\mu(A)=\sup\{\mu(K)\mid K\subset A\text{ compact}\}$. The class of A such that for every ε there are F closed $\subset A\subset G$ open with $\mu(G\setminus F)<\varepsilon$ is a σ-algebra. It contains every closed F (take $G=\{d(\cdot,F)<\frac{1}{m}\}$ and continuity from above), hence all Borel sets. Intersecting F with a large closed ball gives a compact K.

Step 2 (a finitely additive P_0). The general cylinders $Z=\{\omega\mid f_J(\omega)\in A\}$, $J\subset I$ finite, $A\in\mathcal{B}(\mathbb{R}^J)$, form an algebra $\mathcal{Z}$ with $\sigma(\mathcal{Z})=\sigma(\mathcal{C}_I)$. Put $P_0(Z):=\mu_J(A)$. This is well defined: two representations can be lifted to a common finite $J'\supset J$, and (4.3), (4.4), which extend from rectangles to all Borel sets by Theorem 1.9, show that $\mu_J(A)=\mu_{J'}(A\times\mathbb{R}^{J'\setminus J})$. Since finitely many cylinders can be written over one common J, P_0 is finitely additive with $P_0(\mathbb{R}^I)=1$.

Step 3 (σ-additivity). By Theorem 1.15 it suffices that $Z_k \in \mathcal{Z}$, $Z_k \searrow \emptyset$ imply $P_0(Z_k) \to 0$. Suppose $P_0(Z_k) \geq \varepsilon > 0$ for all k, with $Z_k = \{f_{J_k} \in A_k\}$ and $J_1 \subset J_2 \subset \cdots$. By Step 1 choose compact $K_k \subset A_k$ with $\mu_{J_k}(A_k \setminus K_k) \leq \varepsilon 2^{-k-1}$, and put $E_k := \bigcap_{j \leq k} \{f_{J_j} \in K_j\} \subset Z_k$. Then $P_0(Z_k \setminus E_k) \leq \varepsilon/2$, so $P_0(E_k) \geq \varepsilon/2$ and $E_k \neq \emptyset$; pick $\omega_k \in E_k$. On the countable set $J := \bigcup_k J_k$, for each $t \in J_j$ the values $\omega_k(t)$, $k \geq j$, lie in the compact projection of K_j. A diagonal argument gives a subsequence with $\omega_{k_l}(t) \to \omega(t)$ for all $t \in J$; put $\omega(t) := 0$ off J. For each j, $f_{J_j}(\omega_{k_l}) \in K_j$ for $k_l \geq j$ and K_j is closed, so $f_{J_j}(\omega) \in K_j \subset A_j$. Thus $\omega \in \bigcap_j Z_j = \emptyset$, a contradiction.

Step 4. Carathéodory's Theorem 1.11 (on the algebra $\mathcal{Z}$, finite measure) extends P_0 uniquely to $\sigma(\mathcal{C}_I)$, and $P \circ f_{t_1 \dots t_n}^{-1} = \mu_{t_1 \dots t_n}$ by construction. Uniqueness: $\mathcal{Z}$ is a $\cap$-stable generator (Theorem 1.9). ∎

The Brownian finite-dimensional laws. Let $I = \mathbb{R}_+$ and

$$p_t(x,y) := \frac{1}{\sqrt{2\pi t}} e^{-\frac{(y-x)^2}{2t}} \quad (t > 0), \qquad p_0(x,y)\,dy := \delta_x(dy). \tag{4.5}$$

For $0 \leq t_1 < \cdots < t_n$ and $A \in \mathcal{B}(\mathbb{R}^n)$ set, with $t_0 := 0$, $x_0 := 0$,

$$\mu_{t_1 \dots t_n}(A) := \int \cdots \int \mathbb{1}_A(x_1, \dots, x_n) \prod_{i=1}^{n} p_{t_i - t_{i-1}}(x_{i-1}, x_i)\, dx_n \cdots dx_1$$

(if $t_1 = 0$, the first factor means $\delta_0(dx_1)$). This is a probability measure: the integrand is a nonnegative Borel density (Theorem 1.47), and integrating out $x_n, x_{n-1}, \dots$ successively gives total mass 1. By Fubini and Theorem 1.9 it is the unique probability measure with

$$\mu_{t_1 \dots t_n}(A_1 \times \cdots \times A_n) = \int_{A_1} \cdots \int_{A_n} \prod_{i=1}^{n} p_{t_i - t_{i-1}}(x_{i-1}, x_i)\, dx_n \cdots dx_1. \tag{4.6}$$

(Replacing $x_0 = 0$ by $x_0 = x$ gives Brownian motion started at x.) For unordered times, sort them by the unique permutation σ with $t_{\sigma(1)} < \cdots < t_{\sigma(n)}$ and permute the coordinates accordingly. The family is consistent: (4.3) holds by construction, and (4.4) amounts to deleting one time,

$$\mu_{t_1 \dots t_{i-1} t_{i+1} \dots t_n}(\cdots \times A_{i-1} \times A_{i+1} \times \dots) = \mu_{t_1 \dots t_n}(\cdots \times A_{i-1} \times \mathbb{R} \times A_{i+1} \times \dots). \tag{4.7}$$

For $i = n$ this is immediate. For $i < n$ it is the **Chapman–Kolmogorov equation**

$$\int_A p_{t+s}(x,z)\, dz = \int_A \int_{\mathbb{R}} p_s(x,y) p_t(y,z)\, dy\, dz, \qquad \text{i.e.} \quad \int p_s(x,y) p_t(y,z)\, dy = p_{s+t}(x,z),$$

which holds because $N(0,s) * N(0,t) = N(0,s+t)$ (Example 3.44(i)). [Supplement]

The difficulty. Theorem 4.7 gives $\tilde{P}$ on $(\mathbb{R}^{\mathbb{R}_+}, \sigma(\mathcal{C}_{\mathbb{R}_+}))$ with (i)–(iii) of Definition 4.1, but $C(\mathbb{R}_+) \notin \sigma(\mathcal{C}_{\mathbb{R}_+})$ (its sets depend on countably many coordinates). So restrict to $\mathbb{Q}_2^+$: the family is consistent and Theorem 4.7 gives a unique Q on $(\mathbb{R}^{\mathbb{Q}_2^+}, \sigma(\mathcal{C}_{\mathbb{Q}_2^+}))$.

Theorem 4.8

Let $C(\mathbb{Q}_2^+) := \bigcap_{n \geq 1} \{\omega \in \mathbb{R}^{\mathbb{Q}_2^+} \mid \omega|_{[0,n] \cap \mathbb{Q}_2^+} \text{ is uniformly continuous}\}$. Then $C(\mathbb{Q}_2^+) \in \sigma(\mathcal{C}_{\mathbb{Q}_2^+})$ and $Q(C(\mathbb{Q}_2^+)) = 1$; so $(C(\mathbb{Q}_2^+), C(\mathbb{Q}_2^+) \cap \sigma(\mathcal{C}_{\mathbb{Q}_2^+}), Q)$ is a probability space (with the trace σ-algebra).

From Theorem 4.8 to Brownian motion. Let $\Omega := C(\mathbb{R}_+)$ with the metric

$$\rho(\omega, \omega') := \sum_{n \geq 1} 2^{-n} \Big(1 \wedge \sup_{t \in [0,n]} |\omega(t) - \omega'(t)|\Big),$$

under which Ω is a complete separable metric space; $\mathcal{F} := \mathcal{B}(\Omega)$ and $B_t(\omega) := \omega(t)$, which is ρ-continuous and hence measurable, so $\sigma(B_t \mid t \ge 0) \subset \mathcal{F}$. Conversely, for a countable dense $D \subset \Omega$ and $\omega_0 \in D$, $\rho(\omega,\omega_0) = \sum_n 2^{-n}(1 \wedge \sup_{t\in[0,n]\cap\mathbb{Q}_2^+} |B_t(\omega) - \omega_0(t)|)$ is $\sigma(B_t \mid t \in \mathbb{Q}_2^+)$-measurable. Every open set is a countable union of balls $B_\rho(\omega_0, r)$, $\omega_0 \in D$, $r \in \mathbb{Q}_2^+$, so $\mathcal{F} = \sigma(B_t \mid t \ge 0)$.

The map $\psi : C(\mathbb{Q}_2^+) \to \Omega$, $\omega \mapsto$ the unique continuous extension of ω to $\mathbb{R}_+$, is measurable: for $q_m \to t$ in $\mathbb{Q}_2^+$,

$$\psi^{-1}(\{B_t < \alpha\}) = \bigcup_k \bigcup_n \bigcap_{m\ge n} \{\omega \mid \omega(q_m) < \alpha - \tfrac{1}{k}\},$$

and the sets $\{B_t < \alpha\}$ generate $\mathcal{F}$. Let $P := Q \circ \psi^{-1}$. For dyadic t_i, $(\psi\omega)(t_i) = \omega(t_i)$, so

$$P((B_{t_1},\dots,B_{t_n}) \in A_1 \times \cdots \times A_n) = Q((X_{t_1},\dots,X_{t_n}) \in A_1 \times \cdots \times A_n) \qquad (t_i \in \mathbb{Q}_2^+). \tag{4.8}$$

Then $B_t(\omega) = \omega(t)$ is a Brownian motion on $(\Omega, \mathcal{F}, P)$, every path being continuous. The key is that for all $0 \le t_1 < \cdots < t_n$ in $\mathbb{R}_+$, with $t_0 = 0$, $x_0 = 0$,

$$P((B_{t_1},\dots,B_{t_n}) \in A_1 \times \cdots \times A_n) = \int_{A_1} \cdots \int_{A_n} \prod_{i=1}^{n} p_{t_i - t_{i-1}}(x_{i-1}, x_i)\, dx_n \cdots dx_1, \tag{4.9}$$

and that (4.9) is equivalent to Definition 4.1(i)–(iii).

[Supplement] *Proof of* (4.9). Take dyadic $t_i^k \to t_i$ (strictly ordered). For $f \in C_b(\mathbb{R}^n)$, $\mathbb{E}_P[f(B_{t_1^k},\dots,B_{t_n^k})] = \int f\, d\mu_{t_1^k \dots t_n^k}$ by (4.8). As $k \to \infty$ the left side tends to $\mathbb{E}_P[f(B_{t_1},\dots,B_{t_n})]$ by path continuity and the DCT. In the right side substitute $x_i = x_{i-1} + \sqrt{t_i^k - t_{i-1}^k}\, z_i$: it becomes an integral of f evaluated at points depending continuously on k against a fixed Gaussian density, so the DCT gives the limit $\int f\, d\mu_{t_1\dots t_n}$. By Example 2.30 on $\mathbb{R}^n$ the two laws agree.

(4.9) *gives (i)–(iii).* $n = 1$ gives $B_t \sim N(0,t)$ and $B_0 = 0$ a.s. For $t_1 < t_2$, with $\bar{A} = \{(x,y) \mid y - x \le \alpha\}$ and $z = y - x$,

$$P(B_{t_2} - B_{t_1} \le \alpha) = \int_{\mathbb{R}} p_{t_1}(0,x) \int_{-\infty}^{x+\alpha} p_{t_2-t_1}(x,y)\, dy\, dx = \int_{-\infty}^{\alpha} \frac{e^{-z^2/(2(t_2-t_1))}}{\sqrt{2\pi(t_2-t_1)}} dz.$$

More generally, the change of variables $z_i = x_i - x_{i-1}$ turns the density in (4.9) into $\prod_i p_{t_i - t_{i-1}}(0, z_i)$, a product density. So $(B_{t_1}, B_{t_2} - B_{t_1}, \dots, B_{t_n} - B_{t_{n-1}})$ has independent $N(0, t_i - t_{i-1})$ coordinates (Theorem 3.42), i.e.

$$P(B_{t_1} \le \alpha_1, B_{t_2} - B_{t_1} \le \alpha_2, \dots, B_{t_n} - B_{t_{n-1}} \le \alpha_n) = P(B_{t_1} \le \alpha_1) \cdots P(B_{t_n} - B_{t_{n-1}} \le \alpha_n). \tag{4.10}$$

Conversely (i)–(iii) determine the joint law of the increments, hence (4.9).

Proof of Theorem 4.8.

Step 1 (measurability). With $\Omega_n := \{\omega \mid \omega|_{[0,n]\cap\mathbb{Q}_2^+}$ uniformly continuous$\}$,

$$\Omega_n = \bigcap_{M\ge1} \bigcup_{N\ge1} \bigcap_{t,s\in\mathbb{Q}_2^+\cap[0,n],\ |t-s|\le\frac{1}{N}} \Big\{|X_t - X_s| \le \frac{1}{M}\Big\} \in \sigma(\mathcal{C}_{\mathbb{Q}_2^+}),$$

a countable combination of cylinder sets. It remains to show $Q(\Omega_n) = 1$ for each n.

Step 2 (moments). The marginals of $(X_t)_{t\in\mathbb{Q}_2^+}$ satisfy Definition 4.1(i)–(iii), hence scaling invariance: $X_t - X_s \sim \frac{1}{\sqrt{\lambda}}(X_{\lambda t} - X_{\lambda s}) \sim N(0, t-s)$. For $Y_t^n := \frac{1}{\sqrt{n}} X_{nt}$ and $s < t$ in $\mathbb{Q}_2^+ \cap [0,1]$,

$$\mathbb{E}_Q|Y_t^n - Y_s^n|^{2k} = \mathbb{E}_Q|X_t - X_s|^{2k} = \mathbb{E}_Q|\sqrt{t-s}\, X_1|^{2k} = c_k|t-s|^k, \qquad c_k := \mathbb{E}_Q|X_1|^{2k}. \tag{4.11}$$

Step 3. By Theorem 4.9 and Remark 4.10 there is Ω^* with $Q(\Omega^*) = 1$ on which $t \mapsto Y_t^n(\omega)$ is Hölder continuous of order $\gamma \in (0, \frac{1}{2})$ on $[0,1] \cap \mathbb{Q}_2^+$, i.e. $t \mapsto X_t(\omega)$ is Hölder on $[0,n] \cap \mathbb{Q}_2^+$. So $\Omega^* \subset \Omega_n$. ∎

Theorem 4.9 (Kolmogorov–Čentsov, dyadic version)

Let $(X_t)_{t\in\mathbb{Q}_2^+}$ be a process on $(\mathbb{R}^{\mathbb{Q}_2^+}, \sigma(\mathcal{C}_{\mathbb{Q}_2^+}), Q)$ with

$$\mathbb{E}_Q|X_t - X_s|^\alpha \le K|t-s|^{1+\beta} \qquad \forall t,s\in\mathbb{Q}_2^+\cap[0,1]$$

for some $\alpha,\beta,K>0$, and let $\gamma\in(0,\beta/\alpha)$. Then there is $\Omega^*\in\sigma(\mathcal{C}_{\mathbb{Q}_2^+})$ with $Q(\Omega^*)=1$ such that for every $\omega\in\Omega^*$ there is $n^*(\omega)$ with

$$\sup_{t,s\in\mathbb{Q}_2^+\cap[0,1],\ 0<t-s<2^{-n^*(\omega)}} \frac{|X_t(\omega)-X_s(\omega)|}{|t-s|^\gamma} \le C := \frac{2}{1-2^{-\gamma}}.$$

In particular every $\omega\in\Omega^*$ is Hölder continuous of order γ on $\mathbb{Q}_2^+\cap[0,1]$: with $n_0 := n^*(\omega)+1$, splitting $[s,t]$ into 2^{n_0} equal pieces gives $|X_t(\omega)-X_s(\omega)| \le 2^{n_0}C\big((t-s)2^{-n_0}\big)^\gamma = 2^{n_0(1-\gamma)}C|t-s|^\gamma$.

Remark 4.10

By (4.11) we may take $\alpha=2k$, $\beta=k-1$, and $\beta/\alpha=\frac{k-1}{2k}\nearrow\frac12$. So Theorem 4.9 applies to every $\gamma\in(0,\frac12)$: Q-a.e. path is locally Hölder continuous of every order $\gamma<\frac12$. The map ψ preserves local Hölder continuity, so P-a.e. Brownian path in $C(\mathbb{R}_+)$ is locally γ-Hölder for every $\gamma<\frac12$.

Proof of Theorem 4.9.

Step 1 (Borel–Cantelli on dyadic increments). By Markov's inequality and the hypothesis,

$$\begin{aligned} Q\Big(\max_{1\le k\le 2^n}|X_{k2^{-n}}-X_{(k-1)2^{-n}}|\ge 2^{-\gamma n}\Big) &\le \sum_{k=1}^{2^n}\frac{\mathbb{E}_Q|X_{k2^{-n}}-X_{(k-1)2^{-n}}|^\alpha}{2^{-\gamma n\alpha}} \\ &\le 2^n\frac{K2^{-n(1+\beta)}}{2^{-\gamma n\alpha}} = K2^{-n(\beta-\gamma\alpha)}. \end{aligned}$$

This is summable since $\beta>\gamma\alpha$. By Borel–Cantelli, $\Omega^* := \bigcup_l\bigcap_{n\ge l}\{\max_k|X_{k2^{-n}}-X_{(k-1)2^{-n}}|<2^{-\gamma n}\}$ has $Q(\Omega^*)=1$, and for $\omega\in\Omega^*$ there is $n^*(\omega)$ with

$$\max_{1\le k\le 2^n}|X_{k2^{-n}}(\omega)-X_{(k-1)2^{-n}}(\omega)|<2^{-\gamma n} \qquad \forall n\ge n^*(\omega). \tag{4.12}$$

Step 2 (chaining). Fix $\omega\in\Omega^*$, $n\ge n^*(\omega)$, and let $D_m := \{k2^{-m}\mid 0\le k\le 2^m\}$. We claim, for all $m>n$,

$$|X_t(\omega)-X_s(\omega)| \le 2\sum_{j=n+1}^{m}2^{-\gamma j} \qquad \forall t,s\in D_m,\ 0<t-s<2^{-n}. \tag{4.13}$$

For $m=n+1$, t,s are neighbours in D_{n+1}, and (4.12) gives $<2^{-\gamma(n+1)}$. Assume (4.13) for $m=M-1$, and let $s,t\in D_M$, $0<t-s<2^{-n}$. Put $s_1 := \min\{r\in D_{M-1}\mid r\ge s\}$ and $t_1 := \max\{r\in D_{M-1}\mid r\le t\}$. Then $s\le s_1\le t_1\le t$ and $s_1-s, t-t_1\in\{0,2^{-M}\}$, so (4.12) bounds $|X_{s_1}-X_s|$ and $|X_t-X_{t_1}|$ by $2^{-\gamma M}$, and the induction hypothesis bounds $|X_{t_1}-X_{s_1}|$. Summing gives (4.13) for $m=M$.

Step 3. For $t,s\in\mathbb{Q}_2^+\cap[0,1]$ with $0<t-s<2^{-n^*(\omega)}$ choose $m>n\ge n^*(\omega)$ with $t,s\in D_m$ and $2^{-(n+1)}\le t-s<2^{-n}$. By (4.13),

$$|X_t-X_s| \le 2\sum_{j\ge n+1}2^{-\gamma j} = \frac{2\cdot 2^{-\gamma(n+1)}}{1-2^{-\gamma}} \le \frac{2}{1-2^{-\gamma}}|t-s|^\gamma.$$

■

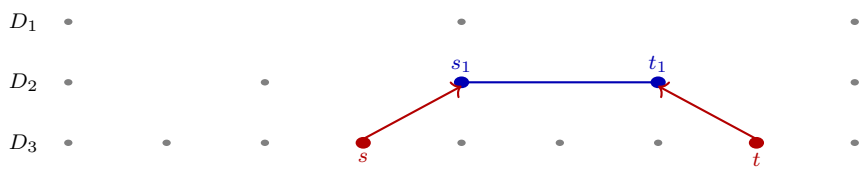


Figure. *Chaining (Theorem 4.9): one 2^{-j}-step per level, increments $\le 2^{-\gamma j}$, a geometric series.*

Remark 4.11

$((B_t)_{t\ge0}, \Omega, \mathcal{F}, P)$ as constructed above is the **canonical model** of Brownian motion; $P = P_0$ is the **Wiener measure** (started at 0).

4.3 The Markov property

Motivation. Given B_s, the future is a Brownian motion from B_s; the filtration $\mathcal{F}_t^{0,+}$ adds Blumenthal's 0–1 law.

Work in the canonical model: $\Omega = C(\mathbb{R}_+)$, $B_t(\omega) = \omega(t)$, $\mathcal{F} = \mathcal{B}(C(\mathbb{R}_+)) = \sigma(B_t \mid t \ge 0)$. For a start in $x \in \mathbb{R}$ there are three equivalent options: (1) (B_t) under P_x, built as P but with $x_0 = x$ in (4.6); (2) $(B_t + x)$ under P_0; (3) (B_t) under $P_0 \circ T_x^{-1}$ with $T_x(\omega) := \omega + x$. All have the same finite-dimensional distributions and continuous paths; in particular $P_x = P_0 \circ T_x^{-1}$. We write $\mathbb{E}_x[Y] := \int Y \, dP_x$.

Let $\mathcal{F}_t^0 := \sigma(B_u \mid u \le t)$. Then $(\mathcal{F}_t^0)_{t\ge0}$ is a **filtration** (information up to t) to which (B_t) is **adapted**. The **shift** $\theta_s(\omega)(t) := \omega(s+t)$ is ρ-continuous and $\hat{\mathcal{F}}_s^0/\mathcal{F}$-measurable, $\hat{\mathcal{F}}_s^0 := \sigma(B_u \mid u \ge s)$; for $Y \in \mathcal{F}_b$, $Y \circ \theta_s$ is a function of the future after s, e.g. $Y = f(B_{t_1}, \dots, B_{t_n})$ gives $Y \circ \theta_s = f(B_{t_1+s}, \dots, B_{t_n+s})$, and $B_t \circ \theta_s = B_{t+s}$.

The canonical model has the **Markov property** w.r.t. $(\mathcal{F}_t^0)$: for $Y \in \mathcal{F}_b$,

$$\mathbb{E}_x[\underbrace{Y \circ \theta_s}_{\text{future}} \mid \underbrace{\mathcal{F}_s^0}_{\text{past}}] = \mathbb{E}_{\underbrace{B_s}_{\text{present}}}[Y] \qquad P_x\text{-a.s., } \forall s \ge 0,\ x \in \mathbb{R}. \tag{4.14}$$

It will follow from the Markov property (4.15) for a larger filtration (Remark 4.13).

Remark 4.12

$x \mapsto \mathbb{E}_x[Y]$ is Borel measurable. [Supplement] *Reason:* such Y form an MVS (MCT) containing the cylinder indicators, whose $\mathbb{E}_x[\mathbb{1}_A]$ are continuous in x ((4.6), DCT); apply Theorem 2.29. So $\mathbb{E}_{B_s}[Y]$ is $\sigma(B_s) \subset \mathcal{F}_s^0$-measurable, and P_x-a.s.

$$\begin{aligned}\mathbb{E}_x[Y \circ \theta_s \mid \sigma(B_s)] &= \mathbb{E}_x\big[\mathbb{E}_x[Y \circ \theta_s \mid \mathcal{F}_s^0] \,\big|\, \sigma(B_s)\big] = \mathbb{E}_x[\mathbb{E}_{B_s}[Y] \mid \sigma(B_s)] \\ &= \mathbb{E}_{B_s}[Y] = \mathbb{E}_x[Y \circ \theta_s \mid \mathcal{F}_s^0],\end{aligned}$$

using Theorem 2.35(iv), (4.14) and Example 2.34(i). The best guess for the future given the present equals the best guess given the whole past.

For technical reasons we enlarge $\mathcal{F}_t^0$ to

$$\mathcal{F}_t^{0,+} := \bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}^0 \ (\supset \mathcal{F}_t^0),$$

which allows an infinitesimal look into the future. It is **right continuous**: $\bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}^{0,+} = \bigcap_n \bigcap_k \mathcal{F}_{t+\frac{1}{n}+\frac{1}{k}}^0 \subset \mathcal{F}_t^{0,+}$, and trivially $\mathcal{F}_t^{0,+} \subset \bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}^{0,+}$. (B_t) is adapted to $(\mathcal{F}_t^{0,+})$ and has the

Markov property w.r.t. it:

$$\mathbb{E}_x[Y \circ \theta_s \mid \mathcal{F}_s^{0,+}] = \mathbb{E}_{B_s}[Y] \qquad P_x\text{-a.s., } \forall s \geq 0,\ x \in \mathbb{R},\ Y \in \mathcal{F}_b. \tag{4.15}$$

Proof of (4.15).

Step 1 (reduction). Let $\mathcal{C}$ be the cylinder sets $A = \{(B_{t_1}, \dots, B_{t_n}) \in A_1 \times \cdots \times A_n\}$, $0 < t_1 < \cdots < t_n$, $A_i \in \mathcal{B}(\mathbb{R})$; $\mathcal{C}$ is $\cap$-stable and generates $\mathcal{F}$. $\mathcal{H} := \{Y \in \mathcal{F}_b \mid (4.15) \text{ holds}\}$ is an MVS (conditional MCT), so by Theorem 2.29 it suffices to treat $Y = \mathbb{1}_A$, $A \in \mathcal{C}$, i.e. to show

$$\mathbb{E}_x[\mathbb{1}_A \circ \theta_s\, \mathbb{1}_C] = \mathbb{E}_x[\mathbb{E}_{B_s}[\mathbb{1}_A]\, \mathbb{1}_C] \qquad \forall C \in \mathcal{F}_s^{0,+}. \tag{4.16}$$

Step 2 (special C). Fix x, s and $0 < h < t_1$, and take $C = \{(B_{s_1}, \dots, B_{s_m}) \in C_1 \times \cdots \times C_m\}$ with $0 \leq s_1 < \cdots < s_m = s + h$. With $s_0 := 0$, $y_0 := x$, (4.9) gives

$$\mathbb{E}_x[\mathbb{1}_A \circ \theta_s\, \mathbb{1}_C] = \int_{C_1} \cdots \int_{C_m} \prod_{j=1}^{m} p_{s_j - s_{j-1}}(y_{j-1}, y_j)\, F(y_m, h)\, dy_m \cdots dy_1 = \mathbb{E}_x[F(B_{s+h}, h) \mathbb{1}_C], \tag{4.17}$$

where, with $x_0 := y$ and $t_0 := h$,

$$F(y, h) := \int_{A_1} \cdots \int_{A_n} p_{t_1 - h}(y, x_1) \prod_{i=2}^{n} p_{t_i - t_{i-1}}(x_{i-1}, x_i)\, dx_n \cdots dx_1.$$

Step 3 (all $C \in \mathcal{F}_{s+h}^0$). The sets C of Step 2 form a $\cap$-stable generator of $\mathcal{F}_{s+h}^0$, and the class of $\tilde{C} \in \mathcal{F}_{s+h}^0$ satisfying (4.17) is a Dynkin system. By Theorem 1.8,

$$\mathbb{E}_x[\mathbb{1}_A \circ \theta_s\, \mathbb{1}_C] = \mathbb{E}_x[F(B_{s+h}, h) \mathbb{1}_C] \qquad \forall C \in \mathcal{F}_{s+h}^0 \supset \mathcal{F}_s^{0,+}. \tag{4.18}$$

Step 4 ($h \to 0$). By the DCT, F is jointly continuous in $(y, h) \in \mathbb{R} \times (0, t_1)$ and bounded by 1. As $h \to 0$, path continuity gives $F(B_{s+h}(\omega), h) \to \int_{A_1} \cdots \int_{A_n} p_{t_1}(B_s(\omega), x_1) \prod_{i \geq 2} p_{t_i - t_{i-1}}(x_{i-1}, x_i)\, dx = \mathbb{E}_{B_s(\omega)}[\mathbb{1}_A]$. The DCT in (4.18) then yields (4.16) for all $C \in \mathcal{F}_s^{0,+}$. ∎

Remark 4.13

(4.15) implies (4.14): $\mathbb{E}_x[Y \circ \theta_s \mid \mathcal{F}_s^0] = \mathbb{E}_x[\mathbb{E}_x[Y \circ \theta_s \mid \mathcal{F}_s^{0,+}] \mid \mathcal{F}_s^0] = \mathbb{E}_x[\mathbb{E}_{B_s}[Y] \mid \mathcal{F}_s^0] = \mathbb{E}_{B_s}[Y]$ by Theorem 2.35(iv), (4.15) and Example 2.34(i).

Theorem 4.14

For $Z \in \mathcal{F}_b$: $\mathbb{E}_x[Z \mid \mathcal{F}_s^{0,+}] = \mathbb{E}_x[Z \mid \mathcal{F}_s^0]$ P_x-a.s., for all $s \geq 0$, $x \in \mathbb{R}$.

Proof. By the monotone class theorem it suffices to take $Z = \mathbb{1}_A$ with $A = \{(B_{t_1}, \dots, B_{t_n}) \in A_1 \times \cdots \times A_n\}$, $0 \leq t_1 < \cdots < t_n$. Then, P_x-a.s., by Theorem 2.35(iii) and (4.14), (4.15) (both equal $\mathbb{E}_{B_s}[\cdot]$),

$$\begin{aligned}
\mathbb{E}_x[\mathbb{1}_A \mid \mathcal{F}_s^{0,+}] &= \prod_{t_i \leq s} \mathbb{1}_{A_i}(B_{t_i})\, \mathbb{E}_x\Big[\prod_{t_i > s} \mathbb{1}_{A_i}(B_{t_i - s}) \circ \theta_s \,\Big|\, \mathcal{F}_s^{0,+}\Big] \\
&= \prod_{t_i \leq s} \mathbb{1}_{A_i}(B_{t_i})\, \mathbb{E}_x\Big[\prod_{t_i > s} \mathbb{1}_{A_i}(B_{t_i - s}) \circ \theta_s \,\Big|\, \mathcal{F}_s^{0}\Big] = \mathbb{E}_x[\mathbb{1}_A \mid \mathcal{F}_s^0].
\end{aligned}$$

∎

Remark 4.15

$\mathcal{F}_s^{0,+}$ and $\mathcal{F}_s^0$ differ only by P_x-null sets (for every x). For $Z \in \mathcal{F}_s^{0,+}$ bounded, $Z = \mathbb{E}_x[Z \mid \mathcal{F}_s^{0,+}] = \mathbb{E}_x[Z \mid \mathcal{F}_s^0]$ P_x-a.s., an $\mathcal{F}_s^0$-measurable r.v. With $Z = \mathbb{1}_{B^c}$, $B \in \mathcal{F}_s^{0,+}$, the set $B = \{\mathbb{1}_{B^c} < 1\}$ agrees up to a P_x-null set with $B_0 := \{\mathbb{E}_x[\mathbb{1}_{B^c} \mid \mathcal{F}_s^0] < 1\} \in \mathcal{F}_s^0$.

Theorem 4.16 (Blumenthal's 0–1 law)

If $A \in \mathcal{F}_0^{0,+}$, then $P_x(A) \in \{0,1\}$ for every $x \in \mathbb{R}$.

Proof. $\mathcal{F}_0^0 = \sigma(B_0)$ and $P_x(B_0 = x) = 1$, so every $\mathcal{F}_0^0$-measurable Y is P_x-a.s. constant, $Y = \mathbb{E}_x[Y]$. For $A \in \mathcal{F}_0^{0,+}$, by Example 2.34(i) and Theorem 4.14, P_x-a.s.

$$\mathbb{1}_A = \mathbb{E}_x[\mathbb{1}_A \mid \mathcal{F}_0^{0,+}] = \mathbb{E}_x[\mathbb{1}_A \mid \mathcal{F}_0^0] = \mathbb{E}_x[\mathbb{E}_x[\mathbb{1}_A \mid \mathcal{F}_0^0]] = P_x(A),$$

so $P_x(A) \in \{0,1\}$. ∎

Remark 4.17

A general BM as in Definition 4.1 (started at x) also has the Markov property w.r.t. $(\mathcal{F}_t^{0,+})$. Since a shift need not exist, it is formulated as

$$\mathbb{E}_x\Big[\prod_{i=1}^n \mathbb{1}_{A_i}(B_{t_i+s}) \,\Big|\, \mathcal{F}_s^{0,+}\Big] = \mathbb{E}_{B_s}\Big[\prod_{i=1}^n \mathbb{1}_{A_i}(B_{t_i})\Big] \qquad P_x\text{-a.s.}, \tag{4.19}$$

for cylinder sets. The proof of (4.16) used only (4.9) and a.s. continuity, which any BM satisfies (Definition 4.1(i)–(iii)). In particular $\mathbb{E}_x[f(B_{t+s}) \mid \mathcal{F}_s^{0,+}] = \mathbb{E}_{B_s}[f(B_t)]$ for $f \in \mathcal{B}(\mathbb{R})_b$.

4.4 The d-dimensional Brownian motion

Motivation. d independent coordinates; everything carries over.

Let $\|\cdot\|$ be the Euclidean norm and

$$p_t(x,y) := \frac{1}{(\sqrt{2\pi t})^d} e^{-\frac{\|y-x\|^2}{2t}} \quad (x,y \in \mathbb{R}^d,\ t>0), \qquad p_0(x,y)\,dy := \delta_x(dy). \tag{4.20}$$

Section 4.2 goes through with this kernel (Kolmogorov extension for Polish-valued processes; Le Gall, Thm. 6.3). Equivalently, if P_0 is the one-dimensional Wiener measure on $(C(\mathbb{R}_+), \mathcal{B}(C(\mathbb{R}_+)))$ and $P^{(i)} = P_0$, $i = 1,\dots,d$, then $\mathbf{P}_0 := \bigotimes_{i=1}^d P^{(i)}$ is the **d-dimensional Wiener measure** on $\Omega := C(\mathbb{R}_+)^d = C(\mathbb{R}_+, \mathbb{R}^d)$, $\mathcal{F} := \mathcal{B}(C(\mathbb{R}_+)^d)$. Under $\mathbf{P}_0$, $W_t(\omega) := \omega(t) = (\omega_1(t),\dots,\omega_d(t))$ is the canonical d-dimensional BM from 0, and under $P_x := \mathbf{P}_0 \circ \Pi_x^{-1}$, $\Pi_x(\omega) = \omega + x$, it starts at x. With $\mathcal{F}_t^0 := \sigma(W_s \mid s \le t)$ and $\mathcal{F}_t^{0,+} := \bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}^0$ (again filtrations to which W is adapted),

$$P_x((W_{t_1},\dots,W_{t_n}) \in A_1 \times \cdots \times A_n) = \int_{A_1} \cdots \int_{A_n} \prod_{i=1}^n p_{t_i - t_{i-1}}(x_{i-1}, x_i)\,dx_1 \cdots dx_n \tag{4.21}$$

for $0 \le t_1 < \cdots < t_n$, $A_i \in \mathcal{B}(\mathbb{R}^d)$, $x_0 = x$, $t_0 = 0$, with d-dimensional Lebesgue measure dx_i. (4.14), (4.15), Theorem 4.14, Remark 4.15, Theorem 4.16 carry over. The W^i are independent BMs: (4.21) factorises, so $P_x(\bigcap_{i=1}^d \tilde{A}_i) = \prod_i P_x(\tilde{A}_i)$ for cylinder sets $\tilde{A}_i = \{(W_{t_1}^i,\dots,W_{t_n}^i) \in B_1^i \times \cdots \times B_n^i\}$, and Theorem 3.10 extends this to the generated σ-algebras.

Definition 4.18 (d-dimensional Brownian motion)

An $\mathbb{R}^d$-valued process $(W_t)_{t\geq 0}$ on some $(\Omega, \mathcal{F}, P_x)$ is a **d-dimensional BM started at x** if (i) $P_x(W_0 = x) = 1$; (ii) for $0 \leq s < t$, $W_t - W_s \sim N(0, (t-s)I_d)$, i.e. $P_x(W_t - W_s \in A) = \int_A p_{t-s}(0, y)\, dy$; (iii) the increments $W_{t_1} - W_0, \ldots, W_{t_n} - W_{t_{n-1}}$ are independent for $0 \leq t_1 < \cdots < t_n$; (iv) almost all paths are continuous.

Remark 4.19

A d-dimensional BM satisfies (4.21) ($\Longleftrightarrow$ Definition 4.18(i)–(iii)), so has the canonical finite-dimensional laws, with only P_x-a.e. continuous paths. As in Remark 4.17 it has the Markov property

$$\mathbb{E}_x\Big[\prod_{i=1}^{n} \mathbb{1}_{A_i}(W_{t_i+s}) \,\Big|\, \mathcal{F}_s^{0,+}\Big] = \mathbb{E}_{W_s}\Big[\prod_{i=1}^{n} \mathbb{1}_{A_i}(W_{t_i})\Big] \quad P_x\text{-a.s.}, \tag{4.22}$$

and in particular $\mathbb{E}_x[f(W_{t+s}) \mid \mathcal{F}_s^{0,+}] = \mathbb{E}_{W_s}[f(W_t)]$ for $f \in \mathcal{B}(\mathbb{R}^d)_b$.

Chapter 5. The Strong Markov Property

5.1 Stopping times and filtrations

Motivation. Restarting at a random time chosen without looking ahead: stopping times, $\mathcal{F}_T$, and measurability of X_T.

Let $(\Omega, \mathcal{F}, P)$ be a probability space with a **filtration** $(\mathcal{F}_t)_{t\geq 0}$ ($\mathcal{F}_s \subset \mathcal{F}_t \subset \mathcal{F}$ for $s \leq t$); $(\Omega, \mathcal{F}, (\mathcal{F}_t), P)$ is a **filtered probability space**. $(\mathcal{F}_t)$ is **right continuous** if $\mathcal{F}_t = \bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}$ for all t. It satisfies the **usual conditions** if it is right continuous, $(\Omega, \mathcal{F}, P)$ is complete, and every P-null set of $\mathcal{F}$ lies in $\mathcal{F}_0$; then all $(\Omega, \mathcal{F}_t, P)$ are complete.

Remark 5.1

Every probability space can be completed. With $\mathcal{N}_P := \{A \subset \Omega \mid A \subset B \in \mathcal{F},\ P(B) = 0\}$, the **completion** $\mathcal{F}^P := \{C \cup A \mid C \in \mathcal{F},\ A \in \mathcal{N}_P\} = \sigma(\mathcal{F}, \mathcal{N}_P)$ is a σ-algebra, and $P(C \cup A) := P(C)$ extends P to it. [Supplement] *Reason:* $\mathcal{N}_P$ is closed under countable unions; $(C \cup A)^c = (C^c \setminus B) \cup (C^c \cap B \setminus A)$ for null $B \supset A$; representations differ by null subsets, and σ-additivity is inherited.

A process $X = (X_t)$ with values in a metric space S is **adapted** to $(\mathcal{F}_t)$ if each X_t is $\mathcal{F}_t$-measurable. Its **natural filtration** $\mathcal{F}_t^0 = \mathcal{F}_t^0(X) := \sigma(X_s \mid s \leq t)$ (also $\mathcal{F}_t^X$) is the smallest filtration making X adapted. X is **continuous** if $P(t \mapsto X_t \text{ continuous}) = 1$, and it has **continuous sample paths** if every path is continuous.

Example 5.2

In the canonical d-dimensional BM $\mathbb{M}_W = (C(\mathbb{R}_+, \mathbb{R}^d), \mathcal{B}(C(\mathbb{R}_+, \mathbb{R}^d)), (W_t), P_x)$, the filtration $(\mathcal{F}_t^{0,+})$ is right continuous and W has continuous sample paths. For a general d-dimensional BM $(\Omega, \mathcal{F}, (W_t), P_x)$ as in Definition 4.18, let $\mathcal{F}_t^0 := \sigma(W_s \mid s \leq t)$, $\mathcal{F}_t^{0,+} := \bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}^0$, $\mathcal{N}_x := \mathcal{N}_{P_x}$, and

$$\mathcal{F}_t^{0,+,x} := \sigma(\mathcal{F}_t^{0,+}, \mathcal{N}_x), \qquad \mathcal{F}_t^{0,x} := \sigma(\mathcal{F}_t^0, \mathcal{N}_x),$$
$$\mathcal{F}_t^{0,x,+} := \bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}^{0,x}, \qquad \mathcal{F}_\infty^{0,+,x} := \sigma(\mathcal{F}_t^{0,+,x} \mid t \geq 0).$$

Then $\mathcal{F}_t^{0,+,x} = \mathcal{F}_t^{0,x,+}$: completing and making right continuous commute (Rogers–Williams, Vol. 1, II.(67.4)). So $(\Omega, \mathcal{F}_\infty^{0,+,x}, (\mathcal{F}_t^{0,+,x}), P_x)$ satisfies the usual conditions and W is continuous. For the canonical model, as Remark 4.15 suggests (Blumenthal's law holds in d dimensions), even $\mathcal{F}_t^{0,x}$ is right continuous (Karatzas–Shreve, Prop. 2.7.7), so $\mathcal{F}_t^{0,x} = \mathcal{F}_t^{0,x,+} = \mathcal{F}_t^{0,+,x}$. Also $\mathcal{F}_t := \bigcap_{x\in\mathbb{R}^d} \mathcal{F}_t^{0,+,x}$, $\mathcal{F}_\infty := \sigma(\mathcal{F}_t \mid t \geq 0)$: right continuous, differing from $(\mathcal{F}_t^{0,+})$ by sets null for every P_x, so the Markov property holds for it too. Continuous paths plus a right continuous filtration settle the measurability questions below.

Definition 5.3 (Stopping time)

$T : \Omega \to [0, \infty]$ is an $(\mathcal{F}_t)$**-stopping time** if

$$\{T \leq t\} \in \mathcal{F}_t \qquad \forall t \geq 0. \tag{5.1}$$

In particular T is $\mathcal{F}$-measurable.

Lemma 5.4

If $(\mathcal{F}_t)$ is right continuous, (5.1) is equivalent to

$$\{T < t\} \in \mathcal{F}_t \qquad \forall t > 0. \tag{5.2}$$

(A T with (5.2) is an **optional** or **weak stopping time**.)

Proof. For a stopping time, $\{T < t\} = \bigcup_n \{T \le t - \frac{1}{n}\} \in \mathcal{F}_t$. Conversely, under (5.2), $\{T \le t\} = \bigcap_{k \ge n} \{T < t + \frac{1}{k}\} \in \mathcal{F}_{t+\frac{1}{n}}$ for every n, hence $\{T \le t\} \in \bigcap_n \mathcal{F}_{t+\frac{1}{n}} = \mathcal{F}_t$ by right continuity. ∎

Example 5.5

(a) If S, T are stopping times, so are $S \vee T$, $S \wedge T$, $S + T$: $\{S \vee T \le t\} = \{S \le t\} \cap \{T \le t\}$ and $\{S \wedge T \le t\} = \{S \le t\} \cup \{T \le t\}$. For the sum, $\tilde{T} \wedge t$ is $\mathcal{F}_t$-measurable for every stopping time $\tilde{T}$ (Lemma 5.6(ii)), and

$$\{S + T \le t\} = \{S \wedge t + \mathbb{1}_{\{S > t\}} + T \wedge t + \mathbb{1}_{\{T > t\}} \le t\} \in \mathcal{F}_t.$$

(b) For stopping times T_n, $\sup_n T_n$ is a stopping time ($\{\sup_n T_n \le t\} = \bigcap_n \{T_n \le t\}$). $\inf_n T_n$, $\liminf_n T_n$, $\limsup_n T_n$ are in general only weak stopping times ($\{\inf_n T_n < t\} = \bigcup_n \{T_n < t\}$, $\limsup_n T_n = \inf_k \sup_{n \ge k} T_n$, $\liminf_n T_n = \sup_k \inf_{n \ge k} T_n$); they are stopping times if $(\mathcal{F}_t)$ is right continuous.

(c) (*Hitting times*) Let X be adapted with values in a metric space (S, d), and assume either (i) X has continuous sample paths and $(\mathcal{F}_t)$ is right continuous, or (ii) X is continuous and $(\mathcal{F}_t)$ satisfies the usual conditions. Under (ii) we may assume (i), since the null set of discontinuous paths and all its subsets lie in $\mathcal{F}_0$. For $A \subset S$ let

$$\sigma_A := \inf\{t > 0 \mid X_t \in A\} \text{ (first hitting time)},$$
$$D_A := \inf\{t \ge 0 \mid X_t \in A\} \text{ (first entry time)},$$

with $\inf \emptyset := \infty$, so $\{\sigma_A \ge t\} = \{X_s \notin A \ \forall s \in (0, t)\}$ and $\{D_A \ge t\} = \{X_s \notin A \ \forall s \in [0, t)\}$.

A open: by continuity $\sigma_A = D_A$, and since A^c is closed, $\{\sigma_A \ge t\} = \bigcap_{s \in [0,t) \cap \mathbb{Q}} \{X_s \notin A\} \in \mathcal{F}_t$.

A closed: $s \mapsto d(X_s, A)$ is continuous, so it is bounded below by a positive constant on $[0, t]$ whenever it is positive there, and

$$\{D_A > t\} = \{d(X_s, A) > 0 \ \forall s \in [0, t]\} = \bigcup_n \bigcap_{s \in [0,t] \cap \mathbb{Q}} \{d(X_s, A) \ge \tfrac{1}{n}\} \in \mathcal{F}_t.$$

[Supplement] Similarly $\{\sigma_A \ge t\} = \bigcap_m \bigcup_n \bigcap_{s \in \mathbb{Q} \cap [\frac{1}{m}, t - \frac{1}{m}]} \{d(X_s, A) \ge \frac{1}{n}\} \in \mathcal{F}_t$.

In all cases σ_A, D_A are weak stopping times, hence stopping times by Lemma 5.4. (For general Borel A and right continuous paths this remains true under extra conditions on S; the proof is hard, see Blumenthal–Getoor, Ch. I, (10.7).)

(d) (*Dyadic approximation*) For a stopping time T let

$$T_n := \sum_{k \ge 0} (k+1) 2^{-n} \mathbb{1}_{[k2^{-n}, (k+1)2^{-n})}(T) \ \text{ on } \{T < \infty\}, \qquad T_n := \infty \ \text{ on } \{T = \infty\}.$$

Then $T_n \searrow T$ and the T_n are stopping times: for $d \in D_n := \{k2^{-n} \mid k \ge 1\}$, $\{T_n = d\} = \{T < d\} \cap \{T \ge d - 2^{-n}\} \in \mathcal{F}_d$, and $\{T_n \le t\} = \bigcup_{d \in D_n,\, d \le t} \{T_n = d\} \in \mathcal{F}_t$.

For a stopping time T, the **σ-algebra of the T-past** is

$$\mathcal{F}_T := \{A \in \mathcal{F} \mid A \cap \{T \le t\} \in \mathcal{F}_t \ \forall t \ge 0\}.$$

[Supplement] It is a σ-algebra: $\Omega \cap \{T \le t\} \in \mathcal{F}_t$; $A^c \cap \{T \le t\} = \{T \le t\} \setminus (A \cap \{T \le t\})$; unions are clear. For $T \equiv t$, $\mathcal{F}_T = \mathcal{F}_t$. T is $\mathcal{F}_T$-measurable, since $\{T \le s\} \cap \{T \le t\} = \{T \le s \wedge t\} \in \mathcal{F}_t$. If $(\mathcal{F}_t)$ is right continuous, then $\mathcal{F}_T = \{A \in \mathcal{F} \mid A \cap \{T < t\} \in \mathcal{F}_t \ \forall t\}$, by the argument of Lemma 5.4 applied to $A \cap \{T \le t\}$.

Lemma 5.6

(i) For stopping times $S \le T$: $\mathcal{F}_S \subset \mathcal{F}_T$.

(ii) For a stopping time T and $t \ge 0$, $T \wedge t$ is $\mathcal{F}_t$-measurable.

(iii) If T_n are decreasing stopping times w.r.t. a right continuous filtration and $T = \inf_n T_n$, then $\mathcal{F}_T = \bigcap_n \mathcal{F}_{T_n}$.

Proof. (i) For $A \in \mathcal{F}_S$: $\{T \le t\} \subset \{S \le t\}$, so $A \cap \{T \le t\} = (A \cap \{S \le t\}) \cap \{T \le t\} \in \mathcal{F}_t$.

(ii) $\{T \wedge t \le t'\}$ is Ω if $t \le t'$, and $\{T \le t'\} \in \mathcal{F}_{t'} \subset \mathcal{F}_t$ if $t' < t$.

(iii) $\mathcal{F}_T \subset \bigcap_n \mathcal{F}_{T_n}$ by (i). Conversely, if $A \in \bigcap_n \mathcal{F}_{T_n}$, then $A \cap \{T_n < t\} \in \mathcal{F}_t$ for all n (right continuity), so $A \cap \{T < t\} = A \cap \bigcup_n \{T_n < t\} \in \mathcal{F}_t$ for all t, i.e. $A \in \mathcal{F}_T$. ∎

For $T(\omega) < \infty$ put $X_T(\omega) := X_{T(\omega)}(\omega)$; for a finite stopping time this defines X_T on Ω. For instance, $T \wedge t$ is a finite stopping time.

Lemma 5.7

Let X be adapted with values in a metric space, under the conditions of Example 5.5(c)(i) or (ii), and T a finite stopping time. Then X_T is $\mathcal{F}_T$-measurable. In particular $X_{t\wedge T}$ is $\mathcal{F}_{t\wedge T} \subset \mathcal{F}_t$-measurable, so $(X_{t\wedge T})_{t\ge 0}$ is adapted.

Proof.

Step 1 (dyadic times). Take $T_n \searrow T$ as in Example 5.5(d) and $B \in \mathcal{B}(S)$. $\{X_{T_n} \in B\} = \bigcup_k \{T_n = (k+1)2^{-n}\} \cap \{X_{(k+1)2^{-n}} \in B\} \in \mathcal{F}$, and

$$\{X_{T_n} \in B\} \cap \{T_n \le t\} = \bigcup_{d \in D_n,\, d \le t} \{X_d \in B\} \cap \{T_n = d\} \in \mathcal{F}_t,$$

so $X_{T_n} \in \mathcal{F}_{T_n}$.

Step 2 (limit). With continuous paths, $X_{T_n} \to X_T$ pointwise. For each n, $X_T = \lim_{k \ge n} X_{T_k}$ is $\mathcal{F}_{T_n}$-measurable (as $\mathcal{F}_{T_k} \subset \mathcal{F}_{T_n}$ for $k \ge n$), so $X_T \in \bigcap_n \mathcal{F}_{T_n} = \mathcal{F}_T$ by Lemma 5.6(iii). ∎

Now let X take values in a Borel set $S \subset \mathbb{R}^d$ with $0 \in S$. For a (not necessarily finite) stopping time T define $\mathbb{1}_{\{T<\infty\}} X_T := X_T$ on $\{T < \infty\}$ and $:= 0$ on $\{T = \infty\}$.

Lemma 5.8

Under the assumptions of Lemma 5.7 (with $S \subset \mathbb{R}^d$ Borel, $0 \in S$), $\mathbb{1}_{\{T<\infty\}} X_T$ is $\mathcal{F}_T$-measurable.

Proof. By Lemmas 5.6(i) and 5.7, $\mathbb{1}_{\{T<\infty\}} X_{T\wedge n} \in \mathcal{F}_T$ for each n, and $\mathbb{1}_{\{T<\infty\}} X_{T\wedge n} \to \mathbb{1}_{\{T<\infty\}} X_T$ pointwise. ∎

Remark 5.9

Example 5.5(c) remains true with "continuous" replaced by "right continuous". Since $T_n \searrow T$ in Example 5.5(d), Lemmas 5.7 and 5.8 hold for right continuous paths as well.

5.2 Markov processes and the strong Markov property

Motivation. Right continuous paths and the Feller property extend the Markov property to stopping times.

Throughout, $S \subset \mathbb{R}^d$ is Borel with $0 \in S$, equipped with $\mathcal{B}(S)$.

Definition 5.10 (Markovian semigroup)

Maps $P_t : S \times \mathcal{B}(S) \to [0,1]$, $t \geq 0$, form a **markovian semigroup of transition kernels** if (i) $P_t(x,\cdot)$ is a probability measure for all t, x; (ii) $x \mapsto P_t(x,A)$ is $\mathcal{B}(S)$-measurable; (iii) (Chapman–Kolmogorov) $P_t P_s = P_{t+s}$, i.e. $\int_S P_s(y,A)\, P_t(x,dy) = P_{t+s}(x,A)$; (iv) $P_0(x,\cdot) = \delta_x$. For bounded (or nonnegative) measurable f put

$$P_t f(x) := \int_S f(y)\, P_t(x,dy), \tag{5.3}$$

so that $P_t \mathbb{1}_A(x) = P_t(x,A)$.

Definition 5.11 (Normal Markov process)

$\mathbb{M} = (\Omega, \mathcal{F}, (X_t)_{t\geq 0}, (P_x)_{x\in S})$ with state space $(S, \mathcal{B}(S))$ is a **normal Markov process** w.r.t. a filtration $(\mathcal{F}_t)$ if

(M1) $x \mapsto P_x(F)$ is $\mathcal{B}(S)$-measurable for every $F \in \mathcal{F}$;

(M2) (X_t) is $(\mathcal{F}_t)$-adapted and $P_x(X_{s+t} \in A \mid \mathcal{F}_s) = P_{X_s}(X_t \in A)$ P_x-a.s. for all $s,t \geq 0$, $A \in \mathcal{B}(S)$, $x \in S$ (Markov property), where $P_x(\cdot \mid \mathcal{F}_s) = \mathbb{E}_x[\mathbb{1}_A(X_{s+t}) \mid \mathcal{F}_s]$;

(M3) $P_x(X_0 = x) = 1$ for all x (normality).

Theorem 5.12

(i) Let $\mathbb{M} = (\Omega, \mathcal{F}, (X_t), (P_x)_{x\in S})$ with $\mathcal{F} = \sigma(X_t \mid t \geq 0)$.

(a) If there is a markovian semigroup (P_t) with, for all $0 \leq t_1 < \cdots < t_n$, $A_i \in \mathcal{B}(S)$, $t_0 := 0$, $x_0 := x$,

$$P_x((X_{t_1}, \ldots, X_{t_n}) \in A_1 \times \cdots \times A_n) = \int_{A_1} \cdots \int_{A_n} \prod_{i=1}^{n} P_{t_i - t_{i-1}}(x_{i-1}, dx_i), \tag{5.4}$$

then $\mathbb{M}$ is a normal Markov process w.r.t. $(\mathcal{F}_t^0) = (\sigma(X_s \mid s \leq t))$.

(b) If $\mathbb{M}$ is a normal Markov process, then

$$P_t(x,A) := \mathbb{E}_x[\mathbb{1}_A(X_t)] = P_x(X_t \in A) \tag{5.5}$$

is a markovian semigroup and (5.4) holds.

(ii) For every markovian semigroup there is a normal Markov process satisfying (5.5), with $\Omega = S^{[0,\infty)}$.

Proof. (i)(a) (M3) is (5.4) with $n = 1$, $t_1 = 0$. (M1) holds for cylinder sets by (5.4) and Definition 5.10(ii) (iterated kernel integrals of measurable functions are measurable), and extends to $\mathcal{F}$ by a Dynkin argument. (M2) is equivalent to

$$\mathbb{E}_x[\mathbb{1}_A(X_{t+s})\mathbb{1}_C] = \mathbb{E}_x[\mathbb{E}_{X_s}[\mathbb{1}_A(X_t)]\mathbb{1}_C] \qquad \forall C \in \mathcal{F}_s^0.$$

For $s = 0$ this is clear ($\mathcal{F}_0^0 = \sigma(X_0)$ is P_x-trivial). For $s > 0$, by the monotone class theorem it suffices to take $C = \{(X_{s_1}, \dots, X_{s_m}) \in C_1 \times \cdots \times C_m\}$, $0 \le s_1 < \cdots < s_m = s$. With $s_{m+1} := t + s$, $y_0 := x$, (5.4) gives

$$\begin{aligned}\mathbb{E}_x[\mathbb{1}_A(X_{t+s})\mathbb{1}_C] &= \int_{C_1} \cdots \int_{C_m} \int_A \prod_{j=1}^{m+1} P_{s_j - s_{j-1}}(y_{j-1}, dy_j) \\ &= \int_{C_1} \cdots \int_{C_m} \mathbb{E}_{y_m}[\mathbb{1}_A(X_t)] \prod_{j=1}^{m} P_{s_j - s_{j-1}}(y_{j-1}, dy_j),\end{aligned}$$

which by (5.4) again equals $\mathbb{E}_x[\mathbb{E}_{X_s}[\mathbb{1}_A(X_t)]\mathbb{1}_C]$.

(i)(b) Definition 5.10(i), (ii),

(iv) are clear from (M1), (M3). By (5.5) and (5.3), $P_t\mathbb{1}_A(x) = \mathbb{E}_x[\mathbb{1}_A(X_t)]$, and by linearity and monotone convergence

$$P_t f(x) = \mathbb{E}_x[f(X_t)] \qquad \text{for bounded or nonnegative measurable } f. \tag{5.6}$$

Chapman–Kolmogorov: with (M2) and (5.6),

$$P_{t+s}(x, A) = \mathbb{E}_x[\mathbb{E}_x[\mathbb{1}_A(X_{t+s}) \mid \mathcal{F}_s]] = \mathbb{E}_x[P_t(X_s, A)] = P_s(P_t\mathbb{1}_A)(x) = \int P_t(y, A)\, P_s(x, dy).$$

For (5.4), it suffices to prove

$$\mathbb{E}_x[f_1(X_{t_1}) \cdots f_n(X_{t_n})] = \int \cdots \int f_1(x_1) \cdots f_n(x_n) \prod_{i=1}^{n} P_{t_i - t_{i-1}}(x_{i-1}, dx_i) \tag{5.7}$$

for bounded measurable f_i, by induction on n; $n = 1$ is (5.6). Conditioning on $\mathcal{F}_{t_{n-1}}$, Theorem 2.35(iii) and (M2) give

$$\mathbb{E}_x\Big[\prod_{i \le n} f_i(X_{t_i})\Big] = \mathbb{E}_x\Big[\prod_{i \le n-1} f_i(X_{t_i})\, P_{t_n - t_{n-1}} f_n(X_{t_{n-1}})\Big],$$

and the induction hypothesis applied to $f_1, \dots, f_{n-2}$, $f_{n-1} \cdot P_{t_n - t_{n-1}} f_n$ yields (5.7).

(ii) The family (5.4) is consistent by Chapman–Kolmogorov. Kolmogorov's extension theorem, whose proof (Theorem 4.7) works verbatim for $\mathbb{R}^d$-valued coordinates (see also Le Gall, Thm. 6.3; Revuz–Yor, Ch. I, (3.2)), gives P_x on the path space. (M1) follows as in (i)(a), and (i)(a) does the rest. ∎

The canonical model. Let $\Omega \subset S^{[0,\infty)}$ (e.g. $S^{[0,\infty)}$ or $C([0,\infty), S)$), $X_t(\omega) := \omega(t)$, $\mathcal{F} := \sigma(X_t \mid t \ge 0)$, $\mathcal{F}_t^0 := \sigma(X_s \mid s \le t)$, and let P_x be probability measures with $P_x(X_0 = x) = 1$. Then $\mathbb{M} = (\Omega, \mathcal{F}, (X_t), (P_x))$ is the **canonical model** of (X_t). The shifts $\theta_s(\omega)(t) := \omega(s + t)$ are $\hat{\mathcal{F}}_s^0/\mathcal{F}$-measurable with $\hat{\mathcal{F}}_t^0 := \sigma(X_s \mid s \ge t)$. If $\mathbb{M}$ is a normal Markov process w.r.t. $(\mathcal{F}_t)$ (hence also w.r.t. $(\mathcal{F}_t^0)$), we call $\mathbb{M} = (\Omega, \mathcal{F}, (\mathcal{F}_t), (X_t), (P_x), (\theta_t))$ a **canonical Markov process**.

Lemma 5.13

Let the canonical model satisfy (M1). Then (M2) holds w.r.t. $\mathcal{F}_t := \mathcal{F}_t^{0,+} = \bigcap_{\varepsilon > 0} \mathcal{F}_{t+\varepsilon}^0$ if and

only if

$$\mathbb{E}_x[Y\circ\theta_s \mid \mathcal{F}_s] = \mathbb{E}_{X_s}[Y] \qquad P_x\text{-a.s., } \forall s\ge 0,\ x\in S,\ Y\in\mathcal{F}_b. \tag{5.8}$$

As in Remark 4.12, (5.8) implies $\mathbb{E}_x[Y\circ\theta_s \mid \mathcal{F}_s] = \mathbb{E}_x[Y\circ\theta_s \mid \sigma(X_s)]$.

Proof. (M3) holds by definition of the canonical model. "⇐": take $Y=\mathbb{1}_A(X_t)$. "⇒" [Supplement]: by Theorem 5.12(i)(b), (5.4) holds with the semigroup (5.5). For $Y=\prod_{i\le n} f_i(X_{t_i})$, $0\le t_1<\cdots<t_n$, condition successively on $\mathcal{F}_{s+t_{n-1}}, \mathcal{F}_{s+t_{n-2}},\dots$ and use (M2) and Theorem 2.35(iii),(iv), exactly as in the induction for (5.7). The result is

$$\mathbb{E}_x[Y\circ\theta_s \mid \mathcal{F}_s] = \Big[P_{t_1}\big(f_1 P_{t_2-t_1}(f_2\cdots P_{t_n-t_{n-1}} f_n)\big)\Big](X_s),$$

and by (5.7) the bracket evaluated at y is $\mathbb{E}_y[Y]$. The class of $Y\in\mathcal{F}_b$ satisfying (5.8) is an MVS containing these products, so Theorem 2.29 gives (5.8). ∎

For a stopping time T put $(\theta_T\omega)(t) := \omega(T(\omega)+t)$ on $\{T<\infty\}$. (In Markov process theory one sets $\theta_T\omega\equiv\Delta$ on $\{T=\infty\}$, with a cemetery point Δ and $f(\Delta):=0$.)

Definition 5.14 (Strong Markov property)

A canonical Markov process $\mathbb{M}$ with right continuous paths and right continuous filtration $(\mathcal{F}_t)$ has the **strong Markov property** (SMP) w.r.t. $(\mathcal{F}_t)$ if for every $(\mathcal{F}_t)$-stopping time T,

$$\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}} Y\circ\theta_T \mid \mathcal{F}_T] = \mathbb{1}_{\{T<\infty\}}\mathbb{E}_{X_T}[Y] \qquad P_x\text{-a.s., } \forall Y\in\mathcal{F}_b,\ x\in S.$$

The right side, $\mathbb{1}_{\{T<\infty\}}\mathbb{E}_{\mathbb{1}_{\{T<\infty\}}X_T}[Y]$, is $\mathcal{F}_T$-measurable by Lemma 5.8, (M1) and Remark 5.9. For finite T the SMP reads $\mathbb{E}_x[Y\circ\theta_T \mid \mathcal{F}_T] = \mathbb{E}_{X_T}[Y]$.

Theorem 5.15

Let $\mathbb{M} = (\Omega, \mathcal{F}=\sigma(X_t \mid t\ge 0), (\mathcal{F}_t), (X_t), (P_x), (\theta_t))$ be a canonical Markov process w.r.t. $\mathcal{F}_t := \mathcal{F}_t^{0,+}$ such that (i) $\mathbb{M}$ has right continuous sample paths, and (ii) the semigroup (5.5) has the **Feller property** $P_t(C_b(S))\subset C_b(S)$. Then $\mathbb{M}$ has the SMP w.r.t. $(\mathcal{F}_t)$.

Proof.

Step 1 (reduction). Fix x. By the monotone class theorem it suffices to take $Y = f_1(X_{t_1})\cdots f_m(X_{t_m})$ with $f_i\in C_b(S)$, $0\le t_1<\cdots<t_m$. Let $T_n\searrow T$ be the dyadic approximations of Example 5.5(d) and $C\in\mathcal{F}_T$.

Step 2 (discretise). By right continuity of paths, $Y\circ\theta_{T_n} = \prod_i f_i(X_{t_i+T_n}) \to Y\circ\theta_T$ on $\{T<\infty\}$, so by the DCT

$$\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}} Y\circ\theta_T \mathbb{1}_C] = \lim_n \sum_{k\ge 1} \mathbb{E}_x\big[\mathbb{1}_{C\cap\{(k-1)2^{-n}\le T<k2^{-n}\}}\, Y\circ\theta_{k2^{-n}}\big].$$

Here $C\cap\{(k-1)2^{-n}\le T<k2^{-n}\}\in\mathcal{F}_{k2^{-n}}$, since $C\cap\{T<k2^{-n}\}\in\mathcal{F}_{k2^{-n}}$ by right continuity and $\{T\ge(k-1)2^{-n}\}\in\mathcal{F}_{(k-1)2^{-n}}$.

Step 3 (Markov at fixed times). By (5.8) (Lemma 5.13) each summand equals

$$\mathbb{E}_x\big[\mathbb{1}_{C\cap\{(k-1)2^{-n}\le T<k2^{-n}\}}\,\mathbb{E}_{X_{k2^{-n}}}[Y]\big],$$

so the sum is $\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}}\mathbb{E}_{X_{T_n}}[Y]\mathbb{1}_C]$.

Step 4 (Feller). $x\mapsto \mathbb{E}_x[Y] = P_{t_1}(f_1 P_{t_2-t_1}(f_2\cdots P_{t_m-t_{m-1}} f_m))(x)$ is continuous by the Feller property, and $X_{T_n}\to X_T$ on $\{T<\infty\}$. The DCT gives the limit $\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}}\mathbb{E}_{X_T}[Y]\mathbb{1}_C]$. ∎

Time-dependent functionals. Let $Y : \mathbb{R} \times \Omega \to \mathbb{R}$ be bounded and $\mathcal{B}(\mathbb{R}) \otimes \mathcal{F}$-measurable, and define $\mathbb{1}_{\{T<\infty\}}\mathbb{E}_{X_T}[Y(T,\cdot)]$ as

$$\omega \mapsto \mathbb{1}_{\{T<\infty\}}(\omega)\, \mathbb{E}_{\mathbb{1}_{\{T<\infty\}}X_T(\omega)}[Y(T(\omega),\cdot)], \tag{5.9}$$

and $(Y(T,\cdot)\circ\theta_T)(\omega) := Y(T(\omega),\theta_T\omega)$ (the shift acts only on the second variable). If Y does not depend on time, these reduce to $\mathbb{1}_{\{T<\infty\}}\mathbb{E}_{X_T}[Y]$ and $Y\circ\theta_T$.

Definition 5.16 (Extended strong Markov property)

$\mathbb{M}$ as in Definition 5.14 has the **extended strong Markov property** (ESMP) w.r.t. $(\mathcal{F}_t)$ if for every stopping time T, $Y \in (\mathcal{B}(\mathbb{R})\otimes\mathcal{F})_b$ and $x \in S$,

$$\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}}Y(T,\cdot)\circ\theta_T \mid \mathcal{F}_T] = \mathbb{1}_{\{T<\infty\}}\mathbb{E}_{X_T}[Y(T,\cdot)] \qquad P_x\text{-a.s.}$$

[Supplement] The right side (5.9) is $\mathcal{F}_T$-measurable. For Y as in (5.10) below it equals $\mathbb{1}_{\{T<\infty\}}\Phi(T,X_T)$ with Φ continuous (Feller), and T, $\mathbb{1}_{\{T<\infty\}}X_T$ are $\mathcal{F}_T$-measurable (Lemma 5.8). The monotone class argument at the end of the next proof extends this to all bounded Y.

Theorem 5.17

Under the assumptions of Theorem 5.15 ($\mathcal{F}_t = \mathcal{F}_t^{0,+}$, right continuous paths, Feller property), $\mathbb{M}$ has the ESMP w.r.t. $(\mathcal{F}_t)$.

Proof.

Step 1 (product functionals). Fix x and first let

$$Y(s,\omega) = g(s)f_1(X_{t_1}(\omega))\cdots f_m(X_{t_m}(\omega)), \qquad g \in C_b(\mathbb{R}),\ f_i \in C_b(S). \tag{5.10}$$

With T_n, $C \in \mathcal{F}_T$ as before, right continuity and the DCT give $\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}}Y(T,\theta_T)\mathbb{1}_C] = \lim_n \mathbb{E}_x[\mathbb{1}_{\{T<\infty\}}Y(T_n,\theta_{T_n})\mathbb{1}_C]$. Splitting according to $\{(k-1)2^{-n} \le T < k2^{-n}\} \in \mathcal{F}_{k2^{-n}}$, on which $Y(T_n,\cdot) = Y(k2^{-n},\cdot) \in \mathcal{F}_b$, the Markov property at the fixed times $k2^{-n}$ turns this into $\lim_n \mathbb{E}_x[\mathbb{1}_{\{T<\infty\}}\mathbb{E}_{X_{T_n}}[Y(T_n,\cdot)]\mathbb{1}_C]$. Since $(s,y) \mapsto \mathbb{E}_y[Y(s,\cdot)] = g(s)P_{t_1}(f_1P_{t_2-t_1}(\cdots P_{t_m-t_{m-1}}f_m))(y)$ is continuous (Feller), the DCT gives $\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}}\mathbb{E}_{X_T}[Y(T,\cdot)]\mathbb{1}_C]$.

Step 2 (monotone class). The $Y \in (\mathcal{B}(\mathbb{R})\otimes\mathcal{F})_b$ satisfying the ESMP form an MVS. The multiplicative class $\mathcal{M}$ of indicators of $U_0 \times \{(X_{t_1},\dots,X_{t_m}) \in U_1\times\cdots\times U_m\}$, U_j open, generates $\mathcal{B}(\mathbb{R})\otimes\mathcal{F}$. Each such indicator is an increasing limit of functions of the form (5.10) (use $\phi(m\,d(\cdot,U^c))$-type cut-offs as in Theorem 3.35, (ii)⇒(iii)), so $\mathcal{M} \subset \mathcal{H}$ and Theorem 2.29 concludes. ∎

Example 5.18

The canonical d-dimensional Brownian motion $\mathbb{M}_W$ has the Markov property w.r.t. $(\mathcal{F}_t^{0,+})$ (Section 4.4), continuous paths, and the Feller property: $P_tf(x) = \int f(y)p_t(x,y)\,dy$ is continuous in x for $f \in C_b$ by the DCT. So it has the SMP and ESMP w.r.t. $(\mathcal{F}_t^{0,+})$ and $(\mathcal{F}_t)$ (Example 5.2).

Remark 5.19

Going through the proofs of Theorems 5.15, 5.17 (cf. Remarks 4.17, 4.19), a d-dimensional BM as in Definition 4.18 satisfies the ESMP for fixed x w.r.t. $(\mathcal{F}_t^{0,+,x})$ (needed for measurability, as it satisfies the usual conditions), for Y of the form (5.10) with $Y(T,\cdot)\circ\theta_T$ replaced by

$g(T)f_1(W_{t_1+T})\cdots f_m(W_{t_m+T})$. In particular, for $f \in \mathcal{B}(\mathbb{R}^d)_b$ and every $(\mathcal{F}_t^{0,+,x})$-stopping time T,

$$\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}} f(W_{t+T}) \mid \mathcal{F}_T^{0,+,x}] = \mathbb{1}_{\{T<\infty\}} \mathbb{E}_{W_T}[f(W_t)] \qquad P_x\text{-a.s.} \tag{5.11}$$

If T is an $(\mathcal{F}_t^{0,+})$-stopping time (hence an $(\mathcal{F}_t^{0,+,x})$-stopping time for every x), (5.11) holds simultaneously for all x.

5.3 Applications to Brownian motion

Motivation. Restart at hitting times plus symmetry: reflection principle, maximum, arcsine law; time inversion for 0 and ∞.

5.3.1 The reflection principle for one-dimensional BM

Let $\mathbb{M}_B = (\Omega, \mathcal{F}, (\mathcal{F}_t^{0,+}), (B_t), P_0, (\theta_t))$ be the canonical one-dimensional BM; it has the ESMP (Example 5.18). For $a, t > 0$ we show

$$P_0(D_{\{a\}} < t,\ B_t > a) = \tfrac{1}{2} P_0(D_{\{a\}} < t). \tag{5.12}$$

Let $Y(s,\omega) := \mathbb{1}_{(-\infty,t)}(s)\mathbb{1}_{(a,\infty)}(B_{(t-s)^+}(\omega))$ (bounded, $\mathcal{B}(\mathbb{R}) \otimes \mathcal{F}$-measurable) and $T := \inf\{0 \le s < t \mid B_s = a\}$ (a stopping time, Example 5.5(c)). Then $\{T < \infty\} = \{T < t\} = \{D_{\{a\}} < t\}$ and $\mathbb{1}_{\{T<\infty\}} Y(T, \theta_T) = \mathbb{1}_{\{D_{\{a\}}<t\}} \mathbb{1}_{\{B_t > a\}}$. Taking expectations in the ESMP (Theorem 5.17), and using $B_T = a$ and $\mathbb{E}_a[\mathbb{1}_{\{B_r>a\}}] = \frac{1}{2}$ for $r > 0$ (symmetry),

$$P_0(D_{\{a\}} < t,\ B_t > a) = \mathbb{E}_0[\mathbb{1}_{\{T<t\}} \mathbb{E}_{B_T}[\mathbb{1}_{\{B_{t-T}>a\}}]] = \tfrac{1}{2} P_0(T < t),$$

which is (5.12). Since $B_0 = 0 < a$, path continuity gives $\{B_t > a\} \subset \{D_{\{a\}} < t\}$, hence

$$P_0(D_{\{a\}} < t) = 2P_0(B_t > a), \tag{5.13}$$

and, by continuity from above,

$$P_0(D_{\{a\}} \le t) = \lim_n P_0(D_{\{a\}} < t + \tfrac{1}{n}) = 2 \lim_n P_0(B_{t+\frac{1}{n}} > a) = \frac{2}{\sqrt{2\pi t}} \int_a^\infty e^{-\frac{x^2}{2t}}\, dx = 2P_0(B_t \ge a). \tag{5.14}$$

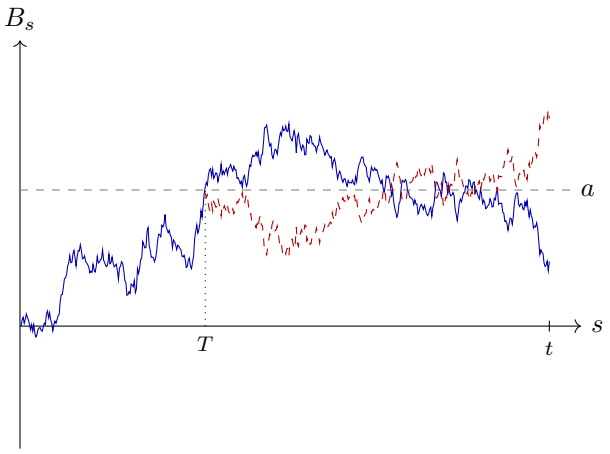


Figure. *Reflection after $T = D_{\{a\}}$:* $P_0(D_{\{a\}} < t) = 2P_0(B_t > a)$.

Proposition 5.20

(i) For $a \in \mathbb{R}\setminus\{0\}$ and $t \geq 0$: $P_0(D_{\{a\}} \leq t) = \int_0^t \frac{|a|}{\sqrt{2\pi s^3}} e^{-\frac{a^2}{2s}}\, ds$. (ii) For $a, t > 0$: $P_0(\sup_{s\leq t} B_s \leq a) = P_0(|B_t| \leq a) = \frac{2}{\sqrt{2\pi t}} \int_0^a e^{-\frac{x^2}{2t}}\, dx$.

Proof. (i) For $a, t > 0$ substitute $x = t^{1/2} a\, s^{-1/2}$ in (5.14):

$$P_0(D_{\{a\}} \leq t) = \frac{2}{\sqrt{2\pi t}} \int_a^\infty e^{-\frac{x^2}{2t}}\, dx = 2\int_0^t \frac{1}{\sqrt{2\pi t}} \frac{t^{1/2} a}{2 s^{3/2}} e^{-\frac{a^2}{2s}}\, ds = \int_0^t \frac{a}{\sqrt{2\pi s^3}} e^{-\frac{a^2}{2s}}\, ds.$$

For $a < 0$ use that $(-B_t)$ is again a canonical BM; $t = 0$ is trivial.

(ii) For $a, t > 0$, $\{D_{\{a\}} > t\} = \{\sup_{s\leq t} B_s < a\}$. By continuity from above and (5.13),

$$P_0\Big(\sup_{s\leq t} B_s \leq a\Big) = \lim_n P_0(D_{\{a+\frac{1}{n}\}} > t) = 1 - 2\lim_n P_0(B_t \geq a + \tfrac{1}{n}) = 1 - 2P_0(B_t > a) = P_0(|B_t| \leq a).$$

■

Remark 5.21

Proposition 5.20 and the arcsine law below hold for every BM $(\tilde{B}_t)$ on $(\tilde{\Omega}, \tilde{\mathcal{F}}, \tilde{P}_0)$ as in Definition 4.1. A shift-free strong Markov property gives this directly (Le Gall, Ch. 2.4); alternatively, the finite-dimensional distributions agree with those of $\mathbb{M}_B$, and the events involved are determined by countably many times:

$$\tilde{P}_0\Big(\sup_{s\leq t} \tilde{B}_s \leq a\Big) = \tilde{P}_0\Big(\sup_{s\in\mathbb{Q}\cap[0,t]} \tilde{B}_s \leq a\Big) = P_0\Big(\sup_{s\in\mathbb{Q}\cap[0,t]} B_s \leq a\Big) = P_0\Big(\sup_{s\leq t} B_s \leq a\Big).$$

Similarly (with a filtration satisfying the usual conditions) $\tilde{P}_0(\tilde{D}_{\{a\}} \leq t) = P_0(D_{\{a\}} \leq t)$, because $\{D_{\{a\}} > t\} = \bigcup_n \bigcap_{s\in[0,t]\cap\mathbb{Q}} \{|B_s - a| \geq \frac{1}{n}\}$, and likewise for $\tilde{B}$.

Example 5.22 (Arcsine law)

Let $L := \sup\{t \leq 1 \mid B_t = 0\}$, the last zero before time 1. L is not a stopping time, but its law can be computed. For $s \in [0,1)$, $\{L \leq s\} = \{B_t \neq 0\ \forall t \in (s,1]\}$, and $P_0(L \leq 1) = 1$. By the Markov property,

$$P_0(L \leq s) = \mathbb{E}_0[\mathbb{E}_0[\mathbb{1}_{\{B_{t-s}\neq 0\ \forall t\in(s,1]\}} \circ \theta_s \mid \mathcal{F}_s^{0,+}]] = \mathbb{E}_0[f(B_s)] = \int p_s(0,y)\, P_y(\sigma_{\{0\}} > 1-s)\, dy,$$

with $f(y) := P_y(B_t \neq 0\ \forall t \in (0, 1-s])$. Since $P_y(\sigma_{\{0\}} > 1-s) = P_y(D_{\{0\}} > 1-s) = P_0(D_{\{-y\}} > 1-s)$, Proposition 5.20(i) and symmetry give

$$\begin{aligned} P_0(L \leq s) &= 2\int_0^\infty \frac{e^{-y^2/(2s)}}{\sqrt{2\pi s}} \Big(\int_{1-s}^\infty \frac{y}{\sqrt{2\pi r^3}} e^{-\frac{y^2}{2r}}\, dr\Big) dy = \frac{1}{\pi}\int_{1-s}^\infty \frac{1}{\sqrt{sr^3}} \int_0^\infty y e^{-\frac{(r+s)y^2}{2rs}}\, dy\, dr \\ &= \frac{1}{\pi}\int_{1-s}^\infty \frac{1}{\sqrt{sr^3}} \frac{rs}{r+s}\, dr = \frac{1}{\pi}\int_{1-s}^\infty \sqrt{\frac{r}{s}}\, \frac{s}{(r+s)^2}\, dr. \end{aligned}$$

The substitution $t = \frac{s}{r+s}$, $dt = -\frac{s}{(r+s)^2} dr$, gives

$$P_0(L \leq s) = \frac{1}{\pi}\int_0^s \frac{dt}{\sqrt{t(1-t)}} = \frac{2}{\pi} \arcsin\sqrt{s}, \qquad s \in [0,1].$$

The density $\frac{1}{\pi\sqrt{t(1-t)}}$ is largest near 0 and 1: the last zero before time 1 is most likely to occur either very early or very late.

5.3.2 Further path properties of BM

Theorem 5.23 (Time inversion)

If (B_t) is a standard BM, so is $X_t := tB_{1/t}$ for $t > 0$, $X_0 := 0$.

Proof.

Step 1 (Definition 4.1(i), (ii)). (i) is clear. For $0 < s < t$, $0 < \frac{1}{t} < \frac{1}{s}$ and

$$X_t - X_s = (t-s)\underbrace{B_{1/t}}_{=:Y_1} - s\underbrace{(B_{1/s} - B_{1/t})}_{=:Y_2},$$

with independent $Y_1 \sim N(0, \frac{1}{t})$, $Y_2 \sim N(0, \frac{1}{s} - \frac{1}{t})$. By Example 3.44(i), $X_t - X_s \sim N\big(0, (t-s)^2\frac{1}{t} + s^2(\frac{1}{s} - \frac{1}{t})\big) = N(0, t-s)$. The case $s = 0$ is similar.

Step 2 (Definition 4.1(iii)). For $s, t > 0$, $\mathbb{E}[X_s X_t] = st(\frac{1}{s} \wedge \frac{1}{t}) = s \wedge t$, and this also holds if s or t is 0. [Supplement] Increments $Z = (X_{t_1}, X_{t_2} - X_{t_1}, \dots)$ are MG, G independent standard normals (normalised increments of B at $1/t_i$), so by Theorem 3.52 $\varphi_Z(u) = \mathbb{E}[e^{i\langle M^\top u, G\rangle}] = e^{-\frac{1}{2}\|M^\top u\|^2} = e^{-\frac{1}{2}u^\top \Sigma u}$ with $\Sigma = MM^\top$ the covariance matrix of Z. From $\mathbb{E}[X_s X_t] = s \wedge t$, Σ is diagonal with entries $t_i - t_{i-1}$. So $\varphi_Z(u) = \prod_i e^{-\frac{1}{2}(t_i - t_{i-1})u_i^2} = \prod_i \varphi_{Z_i}(u_i)$, and Remark 3.50 gives independence.

Step 3 (continuity). By Steps 1–2, X satisfies (4.9), so X and B have the same finite-dimensional distributions. Paths of X are continuous on $(0, \infty)$ a.s. (everywhere if B is canonical). At 0: the laws of X and B on $\mathbb{Q} \cap (0, \infty)$ agree, and $\{\lim_{t \downarrow 0,\, t \in \mathbb{Q}} X_t = 0\}$ is determined by countably many coordinates, so $P(\lim_{t \downarrow 0,\, t \in \mathbb{Q}} X_t = 0) = P(\lim_{t \downarrow 0,\, t \in \mathbb{Q}} B_t = 0) = 1$. With continuity on $(0, \infty)$ this gives $\lim_{t \downarrow 0} X_t = 0$ a.s. ∎

Remark 5.24

Time inversion links the behaviour of B_t as $t \to \infty$ with that of X_t as $t \to 0$. For example $B_t/t = X_{1/t} \to 0$, so paths grow slower than linearly; Theorem 5.26 says more. From now on $\mathbb{M}_B = (\Omega, \mathcal{F}, (B_t), (P_x), (\theta_t))$ is the canonical one-dimensional BM, $\hat{\mathcal{F}}_t^0 := \sigma(B_s \mid s \geq t)$ (decreasing in t) and

$$\mathcal{T} := \bigcap_{t \geq 0} \hat{\mathcal{F}}_t^0$$

is the **tail σ-algebra** of B: the events of the remote future. For example, $\limsup_n A_n \in \mathcal{T}$ for $A_m \in \hat{\mathcal{F}}_m^0$, since $\bigcup_{m \geq n} A_m \in \hat{\mathcal{F}}_n^0$ and $(\hat{\mathcal{F}}_n^0)$ is decreasing.

Theorem 5.25

P_0 is deterministic on $\mathcal{T}$: $P_0(A) \in \{0, 1\}$ for all $A \in \mathcal{T}$.

Proof.

Step 1 (tail of B = germ of X). With X from Theorem 5.23, using $X_0 \equiv 0$ (so adding it changes no σ-algebra),

$$\begin{aligned}\mathcal{T} &= \bigcap_{t>0} \sigma(B_s \mid s \geq t) = \bigcap_{t>0} \sigma(B_{1/s} \mid 0 < s \leq \tfrac{1}{t}) = \bigcap_{t>0} \sigma(X_s \mid 0 < s \leq t) \\ &= \bigcap_{t>0} \sigma(X_s \mid 0 \leq s \leq t) = \mathcal{F}_0^{0,+}(X).\end{aligned}$$

Step 2 (Blumenthal for X) [Supplement]. Transfer from Theorem 4.16 (lectures: Le Gall, Thm. 2.13). Set $X = 0$ on the null set where it is discontinuous at 0; then $X : \Omega \to C(\mathbb{R}_+)$

is measurable, $P_0 \circ X^{-1} = P_0^W$ (Wiener measure, same finite-dimensional laws), and $\sigma(X_s \mid s \le t) = X^{-1}(\mathcal{F}_t^0)$. For $A \in \mathcal{F}_0^{0,+}(X)$ choose $\Gamma_n \in \mathcal{F}_{1/n}^0$ with $A = X^{-1}(\Gamma_n)$. Then $P_0^W(\Gamma_n \triangle \Gamma_m) = P_0(X^{-1}(\Gamma_n \triangle \Gamma_m)) = 0$, so $\Gamma := \limsup_n \Gamma_n \in \mathcal{F}_0^{0,+}$ differs from each Γ_n by a P_0^W-null set. Hence $P_0(A) = P_0^W(\Gamma) \in \{0,1\}$ by Theorem 4.16. ∎

Theorem 5.26

For the canonical one-dimensional BM, P_0-a.s.

$$\limsup_{t\to\infty} \frac{B_t}{\sqrt{t}} = +\infty, \qquad \liminf_{t\to\infty} \frac{B_t}{\sqrt{t}} = -\infty.$$

Proof. Let $K > 0$. By Fatou applied to indicators and scaling invariance (Proposition 4.2),

$$P_0\Big(\limsup_n \Big\{\frac{B_n}{\sqrt{n}} \ge K\Big\}\Big) \ge \limsup_n P_0\Big(\frac{B_n}{\sqrt{n}} \ge K\Big) = P_0(B_1 \ge K) > 0.$$

The event on the left lies in $\mathcal{T}$, so by Theorem 5.25 it has probability 1. As K is arbitrary, $\limsup_t B_t/\sqrt{t} = \infty$ a.s.; the second claim follows by symmetry ($-B$ is a BM). ∎

The next corollary holds for every BM, but we prove it for the canonical one, as a consequence of the SMP w.r.t. $(\mathcal{F}_t^{0,+})$.

Corollary 5.27

Let $N(\omega) := \{s \in [0,1] \mid B_s(\omega) = 0\}$. For P_0-a.e. ω: (i) $N(\omega)$ is closed;
(ii) $N(\omega)$ has Lebesgue measure zero;
(iii) $N(\omega)$ has no isolated points. So $N(\omega)$ is a Cantor-type set: closed, null, and every point is a cluster point.

Proof. (i) ω is continuous and $\{0\}$ is closed.

(ii) By Fubini, $\mathbb{E}_0\big[\int_0^1 \mathbb{1}_{N(\cdot)}(s)\,ds\big] = \int_0^1 P_0(B_s = 0)\,ds = 0$, so $\lambda(N(\omega)) = 0$ for P_0-a.e. ω.

(iii)

Step 1. Let $A_{r,s} := \{N(\omega) \cap (r,s) \text{ is exactly one point}\}$; then

$$J := \{N(\omega) \text{ has isolated points}\} = \bigcup_{r<s\in\mathbb{Q}_+} A_{r,s}.$$

Let $T_r := r + \sigma_{\{0\}} \circ \theta_r = \inf\{t > r \mid B_t = 0\}$.

Step 2 (T_r is a stopping time) [Supplement]. $\{T_r < t\} = \bigcup_n \{\exists s \in [r + \frac{1}{n}, t - \frac{1}{n}] : B_s = 0\} = \bigcup_n \bigcap_m \bigcup_{q\in\mathbb{Q}\cap[r+\frac{1}{n},t-\frac{1}{n}]} \{|B_q| < \frac{1}{m}\} \in \mathcal{F}_t^0$ (continuity and compactness), so T_r is a weak stopping time, hence an $(\mathcal{F}_t^{0,+})$-stopping time (Lemma 5.4).

Step 3 ($P_0(\sigma_{\{0\}} = 0) = 1$) [Supplement]. $A_+ := \bigcap_n \{\sup_{s\le 1/n} B_s > 0\} \in \mathcal{F}_0^{0,+}$ and $P_0(\sup_{s\le 1/n} B_s > 0) \ge P_0(B_{1/n} > 0) = \frac{1}{2}$, so $P_0(A_+) \ge \frac{1}{2}$ and, by Blumenthal (Theorem 4.16), $P_0(A_+) = 1$. Likewise $A_- := \bigcap_n \{\inf_{s\le 1/n} B_s < 0\}$ has probability 1. On $A_+ \cap A_-$ the path takes both signs in every $(0, \frac{1}{n})$, hence vanishes there (intermediate value theorem), so $\sigma_{\{0\}} = 0$.

Step 4 (SMP). Since $B_{T_r} = 0$ on $\{T_r < \infty\}$, by Theorem 2.35(iii) and the SMP,

$$P_0(A_{r,s}) \le P_0(T_r < s,\ \sigma_{\{0\}} \circ \theta_{T_r} > 0) = \mathbb{E}_0\big[\mathbb{1}_{\{T_r<s\}} \mathbb{E}_{B_{T_r}}[\mathbb{1}_{\{\sigma_{\{0\}}>0\}}]\big] = P_0(\sigma_{\{0\}} > 0)\, P_0(T_r < s) = 0.$$

Hence $P_0(J) = 0$. ∎

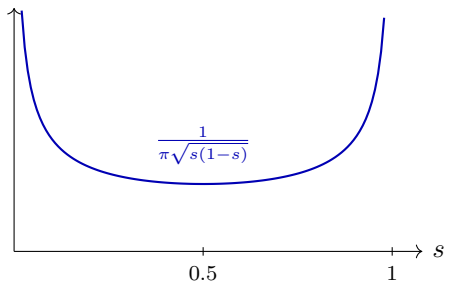


Figure. *Arcsine density of the last zero L before 1 (Example 5.22).*